MSBTE Diploma in Mechanical Engineering

Semester 3 · K-Scheme

Solved Question Papers

Question papers with step-by-step model-style answers

Subject	Code	Papers included in this book
Strength of Materials	313308	Summer 2023 (extra paper, I-scheme)
Fluid Mechanics & Machinery	313309	none received yet
Thermal Engineering	313310	Winter 2024, Summer 2025
Production Drawing	313311	Winter 2024, Summer 2025

Read this first: Only part of the requested set was received: 4 of the 12 papers (Thermal Engineering W24 and S25, Production Drawing W24 and S25). The Strength of Materials file supplied is **Summer 2023, I-scheme (paper code 22306)**, not one of the three sessions you listed, so it is included as a clearly labelled extra paper. No Fluid Mechanics paper was received. Missing papers are listed on the next page and are not guessed.

How the answers were made. The questions are copied from the papers you supplied. The solutions are written by Claude in MSBTE model-answer style; they are **not** the official MSBTE model answers, so the marks split can differ from the official scheme. Every numerical was calculated in a Python script and re-checked by a second method. Where data looked wrong, a **Data note** box states the assumption.

Reading report: which papers were received

You asked for 4 subjects × 3 sessions = 12 papers. This table shows what was actually received and readable. Nothing was guessed for papers that are missing.

Subject	Winter 2024	Summer 2025	Winter 2025
Strength of Materials (313308)	**NOT RECEIVED**	**NOT RECEIVED**	**NOT RECEIVED**
Fluid Mechanics & Machinery (313309)	**NOT RECEIVED**	**NOT RECEIVED**	**NOT RECEIVED**
Thermal Engineering (313310)	Received and read ✓	Received and read ✓	**NOT RECEIVED**
Production Drawing (313311)	Received and read ✓	Received and read ✓	**NOT RECEIVED**

Requested papers received: 4 of 12. Missing: SOM-W24, SOM-S25, SOM-W25, FMM-W24, FMM-S25, FMM-W25, TE-W25, PD-W25.

Page-by-page reading status of the files received

Paper	File	Pages	How read	Unreadable
PD-W24	Production_Drawing_Winter_2024_Question_Paper_First_K_Scheme_Paper.pdf	8	text layer + page images	none
PD-S25	Production_Drawing_Summer_2025_Question_Paper_Second_K_Scheme_Paper.pdf	7	text layer + page images	none
TE-W24	Thermal_Engineering_Winter_2024_Question_Paper_First_K_Scheme_Paper.pdf	3	text layer	none
TE-S25	Thermal_Engineering_Summer_2025_Question_Paper_Second_K_Scheme_Paper.pdf	4	text layer + page images	none
SOM-S23	2023_summer_SOM.pdf	6	scanned images read page by page (watermark over some lines, still legible)	none

Warning: SOM-S23: Summer 2023, paper code 22306 (I-scheme). Not one of the requested sessions and not K-scheme.

Data notes: questions with doubtful data

These questions have data that looks misprinted, missing or inconsistent. The assumption used in the solution is given. Each is repeated in a box at the question itself.

Question	Doubt	Assumption used
SOM-S23 Q.1(d)	The printed term "point of contra-shear" is not a standard term in Strength of Materials.	It is taken to mean the **point of contraflexure** (bending moment changes sign). Note: the point where shear force is zero is the point of maximum bending moment, which is a different thing.
SOM-S23 Q.2(b)	The figure gives lengths in cm (120, 60, 90) but areas in mm².	Lengths are converted to mm (1200, 600, 900). P_1 and P_3 act to the left and P_2, P_4 to the right, as shown in the figure.
SOM-S23 Q.3(a)	The figure does not give the position of the semicircular cut-out explicitly.	As drawn, the diameter lies on the bottom edge and starts at the left edge, so the centre is 150 mm from the left edge and 800 mm from A-A.
SOM-S23 Q.5(a)	The length of the tubes is printed as 140 mm, which equals the steel inside diameter (a typing slip is possible).	It is used as printed. The stresses and loads do not depend on the length; only the shortening (0.0853 mm) does.
SOM-S23 Q.5(b)	The figure shows the 8 kN/m load as a continuous wavy line along the whole beam A to C, and supports at A and B.	The UDL is taken over the full length of 7 m and the supports at A and B (5 m apart), with the beam overhanging 2 m beyond B.
SOM-S23 Q.6(c)	The figure shows a 144 mm dimension but does not say clearly which line it is measured to.	It is taken as the distance e from the load line to the centroidal axis of the vertical member of the clamp.
TE-W24 Q.4(b)	The wall area is not given in the question.	Area A = 1 m² is assumed, so the answer is the heat flow **per square metre** of wall. For any other area, multiply by that area.
TE-W24 Q.4(e)	Steam-table values differ slightly from book to book (for example v_g at 9 bar is 0.2148–0.2149 m³/kg).	Values were taken from the IAPWS-IF97 steam tables (checked in code). Use the values from your own steam table in the exam; the result will differ by about 1 °C.
TE-S25 Q.4(e)	The data are slightly inconsistent: with γ = 1.407 an adiabatic change from 0.14 m³ to 0.057 m³ would not give a final pressure of 4 bar (the end states imply γ = 1.54).	The end-state data are used as printed ($T_2 = p_2V_2/mR$); this is the usual method. The alternative T_2 from the adiabatic law would give a different ΔU.
TE-S25 Q.5(b)	The table prints v_f = 1.127. The correct value is 0.001127 m³/kg (1.127 is in litres/kg, or the decimal is missing). "s_f = 2.140" is also not needed.	v_f, s_f and s_g are not needed for this question, so the answer is not affected. Only t_s, v_g and h_g are used.
TE-S25 Q.6(b)	The thermal conductivities are printed as "0.7, 0.05, 0.20 KW/m°C". Brick, cork and wood have conductivities of 0.7, 0.05 and 0.20 **W**/m°C (a kilowatt value would be 1000 times too large). Also, the interior wall is colder (−5 °C) than the outer wall (15 °C), so heat flows **into** the room (cold store).	k is taken as 0.7, 0.05 and 0.20 W/m°C. "Temperature at inner surfaces" is taken as the temperatures of the brick–cork and cork–wood interfaces.
PD-W24 Q.2(a)	The printed figure is small; a few details (the exact vertical lines in the base of the F.V., and the lines at 30 mm in the S.V.) are hard to read.	The drawing reproduces them as printed (a line at 40 mm from the right end in the F.V.; lines at ± 15 mm in the S.V.). Check against your paper.
PD-W24 Q.2(b)	The figure is small and the angle of the inclined face, and the end radius of the slot are not clear: R12 does not match the slot width of 20 mm (15 + 20 + 15 = 50).	The inclined face is taken as 30° to the XY line and the slot width as 20 mm (end radius 10 mm). No finished view is drawn here because the shape cannot be established with certainty; the method and the calculated sizes are given.
PD-W24 Q.3(d)	The hole size is printed with a lower deviation of −0.04 (below the basic size). For a hole this is unusual (holes normally have a lower deviation of 0 or more).	The sizes are used exactly as printed, giving a transition fit.

Question	Doubt	Assumption used
PD-W24 Q.4(a)	Fig. No. 5 is a small scan; some sizes (the exact outline of the fork ears and the length of the collar) are hard to read.	The drawings use the sizes legible in the figure (Ø88, 86, 44, 28, Ø30, 106, Ø50, Ø6, 30) and are approximate in proportion. Take exact sizes from your paper's Fig. 5 in the exam.
PD-W24 Q.4(b)	Fig. No. 5 is a small scan; some sizes (the exact outline of the fork ears and the length of the collar) are hard to read.	The drawings use the sizes legible in the figure (Ø88, 86, 44, 28, Ø30, 106, Ø50, Ø6, 30) and are approximate in proportion. Take exact sizes from your paper's Fig. 5 in the exam.
PD-W24 Q.4(c)	Fig. No. 5 is a small scan; some sizes (the exact outline of the fork ears and the length of the collar) are hard to read.	The drawings use the sizes legible in the figure (Ø88, 86, 44, 28, Ø30, 106, Ø50, Ø6, 30) and are approximate in proportion. Take exact sizes from your paper's Fig. 5 in the exam.
PD-W24 Q.5(a)	In Fig. 6, the hub diameter is printed as “φ12D” and the flange keyway width as “28H7”. The key is 20 mm wide (20n6) and the shaft keyway is 20H7.	“φ12D” is read as Ø120 (2 × shaft diameter), and the flange keyway width is taken as 20 mm to suit the 20 mm key. The key length 96 is more than the 86 mm bore, so the key is shown projecting 10 mm beyond the hub.
PD-W24 Q.5(b)	In the tolerance chart the entry for 24H7 is printed as “+0.021 / +40.000” and the wedge size “24n8” in the chart but “24n6” on the drawing. The post carries “φ60H7” and the block hole “φ60e8”, which is the reverse of normal practice (capital letters are for holes).	24H7 is taken as +0.021 / 0.000 and 24n6 (+0.036 / +0.015). The Ø60 hole (block) is taken as H7 and the Ø60 post as e8, so that the fit is a clearance fit; if taken literally as printed the hole would be smaller than the shaft.
PD-S25 Q.1(a)	“I-section pipe” is not a standard item; an I-section is a rolled steel section, not a pipe.	It is answered as the conventional representation of a break in an I-section (the usual BIS SP-46 item “break in rolled sections”).
PD-S25 Q.2(a)	The printed figure is small. The exact shape of the arm end (two 10 mm layers and the square hole turned 45°) cannot be read with certainty.	Only the lengths that can be calculated are given (above); no finished view is drawn here, because it would need guessing the shape of the object. Check them against your figure.
PD-S25 Q.3(e)	The question says “plants”; plates are meant.	Read as “plates”.
PD-S25 Q.4	Fig. No. 5 is a rotated, low-resolution scan. The label “20” beside the groove, the exact length of the shaft and key, and the fit label printed on the drawing (“50H6/f6”) differ from the fit chart (“50H8/g8”).	Sizes in the table are those legible in the figure; the fit chart is used for the fits. The shaft and key length are drawn to suit the hub (not given).
PD-S25 Q.5(a)	Fig. 6 is small and several labels are doubtful: the body bore is printed “Ø68Hz” (read as H7); the bush is “68g6” on the part but the tolerance chart gives “68h6”; the bush bore “50H7” has the chart value +0.039 (that is H8); signs of e8 and g6 are lost in the scan.	H7 is used for the body bore; the chart values (68H7/68h6, 50H7 +0.039/0 with 50e8 −0.050/−0.089, 8H7/8g6) are used for the fits. The assembly sketch shows the main arrangement with approximate proportions.
PD-S25 Q.5(b)	The post carries “φ60H7” and the block hole “φ60e8”, which is the reverse of normal practice (capital letters are for holes). The wedge shows “24 h6” in its view while the chart gives “24n6”.	The Ø60 hole (block) is taken as H7 and the Ø60 post as e8 (clearance fit) – if taken literally the hole would be smaller than the shaft. The wedge is taken as 24n6 as in the chart.

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Strength of Materials (313308)

Session	Status
Winter 2024	**Not received – not included**
Summer 2025	**Not received – not included**
Winter 2025	**Not received – not included**
Summer 2023 (extra paper, I-scheme)	Extra paper supplied – included, clearly labelled (I-scheme (paper code 22306))

Marks shown against sub-questions are the paper's group marks divided by the number of sub-questions to attempt (MSBTE equal split); the papers print only the group total.

Papers not received: Winter 2024, Summer 2025, Winter 2025 – not included

Not received, so not included: Winter 2024, Summer 2025, Winter 2025.

Questions and solutions for this paper will be added when the paper is supplied. Nothing has been guessed.

Summer 2023 (extra paper, I-scheme)

Subject	Strength of Materials (313308)	Time / Marks	3 Hours / 70 Marks
Session	Summer 2023 (extra paper, I-scheme)	Paper ID printed	22306 / 22223
Scheme	I-scheme (paper code 22306)	Source file	2023_summer_SOM.pdf

Paper note: This paper is **not** one of the sessions you asked for (Winter 2024, Summer 2025, Winter 2025) and it is from the older **I-scheme** (paper code 22306). It is included because it was the only Strength of Materials paper supplied. Topics may differ slightly from the K-scheme syllabus. The scan has a website watermark over some lines, but all text and figures were readable.

Q.1 Attempt any FIVE of the following: — 10 Marks (each sub-question = 2 marks)

Q.1 (a) [2 Marks]

Define radius of gyration. State its S.I. unit.

Ans.

Radius of gyration (k) of an area about an axis is the distance from that axis at which the whole area can be assumed to be concentrated so that it gives the same moment of inertia as the actual area.

$$k = \sqrt{(I / A)} \text{ or } I = A\,k^2$$

S.I. unit: metre (m); in practice mm.

Marks split	Marks
Definition	1
Formula and unit	1
Total	**2**

Q.1 (b) [2 Marks]

Define shear strain and modulus of rigidity.

Ans.

- **Shear strain (φ):** the angular deformation (in radian) produced in a body, measured as the angle through which a line originally perpendicular to the fixed face is turned under a shear force.
- **Modulus of rigidity (G):** the ratio of shear stress to shear strain within the elastic limit. $G = \tau / \varphi$ (unit N/mm² or GPa).

Marks split	Marks
Shear strain	1
Modulus of rigidity	1
Total	**2**

Q.1 (c) [2 Marks]

Define strain energy. State its S.I. unit.

Ans.

Strain energy is the energy stored in a body when it is deformed by an external load, within the elastic limit. It is equal to the work done by the load in deforming the body.

$$U = \tfrac{1}{2}\,P\,\delta = \sigma^2\,V / (2E)$$

S.I. unit: joule (J) = N·m.

Marks split	Marks
Definition	1
Formula and unit	1
Total	**2**

Q.1 (d) [2 Marks]

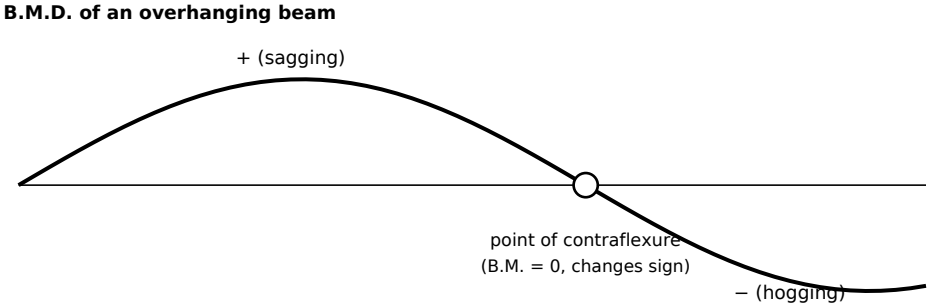
State point of contra-shear with suitable sketch.

Ans.

Data note: The printed term "point of contra-shear" is not a standard term in Strength of Materials.
Assumption used: It is taken to mean the **point of contraflexure** (bending moment changes sign). Note: the point where shear force is zero is the point of maximum bending moment, which is a different thing.

The question says "contra-shear"; the standard term for this idea in beams is the **point of contraflexure** (also called point of inflection).

- It is the point on a beam where the **bending moment changes sign** (from sagging to hogging or the reverse), so the bending moment there is zero.
- Such points occur in overhanging and continuous beams. At this point the curvature of the beam changes from concave to convex.



Point of contraflexure on a B.M.D.

Marks split	Marks
Statement	1
Sketch	1
Total	**2**

Q.1 (e) [2 Marks]

Write the flexural formula and state the meaning of each term used in it.

Ans.

$$M / I = \sigma_b / y = E / R$$

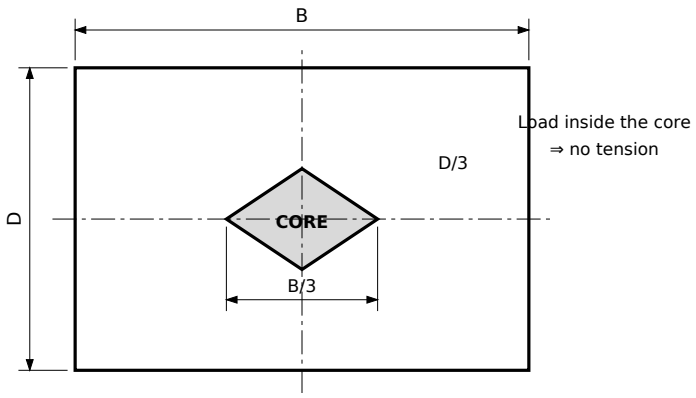
Symbol	Meaning	Unit
M	Bending moment at the section	N·mm
I	Moment of inertia of the cross-section about the neutral axis	mm^4
σ_b	Bending stress at a layer at distance y from the neutral axis	N/mm^2
y	Distance of the layer from the neutral axis	mm
E	Young’s modulus of the beam material	N/mm^2
R	Radius of curvature of the neutral layer	mm

Marks split	Marks
Formula	1
Meaning of terms	1
Total	**2**

Q.1 (f) [2 Marks]

Draw a neat sketch to show core of rectangular section of (B × D) dimensions.

Ans.



Core of a rectangular section (middle-third rule)

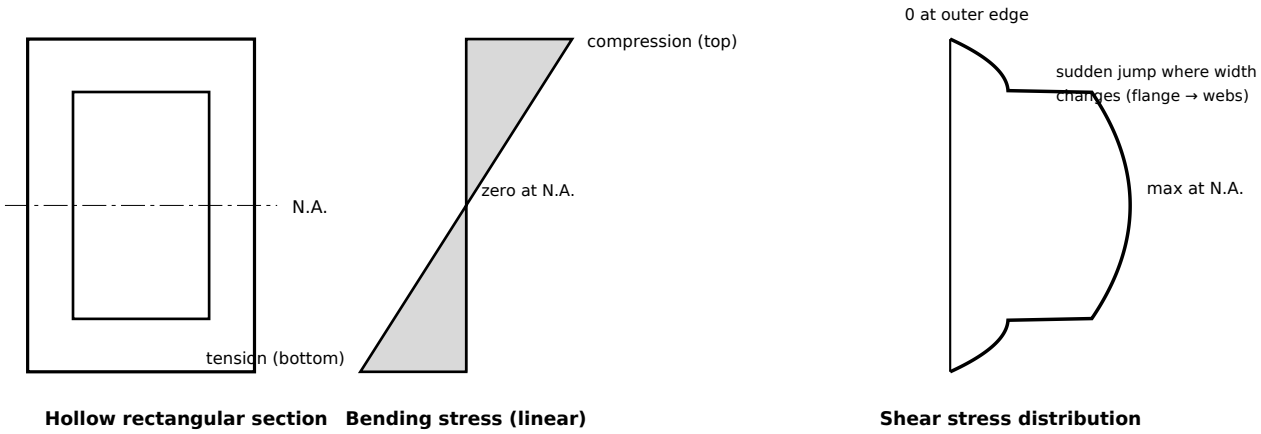
The core is a **rhombus (diamond)** whose diagonals are B/3 along the breadth and D/3 along the depth, symmetrical about the centroid. If the load acts anywhere inside this core, no tensile stress is produced in the section.

Marks split	Marks
Sketch	1
Dimensions B/3 and D/3	1
Total	**2**

Q.1 (g) [2 Marks]

Draw shear stress and bending stress distribution diagram for hollow rectangular beam section.

Ans.



Bending and shear stress distribution – hollow rectangular section

- **Bending stress** varies linearly with distance from the neutral axis: zero at N.A. and maximum (compressive at top, tensile at bottom) at the extreme fibres.
- **Shear stress** is zero at the top and bottom edges, increases parabolically in the flanges, jumps suddenly at the inner face where the width reduces to the two webs, and is maximum at the neutral axis.

Marks split	Marks
Bending stress diagram	1
Shear stress diagram	1
Total	**2**

Q.2 Attempt any THREE of the following: — 12 Marks (each sub-question = 4 marks)

Q.2 (a) [4 Marks]

A hollow circular section having 200 mm external diameter and 100 mm internal diameter. Calculate the moment of inertia of the section about any of the tangent. Also find polar moment of inertia.

Ans.

Numerical: Hollow circular section: MI about tangent and polar MI

Given data	Value	Unit	Remark
External diameter D	200	mm	
Internal diameter d	100	mm	

Formula and calculation:

1. MI about a diameter (centroidal axis)
$I = \pi (D^4 - d^4) / 64 = \pi (200^4 - 100^4) / 64 =$ **7.363×10^7 mm⁴**

2. Area of section
$A = \pi (D^2 - d^2) / 4 = \pi (200^2 - 100^2) / 4 =$ **23561.94 mm²**

3. MI about tangent (parallel-axis theorem)
$I_T = I + A h^2$, $h = D/2 = 100$ mm $= 7.3631 \times 10^7 + 23561.94 \times 100^2 =$ **3.093×10^8 mm⁴**

4. Polar MI
$J = \pi (D^4 - d^4) / 32 = \pi (200^4 - 100^4) / 32 =$ **1.473×10^8 mm⁴**

Final answer
MI about tangent $I_T =$ **3.093×10^8 mm⁴**
Polar MI $J =$ **1.473×10^8 mm⁴**

Checked in code by a second method: ✓ I_T = (outer solid circle) − (inner solid circle) about the same tangent: $3.093 \times 10^8 \approx 3.093 \times 10^8$; ✓ J = 2 I (perpendicular-axis theorem): $1.473 \times 10^8 \approx 1.473 \times 10^8$

Marks split	Marks
Given data and formula	1
Moment of inertia about centroidal axis	1
Parallel-axis theorem – tangent	1
Polar moment of inertia	1
Total	**4**

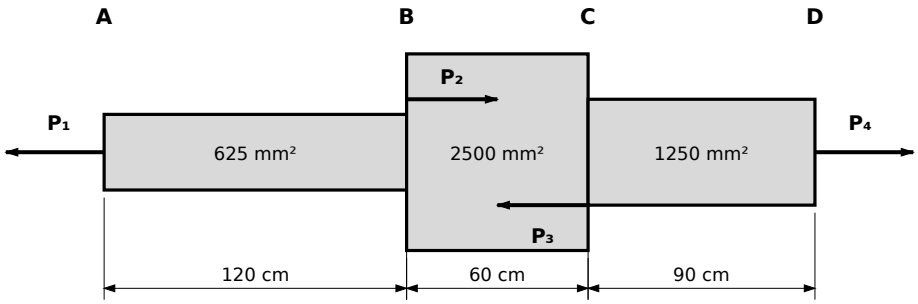
Q.2 (b) [4 Marks]

A member ABCD is subjected to point loads of P_1, P_2, P_3 and P_4 as shown in Figure No. 1. Calculate the force P_2 necessary for equilibrium, if P_1 = 45 kN, P_3 = 450 kN and P_4 = 130 kN. Determine the total elongation of the member, assuming the modulus of elasticity to be 2.1×10^5 N/mm².
[Figure No. 1: member ABCD – AB 625 mm², 120 cm; BC 2500 mm², 60 cm; CD 1250 mm², 90 cm]

Ans.

Data note: The figure gives lengths in cm (120, 60, 90) but areas in mm².
Assumption used: Lengths are converted to mm (1200, 600, 900). P_1 and P_3 act to the left and P_2, P_4 to the right, as shown in the figure.

Axial forces: AB = 45 kN (tension), BC = 320 kN (compression), CD = 130 kN (tension)



Stepped member ABCD (Figure No. 1)

Numerical: Stepped member ABCD in equilibrium

Given data	Value	Unit	Remark
P_1	45	kN	
P_3	450	kN	
P_4	130	kN	
Areas A_{AB}, A_{BC}, A_{CD}	625, 2500, 1250	mm²	
Lengths L_{AB}, L_{BC}, L_{CD}	120, 60, 90	cm	= 1200, 600, 900 mm
E	210000	N/mm²	

Formula and calculation:

1. Equilibrium of whole member (→ positive): $-P_1 + P_2 - P_3 + P_4 = 0$
$P_2 = P_1 + P_3 - P_4 = 45 + 450 - 130 =$ **365.00 kN**

2. Axial force in AB (tension)
$N_{AB} = P_1 = 45 =$ **45.00 kN**

3. Axial force in BC (compression)
$N_{BC} = P_1 - P_2 = 45 - 365 =$ **-320.00 kN**

4. Axial force in CD (tension)
$N_{CD} = P_4 = 130 =$ **130.00 kN**

5. Change in length of AB
$\delta = N\,L / (A\,E) = (45 \times 10^3 \times 1200) / (625 \times 210000) =$ **0.4114 mm**

6. Change in length of BC
$\delta = N\,L / (A\,E) = (-320 \times 10^3 \times 600) / (2500 \times 210000) =$ **-0.3657 mm**

7. Change in length of CD

$\delta = N\,L / (A\,E) = (130 \times 10^3 \times 900) / (1250 \times 210000) =$ **0.4457 mm**

8. Total elongation

$\delta = \delta_{AB} + \delta_{BC} + \delta_{CD} = 0.4114 + (-0.3657) + 0.4457 =$ **0.4914 mm**

Final answer
Force P_2 = **365.00 kN**
Total elongation of member = **0.4914 mm**

Checked in code by a second method: ✓ N_{AB} from the right-hand side equals P_1: 45.0000 ≈ 45.0000; ✓ N_{BC} from the right-hand side ($P_4 - P_3$): -320.0000 ≈ -320.0000; ✓ Σ N L / A E recomputed using right-hand forces: 0.4914 ≈ 0.4914

Marks split	Marks
Equilibrium – value of P_2	1
Axial force in each part	1
Elongation of each part	1
Total elongation	1
Total	**4**

Q.2 (c) [4 Marks]

In a tri-axial stress system, the stresses along the three directions are σ_x = 100 N/mm² (tensile), σ_y = 60 N/mm² (tensile) and σ_z = 30 N/mm² (compressive). Find the strains in each direction. Take E = 2×10^5 N/mm² and μ = 0.25. If X = 400 mm, Y = 150 mm and Z = 300 mm. Also calculate the change in volume.

Ans.

Numerical: Tri-axial stress: strains and volume change

Given data	Value	Unit	Remark
σ_x	100	N/mm²	tensile
σ_y	60	N/mm²	tensile
σ_z	-30	N/mm²	compressive, so negative
E	200000	N/mm²	
μ	0.25		
X, Y, Z	400, 150, 300	mm	

Formula and calculation:

1. Strain in x direction

$\varepsilon_x = [\sigma_x - \mu(\sigma_y + \sigma_z)] / E = [100 - 0.25(60 + (-30))] / 200000 =$ **4.625×10^{-4}**

2. Strain in y direction

$\varepsilon_y = [\sigma_y - \mu(\sigma_x + \sigma_z)] / E = [60 - 0.25(100 + (-30))] / 200000 =$ **2.125×10^{-4}**

3. Strain in z direction

$\varepsilon_z = [\sigma_z - \mu(\sigma_x + \sigma_y)] / E = [-30 - 0.25(100 + 60)] / 200000 =$ **-3.5×10^{-4}**

4. Volumetric strain

$\varepsilon_v = \varepsilon_x + \varepsilon_y + \varepsilon_z = 0.0004625 + 0.0002125 + (-0.00035) =$ **3.25×10^{-4}**

5. Change in volume

$\delta V = \varepsilon_v \times V$, $V = 400 \times 150 \times 300 = 0.000325 \times 1.8 \times 10^7 =$ **5850.0 mm³**

Final answer
ε_x = **4.625×10^{-4}**
ε_y = **2.125×10^{-4}**
ε_z = **-3.5×10^{-4} (negative = contraction)**
Change in volume δV (increase) = **5850.0 mm³**

Checked in code by a second method: ✓ $\varepsilon_v = (1 - 2\mu)(\sigma_x + \sigma_y + \sigma_z) / E$: $3.25 \times 10^{-4} \approx 3.25 \times 10^{-4}$; ✓ δV from changed sides: (X(1+ε_x))(Y(1+ε_y))(Z(1+ε_z)) − XYZ: 5847.5159 ≈ 5850.0000

Marks split	Marks
Formulae for strains	1
Strains ε_x, ε_y, ε_z	1
Volumetric strain	1
Change in volume	1
Total	**4**

Q.2 (d) [4 Marks]

A simply supported beam of span ‘L’ is subjected to downward point load of ‘W’ at a distance ‘a’ from left support and ‘b’ from right support. Draw SF and BM diagrams. Take a > b.

Ans.

Reactions

$$\text{Taking moments about B: } R_A \times L = W \times b \Rightarrow R_A = W\,b / L$$

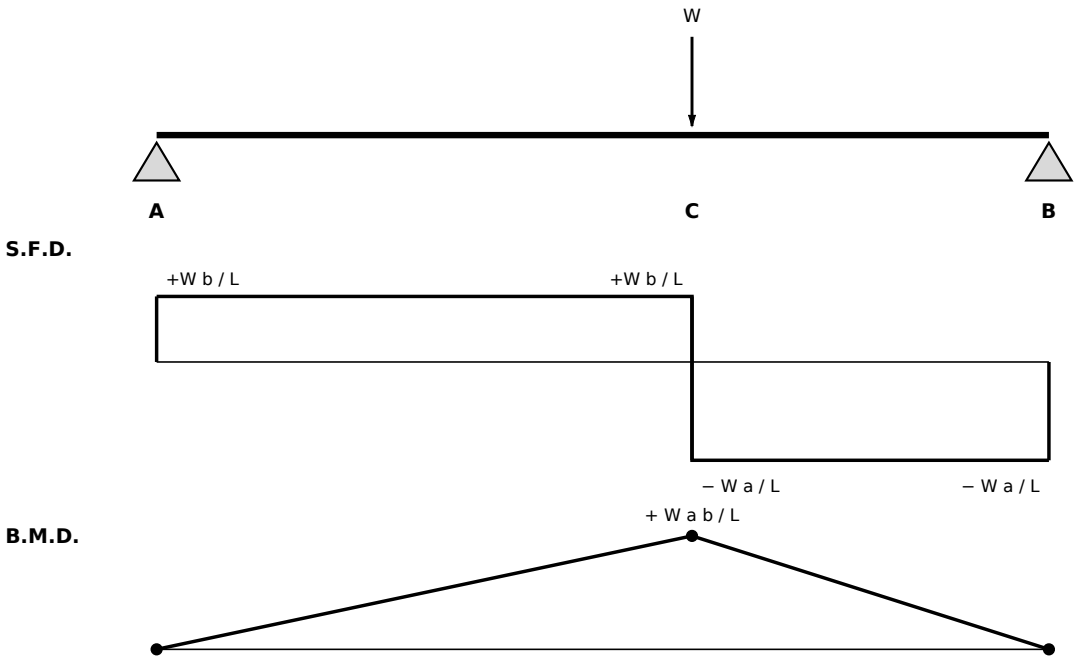
$$R_B = W - R_A = W\,a / L$$

Shear force

- A to C ($0 < x < a$): $F = +R_A = +W\,b / L$ (constant).
- C to B ($a < x < L$): $F = R_A - W = -W\,a / L$ (constant).

Bending moment

- At A and B: $M = 0$.
- At C (under the load): $M_C = R_A \times a =$ **$+W\,a\,b / L$** (maximum, because S.F. changes sign here).



S.F.D. and B.M.D. (the figure is drawn for a > b)

Since a > b, the right reaction W a / L is the larger one and the maximum shear force is W a / L.

Marks split	Marks
Reactions	1
S.F. values and diagram	1
B.M. values and diagram	1
Maximum values stated	1
Total	**4**

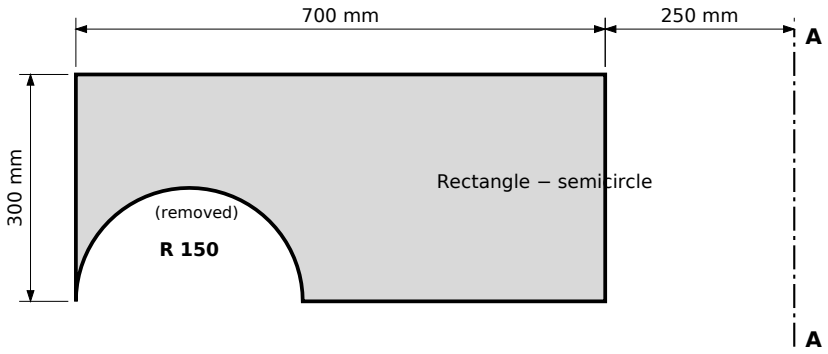
Q.3 Attempt any THREE of the following: — 12 Marks (each sub-question = 4 marks)

Q.3 (a) [4 Marks]

Calculate the moment of inertia about the axis A-A, for the lamina shown in the Figure No. 2.
[Figure No. 2: rectangle 700 mm × 300 mm with a semicircular cut-out of R 150 mm on the bottom edge; axis A-A is 250 mm from the right edge]

Ans.

Data note: The figure does not give the position of the semicircular cut-out explicitly.
Assumption used: As drawn, the diameter lies on the bottom edge and starts at the left edge, so the centre is 150 mm from the left edge and 800 mm from A-A.



Lamina (Figure No. 2)

Numerical: MI of lamina about axis A-A

Given data	Value	Unit	Remark
Rectangle B × H	700 × 300	mm	
Semicircular cut-out radius R	150	mm	diameter on the bottom edge, touching the left edge (as drawn)
Distance of A-A from right edge	250	mm	

Formula and calculation:

1. Rectangle: I about A-A
$I_1 = H\,B^3/12 + A_1\,d_1^2$, $d_1 = 950 - 350 = 600 = 8.5750 \times 10^9 + 210000 \times 600^2 =$ **8.418×10^{10} mm⁴**

2. Semicircle: I about A-A
$I_2 = \pi\,R^4/8 + A_2\,d_2^2$, $d_2 = 950 - 150 = 800 = 1.9880 \times 10^8 + 35342.92 \times 800^2 =$ **2.282×10^{10} mm⁴**

3. Lamina = rectangle − semicircle
$I_{AA} = I_1 - I_2 = 8.4175 \times 10^{10} - 2.2818 \times 10^{10} =$ **6.136×10^{10} mm⁴**

Final answer
I about A-A = **6.136×10^{10} mm⁴**

Checked in code by a second method: ✓ Numerical integration of $\int x^2\,dA$ over the lamina (0.5 mm grid, 0.05 %): $6.135 \times 10^{10} \approx 6.136 \times 10^{10}$

Marks split	Marks
Splitting into rectangle and semicircle	1
I of rectangle about A-A	1
I of semicircle about A-A	1
Subtraction and answer	1
Total	**4**

Q.3 (b) [4 Marks]

A metal bar 20 mm in diameter and 1000 mm long is hung vertically and a collar is attached at the lower end. A weight of 1000 N falls through a height of 250 mm on the collar. Calculate the maximum instantaneous stress, elongation and strain energy stored in a bar. Take E = 2×10^5 N/mm².

Ans.

Numerical: Impact load on a hanging bar

Given data	Value	Unit	Remark
Diameter d	20	mm	
Length L	1000	mm	
Falling weight W	1000	N	
Height of fall h	250	mm	
E	200000	N/mm²	

Formula and calculation:

1. Area of bar

$A = \pi d^2 / 4 = \pi \times 20^2 / 4 =$ **314.16 mm²**

2. Maximum instantaneous stress

$\sigma = (W/A)\,[1 + \sqrt{(1 + 2\,A\,E\,h / (W\,L))}] = (1000/314.16)\,[1 + \sqrt{(1 + 2 \times 314.16 \times 200000 \times 250 / (1000 \times 1000))}] =$ **567.38 N/mm²**

3. Instantaneous elongation

$\delta = \sigma\,L / E = 567.38 \times 1000 / 200000 =$ **2.837 mm**

4. Strain energy stored

$U = \sigma^2 / (2E) \times A\,L = 567.38^2 / (2 \times 200000) \times 314.16 \times 1000 =$ **252.84 N·m**

Final answer
Maximum instantaneous stress = **567.38 N/mm²**
Elongation = **2.837 mm**
Strain energy stored = **252.84 N·m**

Checked in code by a second method: ✓ Energy balance: W (h + δ) = U: 252.8369 ≈ 252.8369; ✓ Equivalent static load P = σ A satisfies P δ / 2 = U: 252.8369 ≈ 252.8369

Marks split	Marks
Area and formula	1
Maximum stress	1
Elongation	1
Strain energy	1
Total	**4**

Q.3 (c) [4 Marks]

Simply supported beam having span 7 m carries two point loads 3 kN at distance of 3 m from left hand support and 5 kN at distance of 2 m from right hand support. In addition to this an udl of 2 kN/m over a span of 3 m from left hand support. Draw SF and BM diagrams.

Ans.

Numerical: Simply supported beam: SF and BM values

Given data	Value	Unit	Remark
Span L	7	m	
Point load W_1	3 kN at 3 m from left support		

Given data	Value	Unit	Remark
Point load W_2	5 kN at 2 m from right support, i.e. 5 m from left		
UDL	2 kN/m over 3 m from left support		total 6 kN, acts at 1.5 m

Formula and calculation:

1. Moments about A ($R_B \times 7$)
$R_B \times 7 = 3\times3 + 5\times5 + (2\times3)\times1.5 = 9 + 25 + 9 = 43 =$ **6.143 kN**

2. Vertical equilibrium
$R_A = (3 + 5 + 6) - R_B = 14 - 6.143 =$ **7.857 kN**

3. SF at A (right)
7.857 kN

4. SF at C (3 m, left of 3 kN)
1.857 kN

5. SF at C (3 m, right of 3 kN)
-1.143 kN

6. SF at D (5 m, left of 5 kN)
-1.143 kN

7. SF at D (5 m, right of 5 kN)
-6.143 kN

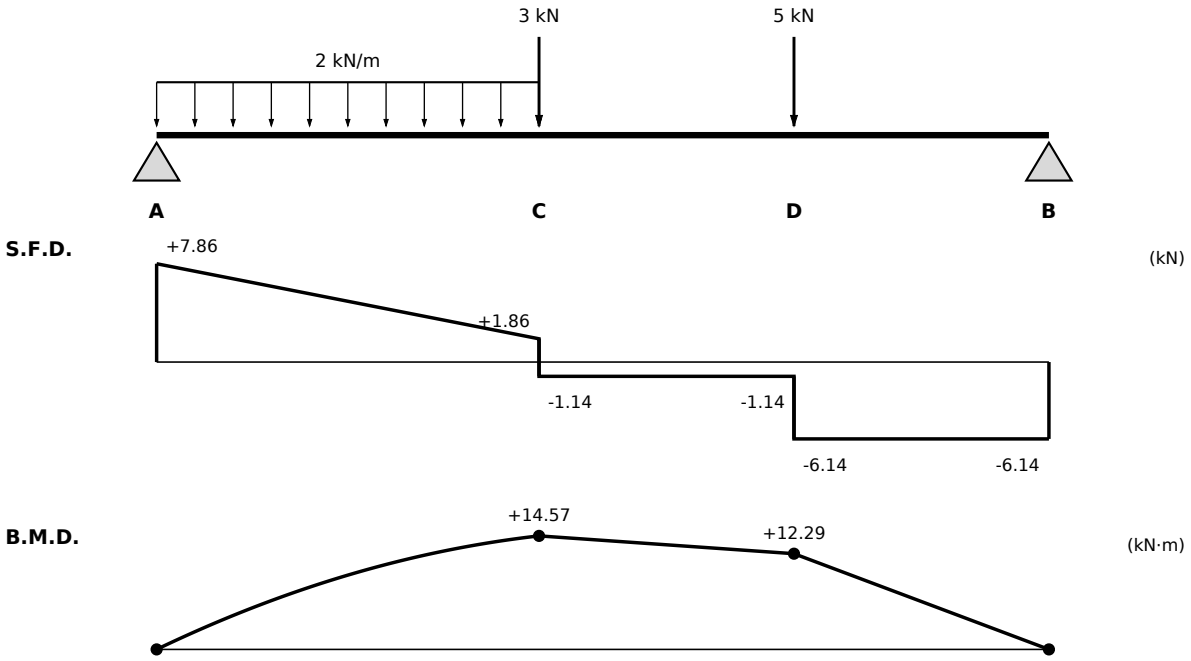
8. SF at B (left)
-6.143 kN

9. BM at C (3 m)
$M_C = R_A \times 3 - 2\times3\times1.5 = 7.857 \times 3 - 9 =$ **14.571 kN·m**

10. BM at D (5 m)
$M_D = R_A \times 5 - 2\times3\times3.5 - 3 \times 2 = 7.857 \times 5 - 21 - 6 =$ **12.286 kN·m**

Final answer
$R_A =$ **7.857 kN**
$R_B =$ **6.143 kN**
Maximum BM (at 3 m from A, where SF changes sign) = **14.571 kN·m**

Checked in code by a second method: ✓ R_A from the beam solver: 7.8571 ≈ 7.8571; ✓ R_B from the beam solver: 6.1429 ≈ 6.1429; ✓ BM at C from the beam solver: 14.5714 ≈ 14.5714; ✓ BM at D from the beam solver: 12.2857 ≈ 12.2857; ✓ BM at B is zero (simple support): 0.0000 ≈ 0.0000



S.F.D. and B.M.D. (values from the beam solver)

Marks split	Marks
Reactions	1
S.F. values	1
B.M. values	1
Diagrams	1
Total	**4**

Q.3 (d) [4 Marks]

A short column of external diameter 40 cm and internal diameter 20 cm carries an eccentric load of 80 kN. Find the greatest eccentricity which the load can have without producing tension in the column section.

Ans.

Numerical: Greatest eccentricity for no tension (hollow circular column)

Given data	Value	Unit	Remark
External diameter D	40	cm	= 400 mm
Internal diameter d	20	cm	= 200 mm
Load P	80	kN	= 80 000 N

Formula and calculation:

1. Core of hollow circular section: $e = (D^2 + d^2)/(8D)$

$e = (D^2 + d^2) / (8\ D) = (400^2 + 200^2) / (8 \times 400) =$ **62.50 mm**

2. Direct stress

$\sigma_d = P / A = 80000 / 94247.78 =$ **0.8488 N/mm²**

3. Bending stress at edge

$\sigma_b = P\ e\ y / I = 80000 \times 62.50 \times 200 / 1.1781 \times 10^9 =$ **0.8488 N/mm²**

Final answer
Greatest eccentricity (radius of core) = **62.50 mm**
Greatest eccentricity = **6.25 cm**

Checked in code by a second method: ✓ Minimum stress $\sigma_d - \sigma_b = 0$ (just no tension): 0.0000 ≈ 0.0000

Marks split	Marks
Condition for no tension	1
Core formula	1
Substitution	1
Answer with unit	1
Total	**4**

Q.4 Attempt any THREE of the following: — 12 Marks (each sub-question = 4 marks)

Q.4 (a) [4 Marks]

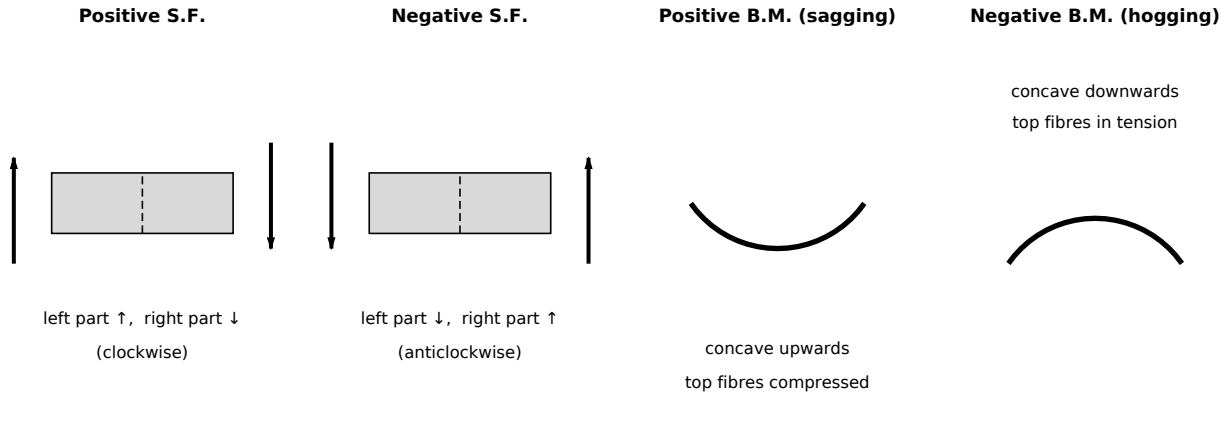
Define shear force and bending moment. Explain sign conventions for shear force and bending moment with neat sketches.

Ans.

- **Shear force (S.F.):** the algebraic sum of all the vertical forces acting on either side of a section of the beam.
- **Bending moment (B.M.):** the algebraic sum of the moments, about the section, of all the forces acting on either side of the section.

Sign conventions

- **S.F.:** positive if the left portion tends to move upward relative to the right portion (clockwise shear); negative otherwise.
- **B.M.:** positive (sagging) if it bends the beam concave upwards (top fibres in compression); negative (hogging) if it bends the beam concave downwards (top fibres in tension).



Sign conventions

Marks split	Marks
Definition of S.F.	1
Definition of B.M.	1
Sign conventions of S.F.	1
Sign conventions of B.M.	1
Total	**4**

Q.4 (b) [4 Marks]

A cast iron pipe having 800 mm external diameter and 700 mm internal diameter is used to pass the hydraulic oil. The pipe is simply supported at both the ends having length 7.5 m. Calculate the intensity of uniformly distributed load that pipe can carry. Assuming bending stress in pipe material is 140 N/mm².

Ans.

Numerical: UDL a cast iron pipe can carry

Given data	Value	Unit	Remark
External diameter D	800	mm	
Internal diameter d	700	mm	
Span L	7.5	m	= 7500 mm

Given data	Value	Unit	Remark
Permissible bending stress σ_b	140	N/mm²	

Formula and calculation:

1. MI of pipe section
$I = \pi (D^4 - d^4) / 64 = \pi (800^4 - 700^4) / 64 =$ **8.32×10^9 mm⁴**

2. Section modulus
$Z = I / y = I / (D/2) = 8.3203 \times 10^9 / 400 =$ **2.08×10^7 mm³**

3. Maximum BM the pipe can resist
$M = \sigma_b Z = 140 \times 2.0801 \times 10^7 =$ **2.912×10^9 N·mm**

4. UDL from $M = w L^2 / 8$
$w = 8 M / L^2 = 8 \times 2.9121 \times 10^9 / 7500^2 =$ **414.17 N/mm**

Final answer
UDL intensity = **414.17 N/mm**
UDL intensity = **414.17 kN/m**

Checked in code by a second method: ✓ Round trip: $\sigma = (w L^2 / 8) / Z$ gives 140 N/mm²: 140.0000 ≈ 140.0000
Note: Self-weight of pipe and oil is ignored (not asked).

Marks split	Marks
Moment of inertia and Z	1
Maximum bending moment M = σZ	1
Equation $M = wL^2/8$	1
Answer	1
Total	**4**

Q.4 (c) [4 Marks]

Write the equation of polar modulus for hollow shaft. State assumptions made in theory of pure torsion.

Ans.

$$\text{Polar modulus } Z_p = J / R = [\pi (D^4 - d^4) / 32] / (D/2) = \pi (D^4 - d^4) / (16\, D)$$

where D = external diameter and d = internal diameter of the shaft. Torque transmitted: $T = (\pi/16)\, \tau\, (D^4 - d^4)/D = \tau\, Z_p$.

Assumptions in the theory of pure torsion

1. The material of the shaft is homogeneous and isotropic.
2. The shaft has a uniform circular cross-section (solid or hollow) throughout its length.
3. Plane transverse sections remain plane and circular after twisting; radii remain straight.
4. The twist is uniform along the length; the shaft is subjected to pure torque only.
5. Stresses do not exceed the elastic limit (Hooke's law applies), so shear stress ∝ shear strain.
6. Shear stress at a point is proportional to its distance from the axis.

Marks split	Marks
Equation of polar modulus	2
Any four assumptions	2
Total	**4**

Q.4 (d) [4 Marks]

A metal bar of 50 mm × 50 mm in cross section, is subjected to axial compressive load of 500 kN. The contraction of a 200 mm gauge length is found to be 0.5 mm and increase in thickness is 0.04 mm. Find the values of Young's modulus and Poisson's ratio.

Ans.

Numerical: Young's modulus and Poisson's ratio from a compression test

Given data	Value	Unit	Remark
Cross-section	50 × 50	mm	
Axial compressive load P	500	kN	
Gauge length	200	mm	

Given data	Value	Unit	Remark
Contraction	0.5	mm	
Increase in thickness	0.04	mm	

Formula and calculation:

1. Stress
$\sigma = P / A = 500000 / (50 \times 50) =$ **200.00 N/mm²**

2. Longitudinal strain
$\varepsilon = \delta L / L = 0.5 / 200 =$ **0.00250**

3. Young's modulus
$E = \sigma / \varepsilon = 200.00 / 0.00250 =$ **80000 N/mm²**

4. Lateral strain
$\varepsilon_{lat} = \delta b / b = 0.04 / 50 =$ **8×10^{-4}**

5. Poisson's ratio
μ = lateral strain / longitudinal strain = 0.000800 / 0.00250 = **0.320**

Final answer
Young's modulus E = **80000 N/mm²**
Young's modulus E = **80.0 GPa**
Poisson's ratio μ = **0.320**

Checked in code by a second method: ✓ Reproduce the contraction: $\delta L = P\,L / (A\,E)$: 0.5000 ≈ 0.5000; ✓ Reproduce the lateral change: $\delta b = \mu\,\varepsilon\,b$: 0.0400 ≈ 0.0400

Marks split	Marks
Stress	1
Young’s modulus	1
Lateral strain	1
Poisson’s ratio	1
Total	**4**

Q.4 (e) [4 Marks]

A hollow shaft of diameter ratio 3/5 is required to transmit maximum torque of 61.5 kNm. The shear stress is not to exceed 63 MPa and twist in a length of 3 m is not to exceed 1.4°. Calculate the minimum external diameter satisfying these conditions. Take G = 84 GPa.

Ans.

Numerical: Minimum external diameter of a hollow shaft (3/5)

Given data	Value	Unit	Remark
Maximum torque T	61.5	kN·m	= 61.5 × 10⁶ N·mm
Shear stress limit τ	63	N/mm²	= 63 MPa
Twist limit θ	1.4	° in 3 m	= 1.4 × π/180 rad
Diameter ratio d/D	3/5 = 0.6		
G	84	GPa	= 84 000 N/mm²

Formula and calculation:

1. Factor $(1 - k^4)$
$1 - (d/D)^4 = 1 - 0.6^4 =$ **0.8704**

2. (a) From strength: $T = (\pi/16)\,\tau\,D^3\,(1 - k^4)$
$D = [16\,T / (\pi\,\tau\,(1 - k^4))]^{1/3} = [16 \times 6.15 \times 10^7 / (\pi \times 63 \times 0.8704)]^{1/3} =$ **178.76 mm**

3. (b) Polar MI needed for twist
$J = T\,L / (G\,\theta) = (6.15 \times 10^7 \times 3000) / (84000 \times 0.024435) =$ **8.989×10^7 mm⁴**

4. Diameter from $J = (\pi/32)\,D^4\,(1 - k^4)$
$D = [32\,J / (\pi\,(1 - k^4))]^{1/4} = [32 \times 8.9890 \times 10^7 / (\pi \times 0.8704)]^{1/4} =$ **180.09 mm**

Final answer
Minimum external diameter (larger of the two) = **180.09 mm**
Internal diameter d = 0.6 D = **108.06 mm**

Checked in code by a second method: ✓ Shear stress at the adopted D does not exceed 63 N/mm²: 61.6072 ≈ ≤ 63.0000; ✓ Twist at D equals 1.4° (stiffness governs): 1.4000 ≈ 1.4000

Marks split	Marks
Diameter from shear stress	1
Diameter from angle of twist	1
Selecting the larger value	1
Answer with units	1
Total	**4**

Q.5 Attempt any TWO of the following: — 12 Marks (each sub-question = 6 marks)

Q.5 (a) [6 Marks]

A compound tube consists of steel and brass, steel tube having 140 mm internal diameter and 160 mm external diameter, an outer side brass tube having 160 mm internal diameter and 180 mm external diameter. Two tubes having same length of 140 mm. The compound tube carries an axial load of 900 kN. Calculate the stresses in the tube materials and load carrying capacity of each material. Also calculate the amount the tubes gets shortens. Take $E_s = 2 \times 10^5$ N/mm² and $E_{BR} = 1 \times 10^5$ N/mm².

Ans.

Data note: The length of the tubes is printed as 140 mm, which equals the steel inside diameter (a typing slip is possible).
Assumption used: It is used as printed. The stresses and loads do not depend on the length; only the shortening (0.0853 mm) does.

Numerical: Compound tube of steel and brass

Given data	Value	Unit	Remark
Steel tube	140 mm ID, 160 mm OD		
Brass tube	160 mm ID, 180 mm OD		
Length of each tube	140	mm	as printed
Axial load P	900	kN	
E_s, E_{br}	2×10^5, 1×10^5	N/mm²	

Formula and calculation:

1. Area of steel
$A_s = \pi (160^2 - 140^2) / 4 =$ **4712.39 mm²**

2. Area of brass
$A_b = \pi (180^2 - 160^2) / 4 =$ **5340.71 mm²**

3. Equal strain $\Rightarrow \sigma_s = (E_s/E_{br})\, \sigma_b = 2\, \sigma_b$; $P = \sigma_s A_s + \sigma_b A_b$
$\sigma_b = P / (2\, A_s + A_b) = 900000 / (2 \times 4712.39 + 5340.71) =$ **60.95 N/mm²**

4. Stress in steel
$\sigma_s = 2\, \sigma_b = 2 \times 60.953 =$ **121.91 N/mm²**

5. Load carried by steel
$P_s = \sigma_s\, A_s = 121.906 \times 4712.39 =$ **574.47 kN**

6. Load carried by brass
$P_b = \sigma_b\, A_b = 60.953 \times 5340.71 =$ **325.53 kN**

7. Shortening of the tubes
$\delta = \sigma_s\, L / E_s = 121.906 \times 140 / 200000 =$ **0.0853 mm**

Final answer
Stress in steel = **121.91 N/mm²**
Stress in brass = **60.95 N/mm²**
Load carried by steel = **574.47 kN**
Load carried by brass = **325.53 kN**
Shortening = **0.0853 mm**

Checked in code by a second method: ✓ $P_s + P_b = 900$ kN: 900.0000 ≈ 900.0000; ✓ Strain in brass = strain in steel: $6.095 \times 10^{-4} \approx 6.095 \times 10^{-4}$

Note: The question gives the tube length as 140 mm (same as the steel ID); it is used as printed. Stresses and loads do not depend on length.

Marks split	Marks
Areas of steel and brass	1
Stresses (equal strain condition)	2
Load shared by each	2
Shortening	1
Total	**6**

Q.5 (b) [6 Marks]

Draw shear force and bending moment diagram for an overhanging beam as shown in the Figure No. 3. Also find point of contra-flexure if any.
[Figure No. 3: beam A–D–B–C; A to D 2 m, D to B 3 m, B to C 2 m; 5 kN at D; 8 kN/m over the whole length; 5 kN at C; supports at A and B]

Ans.

Data note: The figure shows the 8 kN/m load as a continuous wavy line along the whole beam A to C, and supports at A and B.
Assumption used: The UDL is taken over the full length of 7 m and the supports at A and B (5 m apart), with the beam overhanging 2 m beyond B.

Numerical: Overhanging beam: SF, BM and contraflexure

Given data	Value	Unit	Remark
Span AB	5	m	A at 0, D at 2 m, B at 5 m
Overhang BC	2	m	C at 7 m
Point loads	5 kN at D (2 m) and 5 kN at C (7 m)		
UDL	8 kN/m over the whole beam AC		total 56 kN at 3.5 m

Formula and calculation:

1. Moments about A
$R_B \times 5 = 5\times2 + 56\times3.5 + 5\times7 = 10 + 196 + 35 = 241 =$ **48.20 kN**

2. Vertical equilibrium
$R_A = 5 + 56 + 5 - R_B = 66 - 48.20 =$ **17.80 kN**

3. SF at A (right)
17.80 kN

4. SF at D (left)
1.80 kN

5. SF at D (right)
-3.20 kN

6. SF at B (left)
-27.20 kN

7. SF at B (right)
21.00 kN

8. SF at C (left)
5.00 kN

9. SF at C (right)
0.00 kN

10. BM at D
$M_D = R_A \times 2 - 8 \times 2^2/2 = 17.8 \times 2 - 16 =$ **19.60 kN·m**

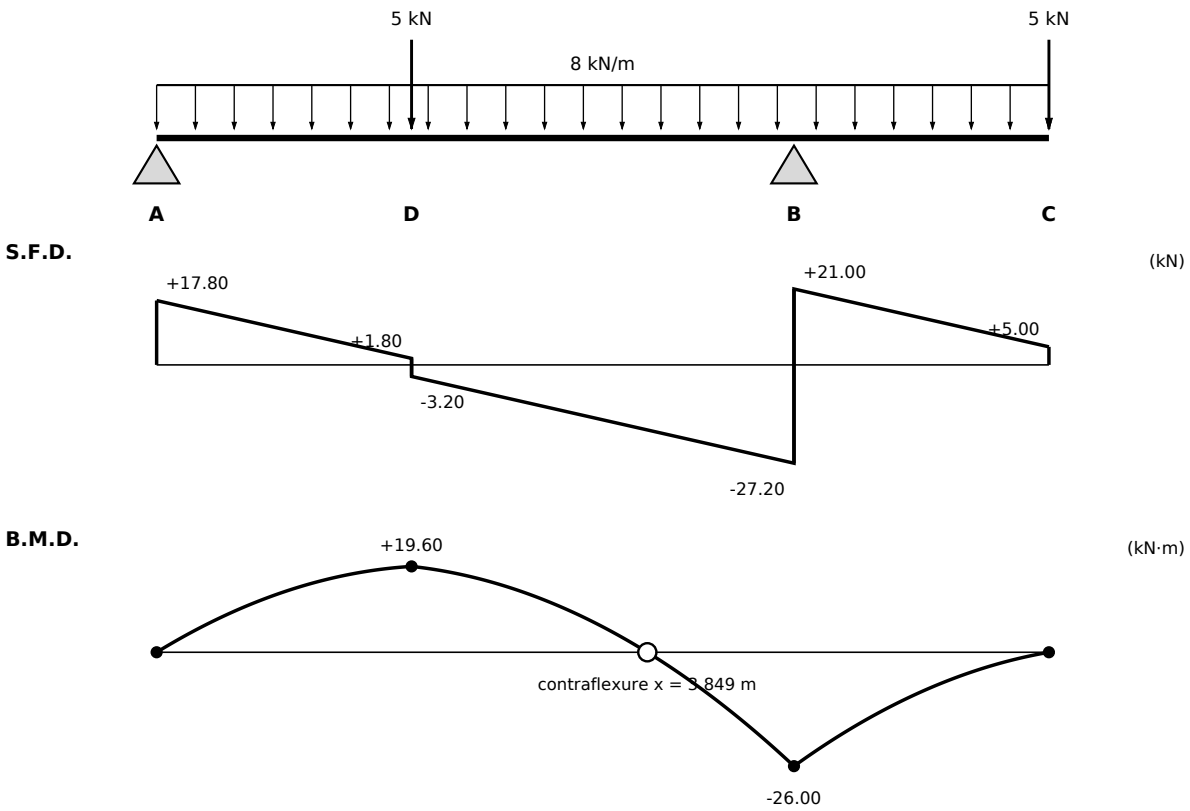
11. BM at B
$M_B = R_A \times 5 - 5 \times 3 - 8 \times 5^2/2 = 17.8 \times 5 - 15 - 100 =$ **-26.00 kN·m**

12. BM at C
M_C free end, no couple = **0.00 kN·m**

13. Point of contraflexure (between D and B): $M = 17.8x - 5(x-2) - 4x^2 = 0$
$4x^2 - 12.8x - 10 = 0$, $x = [12.8 + \sqrt{(12.8^2 + 160)}] / 8 = (12.8 + \sqrt{323.84}) / 8 =$ **3.849 m from A**

Final answer
R_A = **17.80 kN**
R_B = **48.20 kN**
Maximum sagging BM (at D) = **19.60 kN·m**
Maximum hogging BM (at B) = **-26.00 kN·m**
Point of contraflexure from A = **3.849 m**

Checked in code by a second method: ✓ R_A from the beam solver: 17.8000 ≈ 17.8000; ✓ R_B from the beam solver: 48.2000 ≈ 48.2000; ✓ BM at B from the beam solver: -26.0000 ≈ -26.0000; ✓ BM at D from the beam solver: 19.6000 ≈ 19.6000; ✓ BM at the free end C is zero: 0.0000 ≈ 0.0000; ✓ Contraflexure from the beam solver: 3.8494 ≈ 3.8494



S.F.D. and B.M.D. of the overhanging beam

Marks split	Marks
Reactions	1
S.F. values and diagram	2
B.M. values and diagram	2
Point of contraflexure	1
Total	**6**

Q.5 (c) [6 Marks]

A hollow square section 100 mm × 100 mm outside dimensions and 20 mm thick is subjected to a shear force of 75 kN. Calculate the ratio of maximum shear stress to the average shear stress. Draw the shear stress distribution diagram showing all the values on it.

Ans.

Numerical: Hollow square section: shear stress ratio

Given data	Value	Unit	Remark
Outer square	100 × 100	mm	
Thickness t	20	mm	inner square 60 × 60 mm
Shear force F	75	kN	

Formula and calculation:

1. MI about NA

$I = (B^4 - b^4) / 12 = (100^4 - 60^4) / 12 =$ **7 253 333 mm⁴**

2. Area
$A = B^2 - b^2 = 100^2 - 60^2 =$ **6400 mm²**

3. Average shear stress
$\tau_{avg} = F / A = 75000 / 6400 =$ **11.719 N/mm²**

4. First moment of area above NA
$A\,\bar{y} = 100\times50\times25 - 60\times30\times15 = 125\,000 - 27\,000 =$ **98000 mm³**

5. Maximum shear stress (at NA, width 2t = 40 mm)
$\tau_{max} = F\,A\,\bar{y} / (I \times 2t) = 75000 \times 98000 / (7253333 \times 40) =$ **25.333 N/mm²**

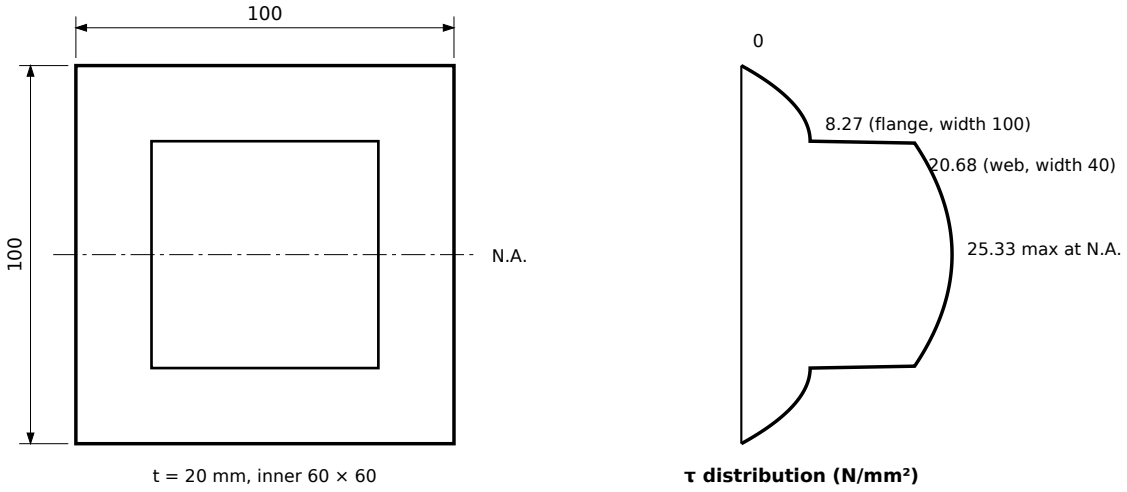
6. Ratio
$\tau_{max} / \tau_{avg} = 25.333 / 11.719 =$ **2.162**

7. τ at the level where flange meets web, width 100 mm
$\tau = F\,Q / (I\,b)$, $Q = 100\times20\times40 = 80\,000 = 75000 \times 80000 / (7253333 \times 100) =$ **8.272 N/mm²**

8. τ just inside the web, width 40 mm
$\tau = F\,Q / (I\,b) = 75000 \times 80000 / (7253333 \times 40) =$ **20.680 N/mm²**

Final answer
τ_{max} (at NA) = **25.333 N/mm²**
τ_{avg} = **11.719 N/mm²**
Ratio τ_{max} / τ_{avg} = **2.162**

Checked in code by a second method: ✓ Integral of τ over the section area equals F (numerical): 75000.0001 ≈ 75000.0000



Shear stress distribution – hollow square section

Marks split	Marks
Moment of inertia and area	1
Average shear stress	1
Maximum shear stress	2
Ratio	1
Diagram with values	1
Total	**6**

Q.6 Attempt any TWO of the following: — 12 Marks (each sub-question = 6 marks)

Q.6 (a) [6 Marks]

A ‘T’ section having dimensions 110 × 110 × 10 mm is used as a simply supported beam with a flange at top. It carries udl of 10 kN/m. If the maximum stress is not to exceed 150 N/mm², calculate the maximum span of beam. Also draw the bending stress distribution diagram.

Ans.

Numerical: T-section simply supported beam: maximum span

Given data	Value	Unit	Remark
T-section	110 × 110 × 10	mm	flange 110 × 10 at top; web 10 × 100
UDL w	10	kN/m	= 10 N/mm
Permissible stress σ	150	N/mm²	

Formula and calculation:

1. Centroid from top

$\bar{y} = (A_1y_1 + A_2y_2) / (A_1 + A_2) = (1100\times5 + 1000\times60) / (1100 + 1000) =$ **31.190 mm**

2. Distance of bottom fibre

$y_b = 110 - \bar{y} = 110 - 31.190 =$ **78.810 mm**

3. MI about NA

$I = [b_1t_1^3/12 + A_1(\bar{y} - y_1)^2] + [t_2d_2^3/12 + A_2(y_2 - \bar{y})^2] = [9166.7 + 1100.0\times26.190^2] + [833333.3 + 1000.0\times28.810^2] =$ **2 427 024 mm⁴**

4. Section modulus (critical = bottom, tension)

$Z = I / y_b = 2427024 / 78.810 =$ **30796 mm³**

5. Maximum BM

$M = \sigma Z = 150 \times 30796 =$ **4 619 411 N·mm**

6. Span: $M = w L^2 / 8$

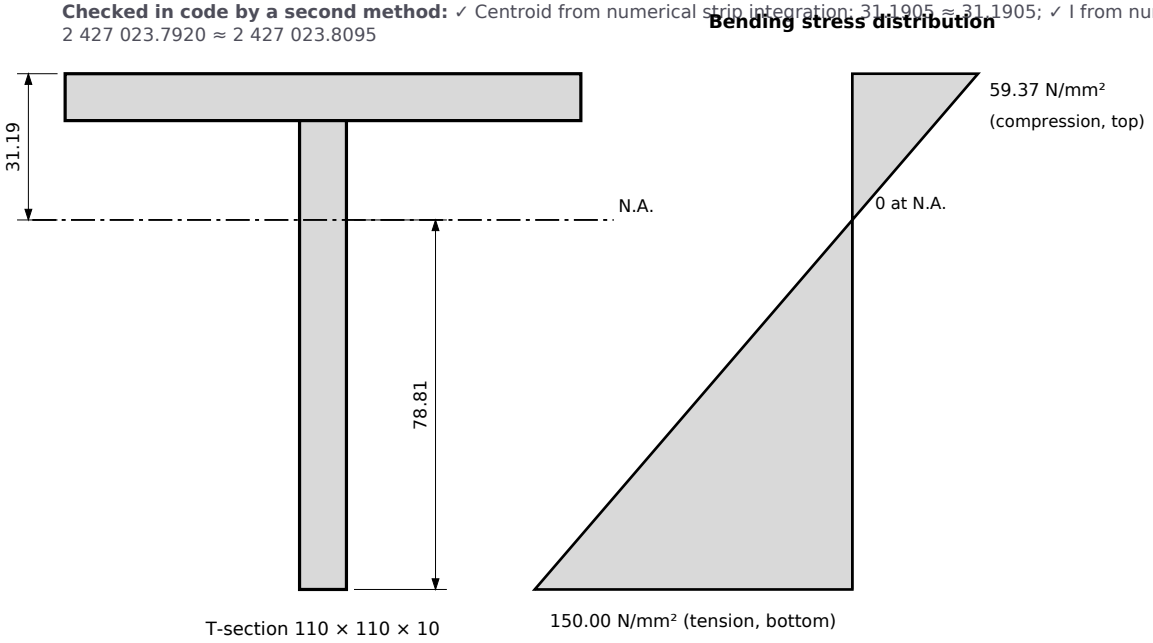
$L = \sqrt{(8 M / w)} = \sqrt{(8 \times 4619411 / 10)} =$ **1.922 m**

7. Maximum compressive stress at top fibre

$\sigma_c = M y_t / I = 4619411 \times 31.190 / 2427024 =$ **59.37 N/mm²**

Final answer
Maximum span = **1.922 m**
Stress at top fibre (compression) = **59.37 N/mm²**
Stress at bottom fibre (tension) = **150.00 N/mm²**

Checked in code by a second method: ✓ Centroid from numerical strip integration: 31.1905 ≈ 31.1905; ✓ I from numerical strip integration: 2 427 023.7920 ≈ 2 427 023.8095



Bending stress distribution in the T-section

Marks split	Marks
Centroid	1
Moment of inertia	2
Section modulus and moment of resistance	1
Span	1
Stress diagram	1
Total	**6**

Q.6 (b) [6 Marks]

A hollow shaft, having an internal diameter 40% of its external diameter, transmits 562.5 kW power at 100 r.p.m. Determine the external diameter of the shaft if the shear stress is not to exceed 60 N/mm² and the twist in a length of 2.5 m should not exceed 1.3°. Assume maximum torque = 1.25 times the mean torque and modulus of rigidity = 9×10^4 N/mm².

Ans.

Numerical: Hollow shaft transmitting 562.5 kW at 100 rpm

Given data	Value	Unit	Remark
Power P	562.5	kW	
Speed N	100	rpm	
d/D	0.4		internal diameter 40 % of external
τ limit	60	N/mm²	
Twist limit	1.3	° in 2.5 m	
T_{max} / T_{mean}	1.25		
G	90000	N/mm²	

Formula and calculation:

1. Mean torque

$T_{mean} = 60\ P / (2\pi\ N) = 60 \times 562500 / (2\pi \times 100) =$ **53714.8 N·m**

2. Maximum torque

$T_{max} = 1.25\ T_{mean} = 1.25 \times 53714.8 =$ **67143.5 N·m**

3. Factor $(1 - k^4)$

$1 - 0.4^4 =$ **0.9744**

4. (a) Strength: $T = (\pi/16)\ \tau\ D^3\ (1 - k^4)$

$D = [16\ T / (\pi\ \tau\ (1 - k^4))]^{1/3} = [16 \times 67143492 / (\pi \times 60 \times 0.9744)]^{1/3} =$ **180.18 mm**

5. (b) Stiffness: $J = T\ L / (G\ \theta)$

$J = (67143492 \times 2500) / (90000 \times 0.022689) =$ **8.22×10^7 mm⁴**

6. Diameter from $J = (\pi/32)\ D^4\ (1 - k^4)$

$D = [32\ J / (\pi\ (1 - k^4))]^{1/4} = [32 \times 8.2202 \times 10^7 / (\pi \times 0.9744)]^{1/4} =$ **171.21 mm**

Final answer
External diameter (larger value) = **180.18 mm**
Internal diameter d = 0.4 D = **72.07 mm**

Checked in code by a second method: ✓ Stress at D from strength equals 60 N/mm² (strength governs): 60.0000 ≈ 60.0000; ✓ Twist at D does not exceed 1.3°: 1.0600 ≈ ≤ 1.3000; ✓ T_{mean} from P = T ω: 562 500.0000 ≈ 562 500.0000

Note: T_{max} is used for both the strength and the twist condition.

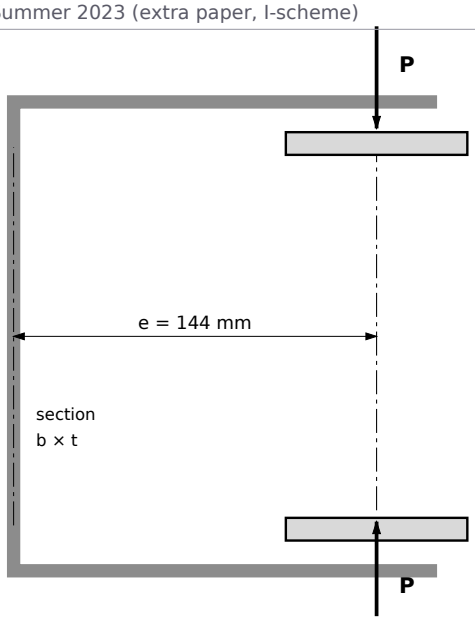
Marks split	Marks
Mean and maximum torque	1
Diameter from shear stress	2
Diameter from twist	2
Selecting larger value	1
Total	**6**

Q.6 (c) [6 Marks]

Figure No. 4. shows a C - clamp carries a load P = 27 kN. Section of clamp is rectangular having width equal to twice the thickness. Assuming that the clamp is made up of steel casting with an allowable stress of 100 N/mm². Find the dimensions of the clamp section.
[Figure No. 4: C-clamp; load line 144 mm from the centroidal axis of the vertical member]

Ans.

Data note: The figure shows a 144 mm dimension but does not say clearly which line it is measured to.
Assumption used: It is taken as the distance e from the load line to the centroidal axis of the vertical member of the clamp.



Section: b = 2t (depth in the plane), thickness t

σ = P/A (direct tension) + P e / Z (bending)

σmax must not exceed 100 N/mm²

C-clamp (Figure No. 4)

Numerical: C-clamp section for P = 27 kN

Given data	Value	Unit	Remark
Load P	27	kN	
Distance of load line from centroidal axis e	144	mm	
Section: depth b = 2t, thickness t	b = 2t		
Allowable stress	100	N/mm²	

Formula and calculation:

Direct tension P/A plus bending P e / Z: $\sigma_{max} = P/(b\,t) + 6\,P\,e/(t\,b^2)$ with b = 2t

A = 2t², Z = t b²/6 = 2t³/3, so σ = 13 500 / t² + 5 832 000 / t³ = 100

1. Solve 100 t³ − 13 500 t − 5 832 000 = 0 for t

t (trial and error / Newton) = **39.94 mm**

2. Adopt (rounded up)

t = **40 mm**

3. Width b = 2t

b = **80 mm**

4. Check: stress at adopted section

σ = P/A + P e/Z = 8.438 + 91.125 = **99.56 N/mm²**

Final answer
Thickness t = **40 mm**
Width b = 2t = **80 mm**

Checked in code by a second method: ✓ Cubic residual at exact root: $-1.863 \times 10^{-9} \approx 0.0000$; ✓ Stress at the adopted section does not exceed 100 N/mm²: 99.5625 ≈ ≤ 100.0000

Marks split	Marks
Direct and bending stress equation	2
Area and section modulus in terms of t	1
Solving for t	2
Final dimensions	1
Total	**6**

Chapter-wise marks table

Marks offered = marks of every sub-question printed in the paper, including the choice ('attempt any …'), so a column total can be more than 70. Chapter names are my own classification of each question (the syllabus was not supplied).

Moment of inertia	**10**	**10**	**9.8 %**
Elastic constants and strain energy	12	**12**	11.8 %
Shear force and bending moment	20	**20**	19.6 %
Bending and shear stresses in beams	20	**20**	19.6 %
Direct and bending stresses (core, eccentric loads)	12	**12**	11.8 %
Simple stresses and strains	14	**14**	13.7 %
Torsion of shafts	14	**14**	13.7 %
Total (marks offered)	**102**	**102**	100 %

Papers counted: Summer 2023 (extra paper, I-scheme).

Top 10 repeated questions

Note: Only one paper was supplied for this subject, so no question can be called 'repeated'. The table below lists the 10 topics with the highest marks in that paper instead.

#	Question / topic	Where asked	Marks	Sessions
1	Compound tube of steel and brass	S23: Q.5(a)	6	1
2	SFD and BMD – overhanging beam	S23: Q.5(b)	6	1
3	Shear stress in hollow square section	S23: Q.5(c)	6	1
4	T-beam maximum span	S23: Q.6(a)	6	1
5	Hollow shaft from power and twist	S23: Q.6(b)	6	1
6	C-clamp section (direct + bending)	S23: Q.6(c)	6	1
7	MI of hollow circular section (tangent, polar)	S23: Q.2(a)	4	1
8	Stepped bar: P_2 and total elongation	S23: Q.2(b)	4	1
9	Tri-axial stress: strains and volume change	S23: Q.2(c)	4	1
10	SFD and BMD – point load (derivation)	S23: Q.2(d)	4	1

Last-minute revision sheet

Properties of sections

- Rectangle: $I = bd^3/12$ about centroid; I about base $= bd^3/3$.
- Circle: $I = \pi d^4/64$; hollow: $\pi(D^4-d^4)/64$; $J = 2I = \pi(D^4-d^4)/32$.
- Semicircle about its axis of symmetry: $\pi R^4/8$.
- Parallel-axis: $I = I_G + A h^2$. Radius of gyration $k = \sqrt{(I/A)}$.
- Composite or lamina: add areas, subtract cut-outs.

Stress, strain, elastic constants

- $\sigma = P/A$; $\varepsilon = \delta L/L$; $E = \sigma/\varepsilon$; $\delta = PL/(AE)$.
- Stepped bar: $\delta = \Sigma\, N\, L/(A\, E)$; find forces from equilibrium first.
- Compound bar: equal strain, $\sigma_1/E_1 = \sigma_2/E_2$; $P = \sigma_1 A_1 + \sigma_2 A_2$.
- μ = lateral strain / longitudinal strain; $E = 2G(1+\mu) = 3K(1-2\mu)$.
- Tri-axial: $\varepsilon_x = [\sigma_x - \mu(\sigma_y + \sigma_z)]/E$; $\varepsilon_v = \Sigma\varepsilon = (1-2\mu)\Sigma\sigma/E$.
- Strain energy $U = \sigma^2 V/(2E)$.

Impact load

- $\sigma = (W/A)[1 + \sqrt{(1 + 2AEh/(WL))}]$; $\delta = \sigma L/E$.
- Energy balance: $W(h + \delta) = \sigma^2 AL/(2E)$.
- Suddenly applied load: $\sigma = 2W/A$; gradually applied: $\sigma = W/A$.

SF and BM

- Reactions first: $\Sigma M = 0$, $\Sigma F = 0$.
- SF changes by the load at a point load; falls linearly under UDL.
- M_{max} where SF = 0 (or changes sign).
- SS beam, central point load: $M = WL/4$; UDL: $M = wL^2/8$. Cantilever: WL, $wL^2/2$.
- Overhanging beam: find the point of contraflexure by putting $M(x) = 0$.
- Sign: sagging +, hogging −; clockwise shear +.

Bending and shear stress in beams

- $M/I = \sigma/y = E/R$; $\sigma_{max} = M/Z$; $Z = I/y_{max}$. Z rectangle $= bd^2/6$; circle $= \pi d^3/32$.
- $\tau = F\, A\, \bar{y} / (I\, b)$; rectangle $\tau_{max} = 1.5\, \tau_{avg}$; circle $4/3\, \tau_{avg}$.
- Hollow box: τ jumps where width changes; maximum at N.A.
- T-beam: find $\bar{y}$, then I, then use the farther fibre for the limit stress.

Torsion

- $T/J = \tau/r = G\theta/L$. Solid: $T = (\pi/16)\tau d^3$. Hollow: $T = (\pi/16)\tau(D^4-d^4)/D$.
- Power $P = 2\pi NT/60$. Use $T_{max} = 1.25\, T_{mean}$ if told.
- Take the larger diameter from the strength and the stiffness (twist) condition; θ in radian = deg × $\pi/180$.
- Polar modulus $Z_p = J/R$.

Direct and bending stress

- $\sigma = P/A \pm M/Z$, with $M = P\, e$.
- No tension condition: $e \le$ core limit. Rectangle: $B/6$ and $D/6$ (core $B/3 \times D/3$). Solid circle: $d/8$. Hollow circle: $(D^2+d^2)/(8D)$.
- C-clamp: direct tension + bending, $\sigma = P/A + Pe/Z \le$ allowable.

Fluid Mechanics & Machinery (313309)

Session	Status
Winter 2024	**Not received – not included**
Summer 2025	**Not received – not included**
Winter 2025	**Not received – not included**

Marks shown against sub-questions are the paper's group marks divided by the number of sub-questions to attempt (MSBTE equal split); the papers print only the group total.

Papers not received: Winter 2024, Summer 2025, Winter 2025 – not included

Not received, so not included: Winter 2024, Summer 2025, Winter 2025.

Questions and solutions for this paper will be added when the paper is supplied. Nothing has been guessed.

Chapter-wise marks table

No paper was received for this subject, so no table can be made from the papers.

Thermal Engineering (313310)

Session	Status
Winter 2024	Included
Summer 2025	Included
Winter 2025	**Not received – not included**

Marks shown against sub-questions are the paper's group marks divided by the number of sub-questions to attempt (MSBTE equal split); the papers print only the group total.

Winter 2024

Subject	Thermal Engineering (313310)	Time / Marks	3 Hours / 70 Marks
Session	Winter 2024	Paper ID printed	12425
Scheme	K-scheme	Source file	Thermal_Engineering_Winter_2024_Question_Paper_First_K_Scheme_Paper.pdf

Q.1 Attempt any FIVE of the following: — 10 Marks (each sub-question = 2 marks)

Q.1 (a) [2 Marks]

Define Zeroth law of thermodynamics.

Ans.

Zeroth law

Statement: If two bodies (systems) are each in thermal equilibrium with a third body, then they are also in thermal equilibrium with each other.

That is, if A is in equilibrium with C and B is in equilibrium with C, then A and B are in equilibrium with each other (their temperatures are equal).

Use: This law is the basis of temperature measurement – the third body is the thermometer.

Marks split	Marks
Statement	1
Explanation / example	1
Total	**2**

Q.1 (b) [2 Marks]

Define Dryness fraction of steam.

Ans.

Dryness fraction (x) is the ratio of the mass of dry steam present in a given mass of wet steam to the total mass of the wet steam.

$$x = m_g / (m_g + m_f)$$

where m_g = mass of dry (vapour) steam and m_f = mass of water particles in suspension. For dry saturated steam x = 1; for saturated water x = 0.

Marks split	Marks
Definition	1
Formula with meaning of terms	1
Total	**2**

Q.1 (c) [2 Marks]

What is universal gas constant? State it’s unit.

Ans.

Universal gas constant (R_u) is the product of the molecular mass (M) of a gas and its characteristic gas constant (R). Its value is the same for all gases.

$$R_u = M \times R = 8.314 \text{ kJ/kmol K } (= 8314 \text{ J/kmol K})$$

Unit: kJ/kmol K (or J/kmol K).

Marks split	Marks
Definition	1
Value and unit	1
Total	**2**

Q.1 (d) [2 Marks]

State the Datton's Law of partial pressure.

Ans.

Dalton's law: The total pressure exerted by a mixture of non-reacting gases is equal to the sum of the partial pressures of the individual gases.

$$p = p_1 + p_2 + p_3 + \ldots$$

Partial pressure of a gas is the pressure it would exert if it alone occupied the whole volume of the mixture at the same temperature.

Marks split	Marks
Statement	1
Equation / partial pressure	1
Total	**2**

Q.1 (e) [2 Marks]

Write basic function of condenser.

Ans.

- A condenser **condenses the exhaust steam** from the turbine into water by removing its latent heat with cooling water.
- It creates a **low pressure (vacuum)** at the turbine exhaust, so the pressure drop across the turbine increases and the work output and efficiency increase.
- The condensate is pure water and is reused as **feed water** for the boiler.

Marks split	Marks
Function – condensing the steam	1
Vacuum / feed water recovery	1
Total	**2**

Q.1 (f) [2 Marks]

Define Thermal conductivity. Give it's unit.

Ans.

Thermal conductivity (k) is the quantity of heat that flows per second through a unit area of a material of unit thickness when a unit temperature difference is maintained between its two faces.

$$k = Q\ L / (A\ \Delta T)$$

Unit: W/m K (or W/m °C).

Marks split	Marks
Definition	1
Unit	1
Total	**2**

Q.1 (g) [2 Marks]

Define mean effective pressure for I.C. Engine.

Ans.

Mean effective pressure (m.e.p.) is the hypothetical constant pressure which, acting on the piston during the whole working stroke, would produce the same net work as is actually produced by the engine cycle.

$$\text{m.e.p.} = \text{Net work done per cycle / Swept volume} = W_{net} / V_s$$

Unit: bar or N/m².

Marks split	Marks
Definition	1
Formula and unit	1
Total	**2**

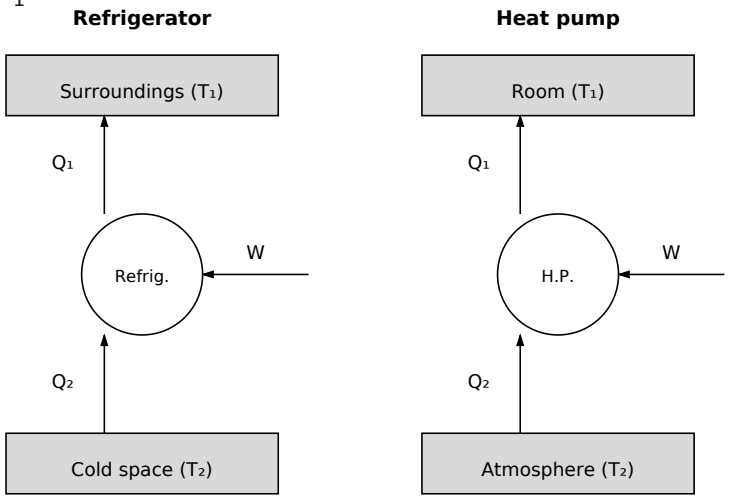
Q.2 Attempt any THREE of the following: — 12 Marks (each sub-question = 4 marks)

Q.2 (a) [4 Marks]

Prove that (COP)Heat pump = (COP)Refrigerator + 1

Ans.

Consider a reversible cycle working as a refrigerator or as a heat pump. Let Q_2 = heat absorbed from the low-temperature body, Q_1 = heat rejected to the high-temperature body, and W = work supplied.



Refrigerator and heat pump (block diagram)

By the first law for the cycle:

$$W = Q_1 - Q_2$$

$$(COP)_R = \text{Refrigerating effect / Work} = Q_2 / W = Q_2 / (Q_1 - Q_2)$$

$$(COP)_{HP} = \text{Heat delivered / Work} = Q_1 / W = Q_1 / (Q_1 - Q_2)$$

Now write $Q_1 = Q_2 + (Q_1 - Q_2)$:

$$(COP)_{HP} = [Q_2 + (Q_1 - Q_2)] / (Q_1 - Q_2) = Q_2/(Q_1 - Q_2) + (Q_1 - Q_2)/(Q_1 - Q_2)$$

$$\mathbf{(COP)_{HP} = (COP)_R + 1}$$ Hence proved.

Marks split	Marks
Neat block diagram	1
Definitions of COP	1
Steps of proof	2
Total	**4**

Q.2 (b) [4 Marks]

State Kelvin-plank and clausius statement of second law of thermodynamics.

Ans.

Kelvin–Planck statement: It is impossible to construct a device that, operating in a cycle, produces no effect other than the transfer of heat from a single reservoir and the performance of an equal amount of work. (In other words, no heat engine can have 100 % efficiency; some heat must be rejected to a sink.)

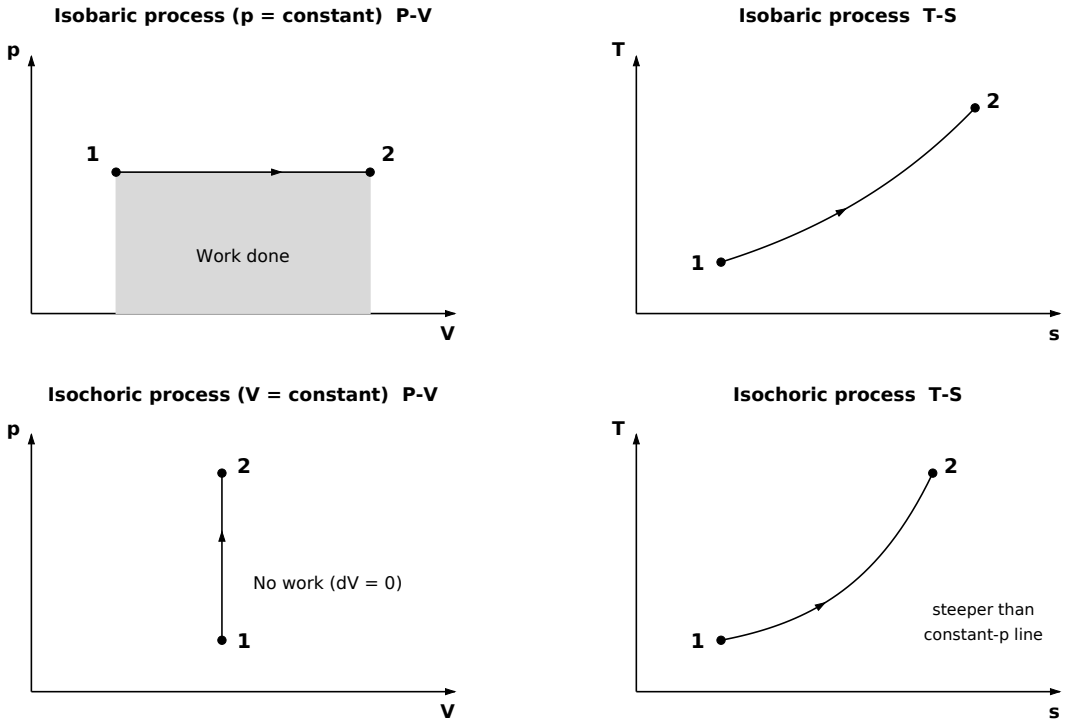
Clausius statement: It is impossible to construct a device that, operating in a cycle, produces no effect other than the transfer of heat from a colder body to a hotter body. (Heat cannot flow from cold to hot by itself; external work is needed, as in a refrigerator.)

Marks split	Marks
Kelvin–Planck statement	2
Clausius statement	2
Total	**4**

Q.2 (c) [4 Marks]

Represent Isobaric process and Isochoric process on P-V and T-S diagram.

Ans.



Isobaric and isochoric processes on P-V and T-S diagrams

Item	Isobaric (p = constant)	Isochoric (V = constant)
Law	V / T = constant	p / T = constant
Work done	$W = p\,(V_2 - V_1)$	W = 0
Heat transfer	$Q = m\,C_p\,(T_2 - T_1)$	$Q = m\,C_v\,(T_2 - T_1)$
Change in U	$m\,C_v\,(T_2 - T_1)$	$m\,C_v\,(T_2 - T_1)$

Marks split	Marks
Isobaric: P-V and T-S diagrams	1
Isochoric: P-V and T-S diagrams	1
Relations / explanation	2
Total	**4**

Q.2 (d) [4 Marks]

A volume of 0.5 m3 of gas at a pressure of 10 bar and 200°C is expanded in a cylinder to 1.2 m3 at a constant pressure. Calculate the amount of workdone by the gas and the increase in internal energy. Assume (Cp = 1.005 kJ/kg K and Cv = 0.712 kJ/kg K).

Ans.

The mass of gas is not given, so it is found from the characteristic gas equation.

Numerical: Constant-pressure expansion of a gas

Given data	Value	Unit	Remark
Initial volume V_1	0.5	m^3	
Final volume V_2	1.2	m^3	
Pressure p (constant)	10	bar	= 10 × 100 = 1000 kPa
Initial temperature t_1	200	°C	T_1 = 200 + 273 = 473 K
C_p	1.005	kJ/kg K	
C_v	0.712	kJ/kg K	

Formula and calculation:

1. Characteristic gas constant
$R = C_p - C_v = 1.005 - 0.712 =$ **0.293 kJ/kg K**

2. Mass of gas
$m = p V_1 / (R T_1) = (1000 \times 0.5) / (0.293 \times 473) =$ **3.6078 kg**

3. Work done (constant pressure)
$W = p (V_2 - V_1) = 1000 \times (1.2 - 0.5) =$ **700.00 kJ**

4. Final temperature (V/T = constant)
$T_2 = T_1 \times V_2 / V_1 = 473 \times 1.2 / 0.5 =$ **1135.20 K**

5. Increase in internal energy
$\Delta U = m C_v (T_2 - T_1) = 3.6078 \times 0.712 \times (1135.20 - 473) =$ **1701.02 kJ**

Final answer
Work done by the gas = **700.00 kJ**
Increase in internal energy = **1701.02 kJ**

Checked in code by a second method: ✓ First law: Q − W (Q = m C_p ΔT) equals ΔU: 1701.0239 ≈ 1701.0239; ✓ ΔU = (C_v/R) × p ΔV: 1701.0239 ≈ 1701.0239

Marks split	Marks
Given data and $R = C_p - C_v$	1
Mass of gas and final temperature	1
Work done	1
Increase in internal energy	1
Total	**4**

Q.3 Attempt any THREE of the following: — 12 Marks (each sub-question = 4 marks)

Q.3 (a) [4 Marks]

A vessel of 0.03 m3 capacity contains gas at 350 kN/m2. pressure and 35°C temperature. Determine the mass of the gas in the vessel. If the pressure of this gas is increased to 1.05×10^6 N/m2 while the volume remains constant, what will be the temperature of the gas? Take R = 290 J/kg K for the gas.

Ans.

Numerical: Gas in a closed vessel

Given data	Value	Unit	Remark
Volume V	0.03	m^3	
Initial pressure p_1	350	kN/m^2	= 350 000 N/m^2
Initial temperature t_1	35	°C	$T_1 = 35 + 273 = 308$ K
Final pressure p_2	1050000	N/m^2	
Gas constant R	290	J/kg K	

Formula and calculation:

1. Mass of gas
$m = p_1 V / (R T_1) = (350000 \times 0.03) / (290 \times 308) =$ **0.1176 kg**

2. Final temperature (V constant: p/T = constant)
$T_2 = T_1 \times p_2 / p_1 = 308 \times 1050000 / 350000 =$ **924.00 K**

3. In °C
$t_2 = T_2 - 273 = 924.00 - 273 =$ **651.00 °C**

Final answer
Mass of gas = **0.1176 kg**
Final temperature = **924.00 K**
Final temperature = **651.00 °C**

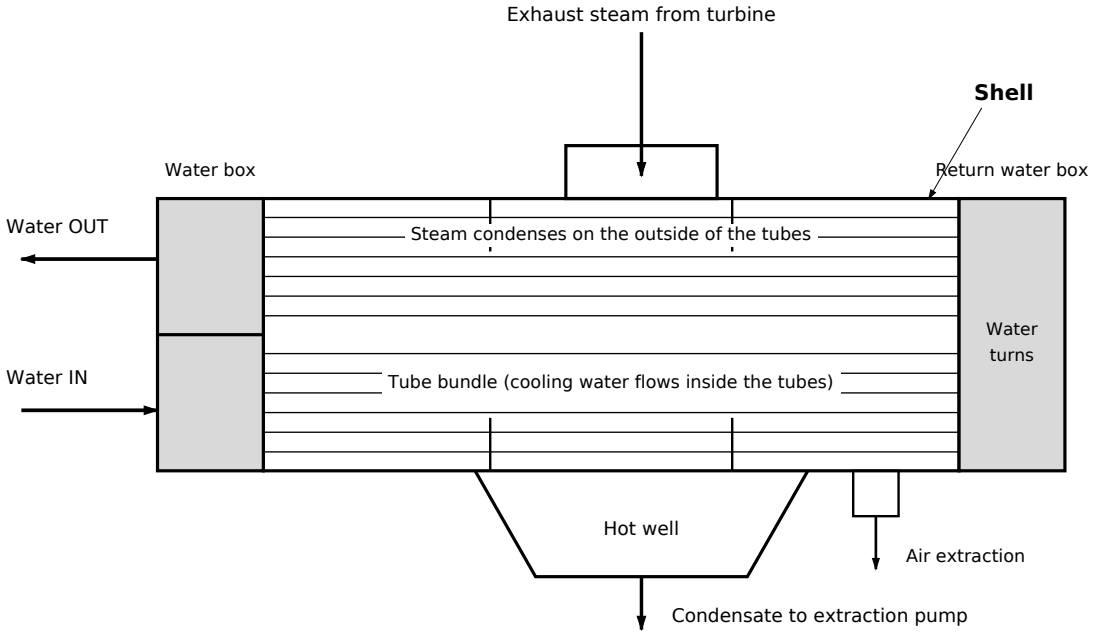
Checked in code by a second method: ✓ $p_2 V = m R T_2$ (characteristic equation at final state): 31500.0000 ≈ 31500.0000

Marks split	Marks
Given data / unit conversion	1
Mass of gas	1
Final temperature (p/T = constant)	1
Answer with units	1
Total	**4**

Q.3 (b) [4 Marks]

Explain construction and working of surface condenser with neat sketch.

Ans.



Surface condenser (two-pass, shell and tube type)

Construction

- A **closed cylindrical shell** contains a large number of small-diameter brass tubes held between two tube plates.
- Cooling water flows **inside the tubes**; the exhaust steam from the turbine enters at the top of the shell and flows **outside the tubes**.
- Water boxes at the ends give a two-pass flow of cooling water. Baffles guide the steam over the tubes.
- A hot well at the bottom collects condensate; an air extraction pump removes air and non-condensable gases from the cold-end of the shell.

Working

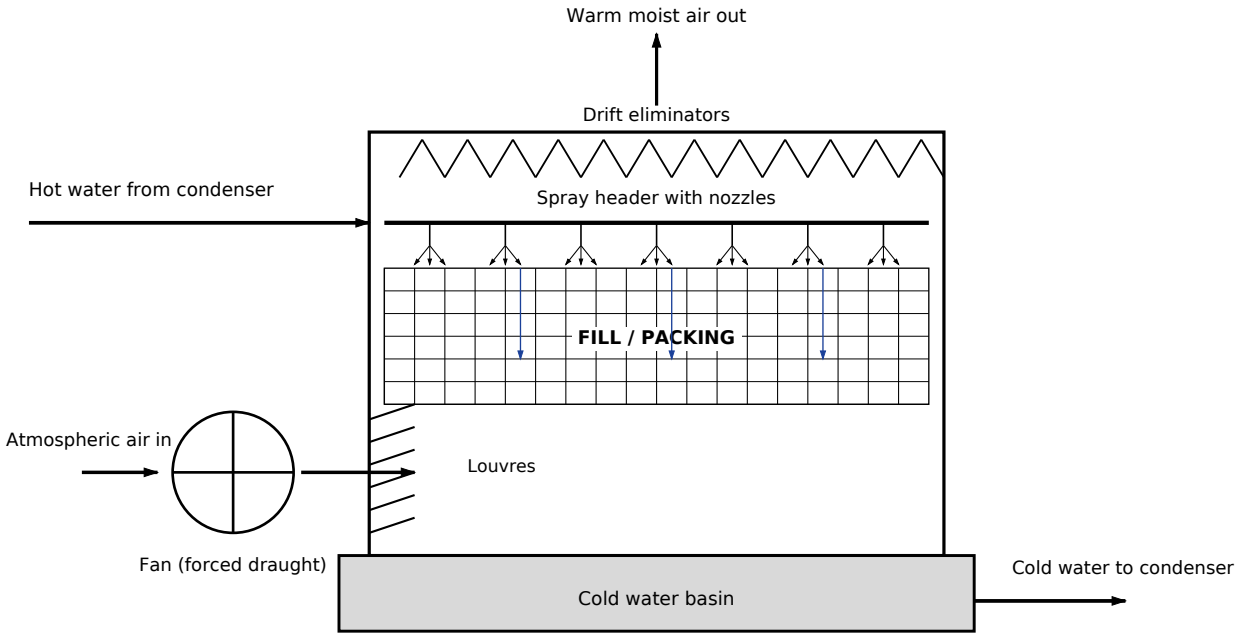
1. Exhaust steam enters the shell and comes in contact with the cold outer surface of the tubes.
2. The steam gives up its latent heat to the cooling water and condenses; the volume reduces, so a vacuum is formed.
3. The condensate collects in the hot well and is pumped to the boiler as feed water (the condensate is not mixed with cooling water).
4. Air and other gases are drawn out by the air extraction pump to maintain the vacuum.
5. Warm cooling water leaves from the outlet and goes to the cooling tower / river.

Marks split	Marks
Neat sketch with labels	2
Construction	1
Working	1
Total	**4**

Q.3 (c) [4 Marks]

Explain construction and working of forced draught cooling tower with neat sketch.

Ans.



Forced draught (mechanical draught) cooling tower

Construction

- A tall rectangular tower with louvres on the side or base for entry of air.
- A **fan at the air inlet (base)** forces air into the tower – hence the name forced draught.
- Inside, a **spray header with nozzles** at the top distributes hot water over the **fill (packing)** made of wooden or plastic slats. Drift eliminators at the top stop water droplets from leaving. A cold water basin is at the bottom.

Working

1. Hot water from the condenser is pumped to the top and sprayed over the fill.
2. The fan blows atmospheric air up through the fill, against the falling water (counter flow).
3. A small part of the water evaporates, taking its latent heat from the remaining water, so the water is cooled.
4. Cooled water collects in the basin and is pumped back to the condenser; make-up water replaces the evaporation loss; the warm moist air leaves at the top.

Marks split	Marks
Neat sketch	2
Construction	1
Working	1
Total	**4**

Q.3 (d) [4 Marks]

Give classification of Boiler.

Ans.

Basis of classification	Types	Examples
Contents of tubes	Fire tube boiler; Water tube boiler	Cochran, Lancashire; Babcock & Wilcox
Position of furnace	Internally fired; Externally fired	Cochran; Babcock & Wilcox
Axis of shell	Vertical; Horizontal; Inclined	Cochran; Lancashire; Babcock
Circulation of water	Natural circulation; Forced circulation	Babcock; La-Mont, Benson
Pressure	Low (below 10 bar); Medium (10–80 bar); High (above 80 bar)	Cochran; Babcock; Benson
Use	Stationary; Portable (mobile) – locomotive, marine	Lancashire; Locomotive
Number of tubes	Single-tube; Multi-tubular	Simple vertical; Cochran

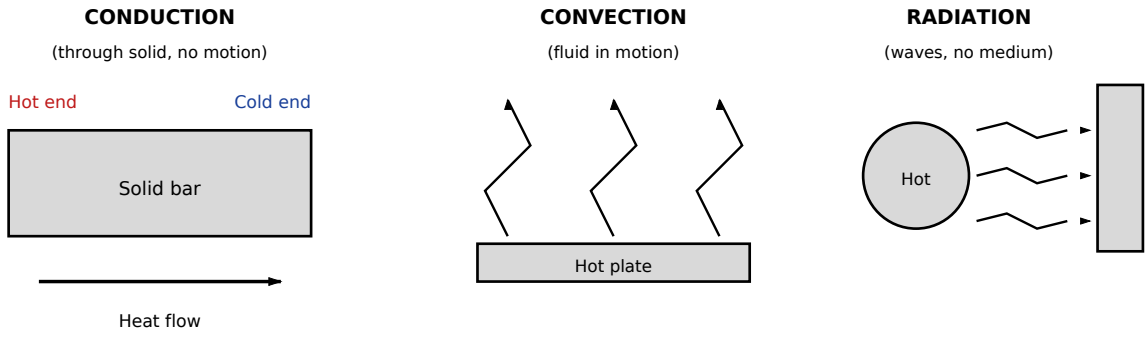
Marks split	Marks
Any four bases of classification, 1 mark each (with examples)	4
Total	**4**

Q.4 Attempt any THREE of the following: — 12 Marks (each sub-question = 4 marks)

Q.4 (a) [4 Marks]

Define various modes of heat transfer with suitable example.

Ans.



Three modes of heat transfer

- **Conduction:** transfer of heat from one part of a body to another part of the same body, or between bodies in contact, by molecular vibration without any visible motion of the material. Law: Fourier, $Q = -k\,A\,dT/dx$. Example: heat flows along a metal rod heated at one end; heat flow through a boiler wall.
- **Convection:** transfer of heat by the actual movement of the fluid particles. It may be natural (due to density difference) or forced (by a fan or pump). Law: Newton, $Q = h\,A\,(T_s - T_f)$. Example: heating of water in a kettle; cooling of an engine radiator by air.
- **Radiation:** transfer of heat in the form of electromagnetic waves without any medium. Law: Stefan–Boltzmann, $Q = \sigma\,A\,T^4$. Example: heat from the sun to the earth; heat from a furnace to boiler tubes.

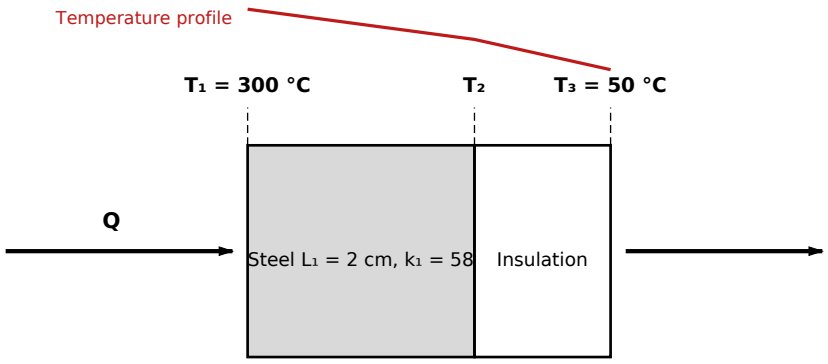
Marks split	Marks
Conduction with example	1
Convection with example	1
Radiation with example	1
Sketch / laws	1
Total	**4**

Q.4 (b) [4 Marks]

Determine the rate of heat flow through the boiler wall made of 2 cm thick steel and covered with an insulating material of 0.5 cm thick.
K(steel) = 58 w/mk,
K(insulation) = 0.116 w/mk.
the temperature at the inner and outer surfaces of the wall are 300°C and 50°C respectively.

Ans.

Data note: The wall area is not given in the question.
Assumption used: Area A = 1 m² is assumed, so the answer is the heat flow **per square metre** of wall. For any other area, multiply by that area.



Composite wall of steel and insulation

Numerical: Heat flow through composite boiler wall

Given data	Value	Unit	Remark
Steel thickness L_1	2	cm	= 0.02 m
Steel conductivity k_1	58	W/m K	
Insulation thickness L_2	0.5	cm	= 0.005 m
Insulation conductivity k_2	0.116	W/m K	
Inner surface temperature T_1	300	°C	
Outer surface temperature T_3	50	°C	
Area A	1	m²	assumed – not given in the question

Formula and calculation:

1. Resistance of steel

$R_1 = L_1 / (k_1 A) = 0.02 / (58 \times 1) =$ **3.448×10^{-4} K/W**

2. Resistance of insulation

$R_2 = L_2 / (k_2 A) = 0.005 / (0.116 \times 1) =$ **0.043103 K/W**

3. Total resistance

$R = R_1 + R_2 = 0.000345 + 0.043103 =$ **0.043448 K/W**

4. Rate of heat flow

$Q = (T_1 - T_3) / R = (300 - 50) / 0.043448 =$ **5753.97 W**

5. Interface temperature (for check)

$T_2 = T_1 - Q R_1 = 300 - 5753.97 \times 0.000345 =$ **298.02 °C**

Final answer
Rate of heat flow (per m² of wall) = **5753.97 W**

Checked in code by a second method: ✓ Q through insulation = $k_2 A (T_2 - T_3) / L_2$: 5753.9683 ≈ 5753.9683

Marks split	Marks
Given data / sketch	1
Thermal resistances	1
Total resistance and formula	1
Answer with units	1
Total	**4**

Q.4 (c) [4 Marks]

Differentiate two stroke and four stroke I.C. engine.

Ans.

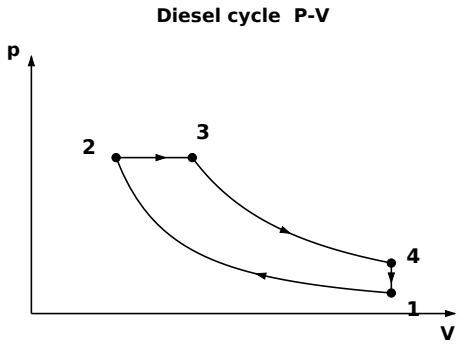
Sr.	Two stroke engine	Four stroke engine
1	One working cycle is completed in two strokes (one crankshaft revolution).	One working cycle is completed in four strokes (two crankshaft revolutions).
2	One power stroke in every revolution.	One power stroke in every two revolutions.
3	Ports (inlet, transfer, exhaust) are used; no valves.	Inlet and exhaust valves with a cam and camshaft are used.
4	Turning moment is more uniform; lighter flywheel.	Turning moment is less uniform; heavier flywheel.
5	Power output is higher for the same size (theoretically double).	Power output is lower for the same size.
6	Thermal efficiency is lower (some fresh charge escapes with exhaust).	Thermal efficiency is higher.
7	Lubrication: oil mixed with petrol; more fuel consumption.	Separate lubrication system; less fuel consumption.
8	Used in scooters, mopeds, small outboard motors.	Used in cars, buses, trucks, tractors.

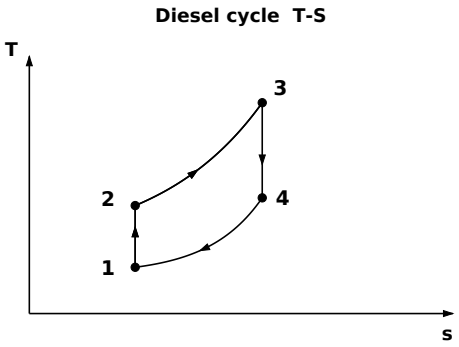
Marks split	Marks
Any four points, 1 mark each	4
Total	**4**

Q.4 (d) [4 Marks]

Explain Diesel cycle with P-V and T-S diagram.

Ans.





1-2 Isentropic compression

2-3 Heat added at constant pressure

3-4 Isentropic expansion

4-1 Heat rejected at constant volume

Diesel cycle

Explanation

- The Diesel cycle is the ideal air-standard cycle for the compression-ignition engine. Heat is added at **constant pressure** and rejected at constant volume.
- **1–2:** Isentropic (reversible adiabatic) compression of air; the volume reduces from V_1 to V_2.
- **2–3:** Heat addition at constant pressure (fuel injected and burnt); volume increases from V_2 to V_3.
- **3–4:** Isentropic expansion of the hot gases (power stroke).
- **4–1:** Heat rejection at constant volume (exhaust).

Air standard efficiency

$$\eta = 1 - (1 / r^{\gamma-1}) \times [\ (\rho^{\gamma} - 1) / (\gamma\ (\rho - 1))\ ]$$

where r = compression ratio = V_1/V_2, ρ = cut-off ratio = V_3/V_2, γ = C_p/C_v.

Marks split	Marks
P-V diagram	1
T-S diagram	1
Processes explanation	1
Efficiency formula	1
Total	**4**

Q.4 (e) [4 Marks]

Determine if the steam is wet or superheated and calculate the dryness fraction or the superheated steam temperature for P = 9 bar and V = 0.30 m3/kg.

Ans.

Data note: Steam-table values differ slightly from book to book (for example v_g at 9 bar is 0.2148–0.2149 m³/kg). **Assumption used:** Values were taken from the IAPWS-IF97 steam tables (checked in code). Use the values from your own steam table in the exam; the result will differ by about 1 °C.

Numerical: State of steam at 9 bar, v = 0.30 m³/kg

Given data	Value	Unit	Remark
Pressure p	9	bar	
Specific volume v	0.3	m³/kg	

Formula and calculation:

From steam tables at 9 bar: t_s = 175.4 °C, v_g = 0.2149 m³/kg

1. Compare v with v_g

v > v_g ? 0.3 > 0.2149

Since v is greater than v_g, the steam is SUPERHEATED.

2. Superheat table at 9 bar, 300 °C
v_{300} = **0.2874 m³/kg**

3. Superheat table at 9 bar, 350 °C
v_{350} = **0.3145 m³/kg**

4. Interpolate for v = 0.30
$t = 300 + 50 \times (v - v_{300})/(v_{350} - v_{300}) = 300 + 50 \times (0.3 - 0.2874) / (0.3145 - 0.2874)$ = **323.3 °C**

Final answer
State of steam = **Superheated**
Temperature of superheated steam = **323.3 °C**
Degree of superheat = **147.9 K**

Checked in code by a second method: ✓ Exact solution of v(t) = 0.30 (IAPWS-IF97) vs interpolation, within 1 °C: 323.2929 ≈ 323.1743
Note: Exact IF97 value: 323.2 °C. Table interpolation can differ by about 1 °C from your steam table.

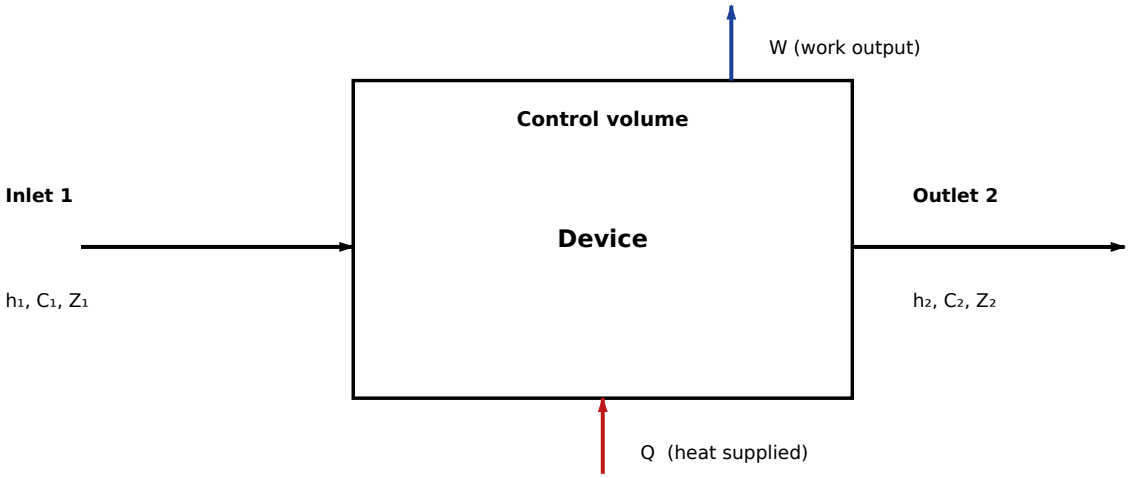
Marks split	Marks
Steam table data at 9 bar	1
Comparison of v with v_g – state of steam	1
Interpolation	1
Superheat temperature with unit	1
Total	**4**

Q.5 Attempt any TWO of the following: — 12 Marks (each sub-question = 6 marks)

Q.5 (a) [6 Marks]

Write steady flow energy equation and apply it to steam turbine and steam boiler.

Ans.



Control volume for steady flow

Steady flow energy equation (S.F.E.E.) per kg of fluid

For a steady flow system, the energy entering with the fluid and as heat equals the energy leaving with the fluid and as work:

$$h_1 + C_1^2/2 + g\,Z_1 + Q = h_2 + C_2^2/2 + g\,Z_2 + W \text{ (J/kg or kJ/kg)}$$

h = specific enthalpy, C = velocity, Z = height above datum, Q = heat supplied to the system, W = work done by the system.

(i) Steam turbine

The turbine is well insulated, so Q = 0; change in potential energy is zero; kinetic energy change is small and neglected:

$$W = h_1 - h_2 \text{ (work output = drop in enthalpy of steam)}$$

(ii) Steam boiler

No work is done (W = 0) and the changes in kinetic and potential energy are negligible:

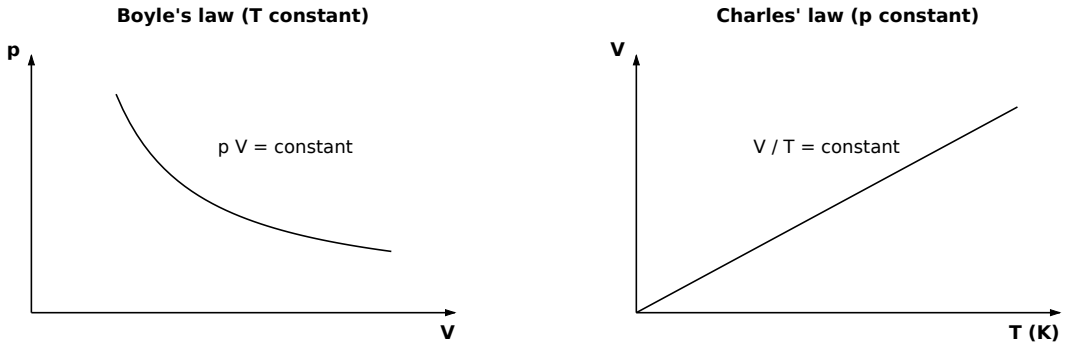
$$Q = h_2 - h_1 \text{ (heat supplied = increase in enthalpy of water/steam)}$$

Marks split	Marks
S.F.E.E. with meaning of terms	2
Assumptions	1
Steam turbine	1
Steam boiler	1
Sketch	1
Total	**6**

Q.5 (b) [6 Marks]

Derive characteristics gas equation using Boyle's law and Charle's law.

Ans.



Boyle's law and Charles' law

- **Boyle's law:** at constant temperature the absolute pressure of a given mass of gas varies inversely with its volume: $p \propto 1/V$, i.e. $p\,V = \text{constant}$.
- **Charles' law:** at constant pressure the volume of a given mass of gas varies directly with its absolute temperature: $V \propto T$, i.e. $V / T = \text{constant}$.

Let 1 kg of gas change from state 1 (p_1, V_1, T_1) to state 2 (p_2, V_2, T_2) in two steps:

1. **Process 1–A at constant temperature T_1** to the final pressure p_2 (Boyle's law): $p_1 V_1 = p_2 V_A$, so $V_A = p_1 V_1 / p_2$.
2. **Process A–2 at constant pressure p_2** (Charles' law): $V_A / T_1 = V_2 / T_2$.

Substituting V_A from step 1 into step 2:

$$p_1 V_1 / (p_2 T_1) = V_2 / T_2 \Rightarrow p_1 V_1 / T_1 = p_2 V_2 / T_2 = \text{constant}$$

The constant is called the characteristic gas constant R (per kg). For m kg of gas:

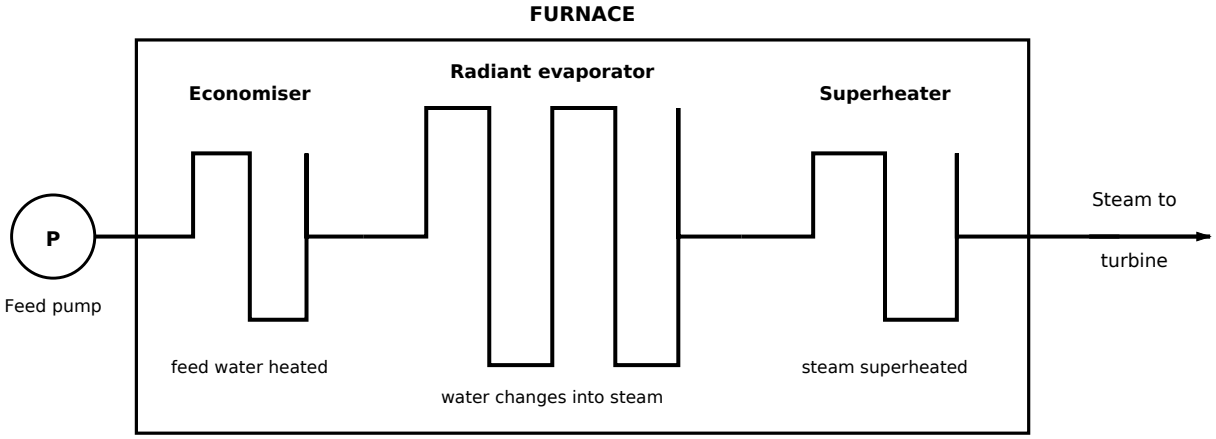
$$\mathbf{p\,V = m\,R\,T} \text{ (characteristic gas equation)}$$

Marks split	Marks
Statement of the two laws	1
Sketch	1
Derivation steps	3
Final equation	1
Total	**6**

Q.5 (c) [6 Marks]

Explain Benson boiler with neat sketch.

Ans.



One continuous tube, no steam drum.

Feed water enters at about 220 bar (above the critical pressure) and leaves as superheated steam.

Benson boiler (once-through, supercritical)

Construction

- The Benson boiler is a **high-pressure, water-tube, once-through** boiler working at or above the critical pressure (about 225 bar).
- It has **no steam drum**. A feed pump forces water through one continuous system of tubes: economiser, radiant evaporator and superheater placed in the furnace.

Working

1. At the critical pressure (221.2 bar) the latent heat of vaporisation is zero, so water changes to steam without boiling and without bubbles.
2. Feed water from the pump is heated in the economiser, then in the evaporator tubes (radiant heating) where it turns into steam, and the steam is superheated in the superheater.
3. The superheated steam is sent to the turbine.

Advantages

- No drum, so light in weight and less cost; quick starting.
- High efficiency; can be run at supercritical pressure.
- Tube arrangement is flexible; no scale-deposit problem in the drum.

Marks split	Marks
Neat sketch	2
Construction	2
Working	2
Total	**6**

Q.6 Attempt any TWO of the following: — 12 Marks (each sub-question = 6 marks)

Q.6 (a) [6 Marks]

Differentiate between Impulse turbine and Reaction turbine.

Ans.

Sr.	Impulse turbine	Reaction turbine
1	Steam expands completely in the nozzle; pressure drop takes place only in the fixed nozzles.	Steam expands partly in fixed blades and partly in moving blades; pressure drops in both.
2	Pressure of steam is constant while passing over moving blades.	Pressure of steam falls continuously while passing over moving blades.
3	Moving blades are symmetrical (bucket shaped).	Moving blades are aerofoil / asymmetrical; same shape as fixed blades.
4	Relative velocity of steam remains same (ignoring friction) over the blades.	Relative velocity of steam increases over the moving blades.
5	Partial admission of steam is possible.	Full (all-round) admission of steam is required.
6	Blade efficiency is lower; used for high-pressure stages.	Blade efficiency is higher; used for low-pressure stages.
7	Fewer stages, so space required is less.	More stages, so more space is required.
8	Examples: De-Laval, Curtis, Rateau.	Example: Parsons turbine.

Marks split	Marks
Any six points, 1 mark each	6
Total	**6**

Q.6 (b) [6 Marks]

Classify heat exchangers and state their applications.

Ans.

Classification of heat exchangers

Basis	Types
Nature of heat exchange	Direct contact (open) type – e.g. cooling tower, open feed water heater; Indirect contact (closed) type – recuperative (shell and tube) and regenerative
Direction of flow	Parallel flow; Counter flow; Cross flow
Constructional features	Shell and tube; Double pipe; Plate type; Finned tube (compact)
Change of phase	Condensers; Evaporators; Boilers (with phase change) / Heaters, coolers (no phase change)

Applications

- Condenser and feed water heater in steam power plants; economiser, air pre-heater and superheater in boilers.
- Radiators of automobiles and oil coolers in engines.
- Condenser and evaporator in refrigeration and air-conditioning plants.
- Intercoolers in compressors; chemical, food (pasteuriser) and dairy industries.

Marks split	Marks
Classification (any three bases)	3
Applications (any three)	3
Total	**6**

Q.6 (c) [6 Marks]

Define:
i) Compression Ratio (Rc)
ii) Swept Volume (Vs)
iii) Cut off ratio.

Ans.

- **Compression ratio (r):** the ratio of the total cylinder volume when the piston is at B.D.C. to the clearance volume (volume when the piston is at T.D.C.). $r = (V_s + V_c) / V_c$.

- **Swept volume (V_s):** the volume displaced by the piston in moving from T.D.C. to B.D.C. $V_s = (\pi/4)\ D^2 \times L$, where D = cylinder bore and L = stroke length.
- **Cut-off ratio (ρ):** in the Diesel cycle, the ratio of the volume at the end of heat addition (cut-off) to the volume at the beginning of heat addition (clearance volume). $\rho = V_3 / V_2$.

Marks split	Marks
Compression ratio	2
Swept volume	2
Cut-off ratio	2
Total	**6**

Summer 2025

Subject	Thermal Engineering (313310)	Time / Marks	3 Hours / 70 Marks
Session	Summer 2025	Paper ID printed	24225
Scheme	K-scheme	Source file	Thermal_Engineering_Summer_2025_Question_Paper_Second_K_Scheme_Paper.pdf

Q.1 Attempt any FIVE of the following : — 10 Marks (each sub-question = 2 marks)

Q.1 (a) [2 Marks]

Define the terms:
i) Thermodynamic System
ii) Thermodynamic Property.

Ans.

- **Thermodynamic system:** a definite quantity of matter or a region in space, chosen for study, which is separated from its surroundings by a boundary (real or imaginary). Types: closed, open and isolated system.
- **Thermodynamic property:** any measurable characteristic that describes the state (condition) of a system, such as pressure, temperature, volume, internal energy and enthalpy. Properties are of two kinds – intensive (independent of mass, e.g. p, T) and extensive (depend on mass, e.g. V, U).

Marks split	Marks
Thermodynamic system	1
Thermodynamic property	1
Total	**2**

Q.1 (b) [2 Marks]

Define the term Universal Gas Constant. State its unit.

Ans.

Universal gas constant (R_u) is the product of the molecular mass (M) of a gas and its characteristic gas constant (R). It has the same value for all gases.

$$R_u = M \times R = 8.314 \text{ kJ/kmol K } (= 8314 \text{ J/kmol K})$$

Marks split	Marks
Definition	1
Value and unit	1
Total	**2**

Q.1 (c) [2 Marks]

Define the terms:
i) Sensible Heat
ii) Latent Heat

Ans.

- **Sensible heat (h_f)** is the heat required to raise the temperature of 1 kg of water from 0 °C to the saturation (boiling) temperature at the given pressure. It changes the temperature.
- **Latent heat (h_{fg})** is the heat required to convert 1 kg of water at saturation temperature into dry saturated steam at the same temperature and pressure. It changes only the phase, not the temperature.

Marks split	Marks
Sensible heat	1
Latent heat	1
Total	**2**

Q.1 (d) [2 Marks]

Classify the boilers according to -
i) Relative position of flow of water and hot gases
ii) Position of furnace.

Ans.

- **(i) Relative position of flow of water and hot gases:** (a) Fire tube boiler – hot gases flow inside the tubes and water around them (Cochran, Lancashire, Locomotive); (b) Water tube boiler – water flows inside the tubes and hot gases outside (Babcock & Wilcox, La-Mont, Benson).
- **(ii) Position of furnace:** (a) Internally fired boiler – the furnace is inside the boiler shell (Cochran, Lancashire); (b) Externally fired boiler – the furnace is outside the shell (Babcock & Wilcox).

Marks split	Marks
Classification (i) with examples	1
Classification (ii) with examples	1
Total	**2**

Q.1 (e) [2 Marks]

State the Dalton's law of partial pressure.

Ans.

Dalton’s law: The total pressure exerted by a mixture of non-reacting gases is equal to the sum of the partial pressures of the individual gases, each gas occupying the whole volume of the mixture at the same temperature.

$$p = p_1 + p_2 + p_3 + \ldots$$

Marks split	Marks
Statement	1
Equation	1
Total	**2**

Q.1 (f) [2 Marks]

Define the terms –
i) Black body
ii) Abosoptivity.

Ans.

- **Black body:** an ideal body which absorbs all the thermal radiation falling on it, whatever the wavelength or direction. It is also the best emitter of radiation (emissivity $\varepsilon = 1$).
- **Absorptivity (α):** the ratio of the radiant energy absorbed by a body to the total radiant energy falling on it. For a black body $\alpha = 1$; for a real body $\alpha < 1$.

Marks split	Marks
Black body	1
Absorptivity	1
Total	**2**

Q.1 (g) [2 Marks]

Classify the I.C. engines according to –
i) Method of igniting the fuel
ii) Cycle of operation.

Ans.

- **(i) Method of igniting the fuel:** Spark ignition (S.I.) engine – petrol engine, fuel ignited by a spark plug; Compression ignition (C.I.) engine – diesel engine, fuel ignited by heat of compressed air.
- **(ii) Cycle of operation:** Otto cycle engine (constant-volume heat addition), Diesel cycle engine (constant-pressure heat addition), Dual cycle engine (part at constant volume and part at constant pressure).

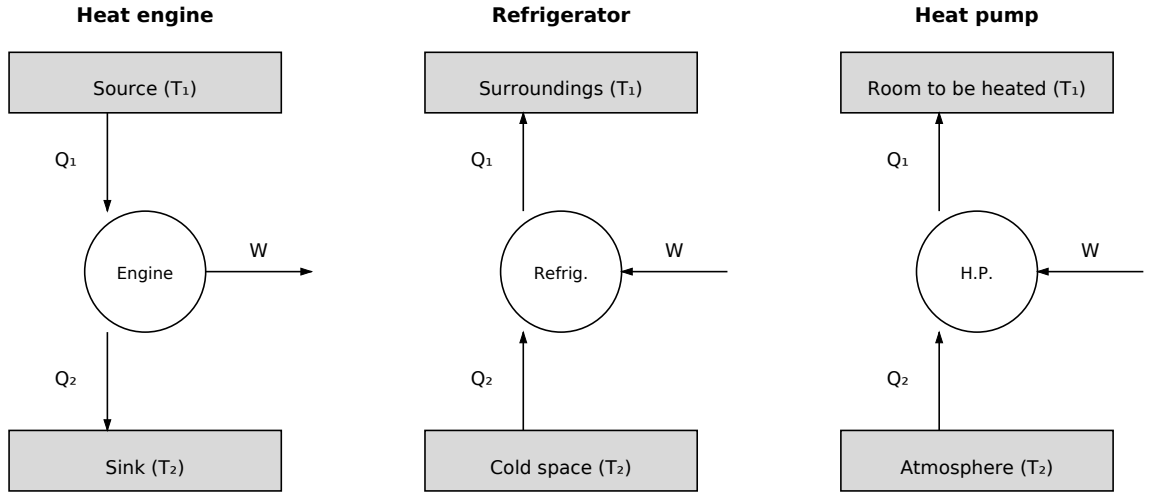
Marks split	Marks
Method of ignition	1
Cycle of operation	1
Total	**2**

Q.2 Attempt any THREE of the following : — 12 Marks (each sub-question = 4 marks)

Q.2 (a) [4 Marks]

Explain the concept of Heat engine, Heat pump and Refrigerator with block diagram.

Ans.



Heat engine, refrigerator and heat pump

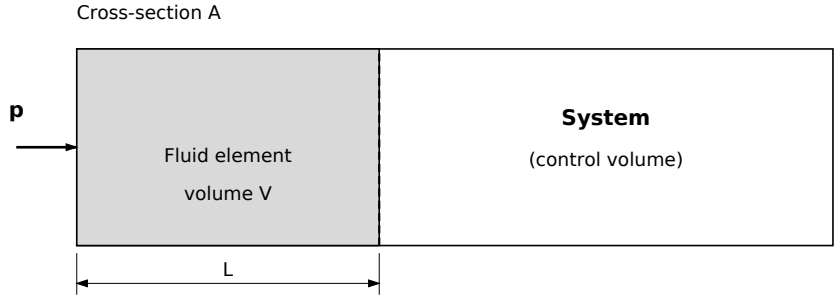
- **Heat engine:** a device that continuously converts heat into work while working in a cycle. It receives heat Q_1 from a high-temperature source (T_1), converts part of it into work W and rejects the rest Q_2 to a sink (T_2). $W = Q_1 - Q_2$; $\eta = W / Q_1 = 1 - Q_2/Q_1$.
- **Refrigerator:** a device that works in a cycle to remove heat Q_2 from a low-temperature space (T_2) and reject Q_1 to the surroundings (T_1) with the help of work input W. $(COP)_R = Q_2 / W$.
- **Heat pump:** a device that works in a cycle to supply heat Q_1 to a space to be kept warm (T_1) by taking Q_2 from the cold surroundings (T_2) with work input W. $(COP)_{HP} = Q_1 / W$.
- Relation: $W = Q_1 - Q_2$ and $(COP)_{HP} = (COP)_R + 1$.

Marks split	Marks
Heat engine – diagram and explanation	1
Refrigerator – diagram and explanation	1
Heat pump – diagram and explanation	1
Formulae / relation	1
Total	**4**

Q.2 (b) [4 Marks]

Explain the concept of flow work with neat sketch.

Ans.



Force on the element F = p × A

Distance moved = L

Flow work = F × L = p A L

= p V

Flow work

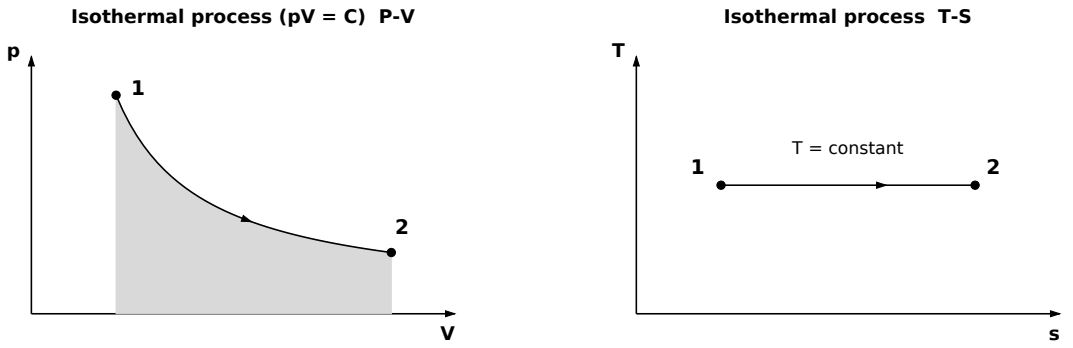
- **Flow work** is the work required to push a fluid element into, or out of, a control volume (an open system) against the pressure existing at the boundary.
- Consider a fluid element of volume V pushed into the system through a pipe of cross-section A at pressure p. Force on the element F = p × A. If the element moves a distance L, work = F × L = p A L.
- Since A × L = V, **flow work = p V** (for 1 kg: p v, in kJ/kg).
- Flow work is the reason for the term p v in enthalpy: h = u + p v.

Marks split	Marks
Sketch	2
Explanation	1
Expression for flow work	1
Total	**4**

Q.2 (c) [4 Marks]

Represent the isothermal process on P-V and T-S chart. State the relation to calculate the workdone during isothermal process.

Ans.



Isothermal process on P-V and T-S diagrams

- In an isothermal process the temperature remains constant: p V = constant (Boyle's law), so ΔU = 0 and all the heat supplied is converted into work (Q = W).
- On the P-V diagram it is a rectangular hyperbola; on the T-S diagram it is a horizontal straight line.

Work done

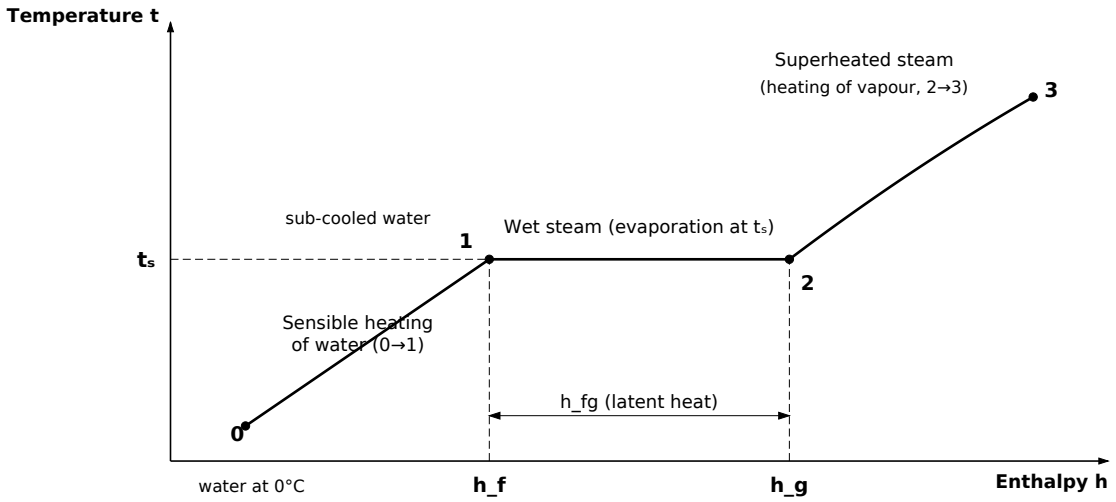
$$W = p_1 V_1 \ln (V_2 / V_1) = m R T \ln (V_2 / V_1) = p_1 V_1 \ln (p_1 / p_2)$$

Marks split	Marks
P-V diagram	1
T-S diagram	1
Characteristics	1
Work formula	1
Total	**4**

Q.2 (d) [4 Marks]

Explain the process of steam generation with the help of T-H chart.

Ans.



Steam generation at constant pressure on the T–h chart

1. **0 → 1 (sensible heating):** water at 0 °C is heated at constant pressure; its temperature rises to the saturation temperature t_s. Heat added = h_f (sensible heat).
2. **1 → 2 (evaporation):** further heat added at constant temperature t_s is the latent heat h_{fg}; the water turns into steam. In between, the mixture is **wet steam** with dryness fraction x. At point 2 all water is converted: **dry saturated steam** (x = 1). $h_{wet} = h_f + x\, h_{fg}$; $h_g = h_f + h_{fg}$.
3. **2 → 3 (superheating):** heating dry steam at constant pressure raises its temperature above t_s; this is **superheated steam**. $h_{sup} = h_g + C_{ps}\,(t_{sup} - t_s)$.

Marks split	Marks
T-h chart with labels	2
Explanation of three stages	2
Total	**4**

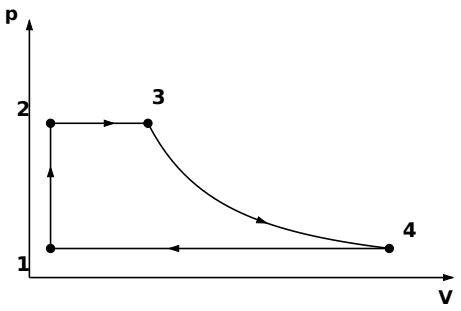
Q.3 Attempt any THREE of the following : — 12 Marks (each sub-question = 4 marks)

Q.3 (a) [4 Marks]

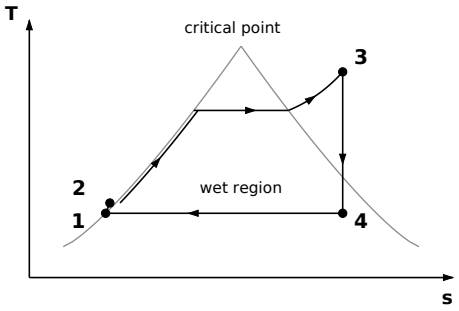
Represent the Rankine cycle on P-V and T-S chart. Name the various processes involved in it.

Ans.

Rankine cycle P-V



Rankine cycle T-S



1-2 Pump work (isentropic)

2-3 Heat added in boiler (constant p)

3-4 Isentropic expansion in turbine

4-1 Condensation (constant p, T)

Rankine cycle

Process	Name	Where it occurs
1 – 2	Reversible adiabatic (isentropic) compression	Feed pump
2 – 3	Constant-pressure heat addition (water → steam → superheated steam)	Boiler and superheater
3 – 4	Reversible adiabatic (isentropic) expansion	Steam turbine
4 – 1	Constant-pressure heat rejection (condensation)	Condenser

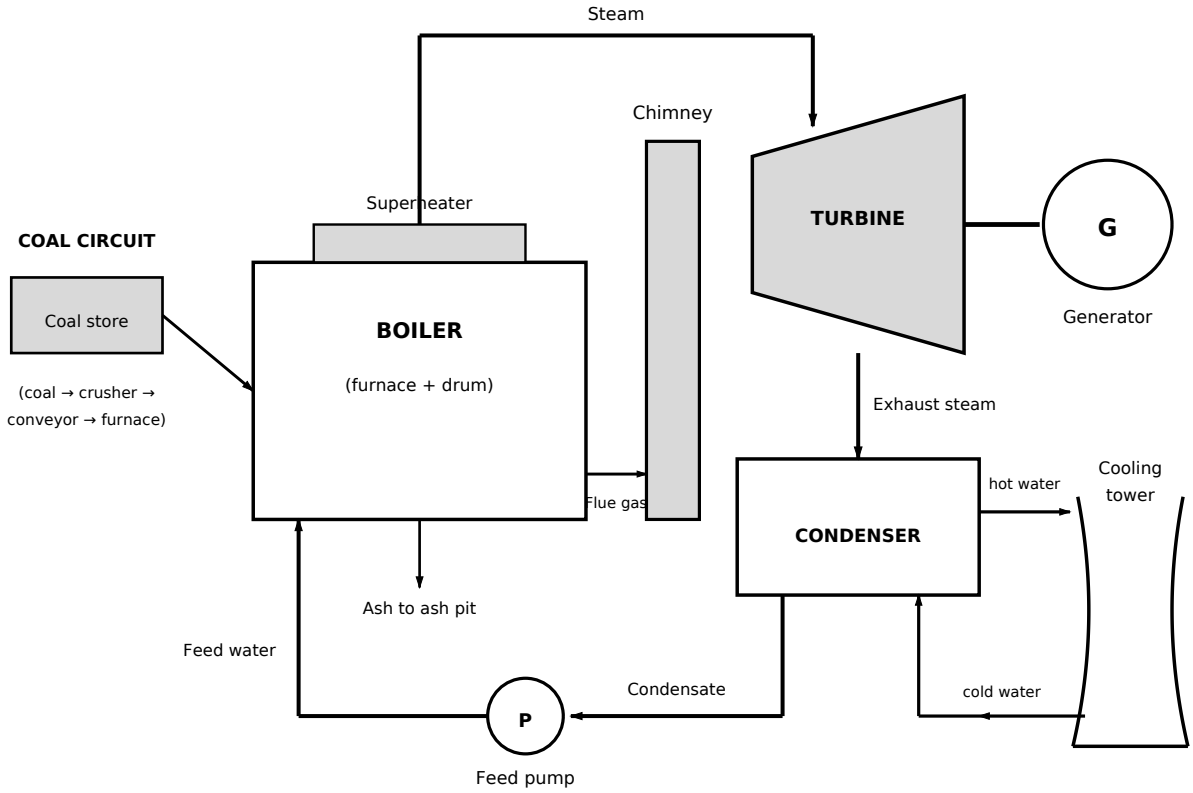
$$\eta_{Rankine} = [(h_3 - h_4) - (h_2 - h_1)] / (h_3 - h_2)$$

Marks split	Marks
P-V diagram	1
T-S diagram	1
Names of processes	2
Total	**4**

Q.3 (b) [4 Marks]

Draw the layout of steam power plant and label the following components –
i) Boiler
ii) Turbine
iii) Condenser
iv) Generator
v) Cooling tower
vi) Chimney
vii) Feed Pump
viii) Coal Circuit

Ans.



Layout of a steam power plant

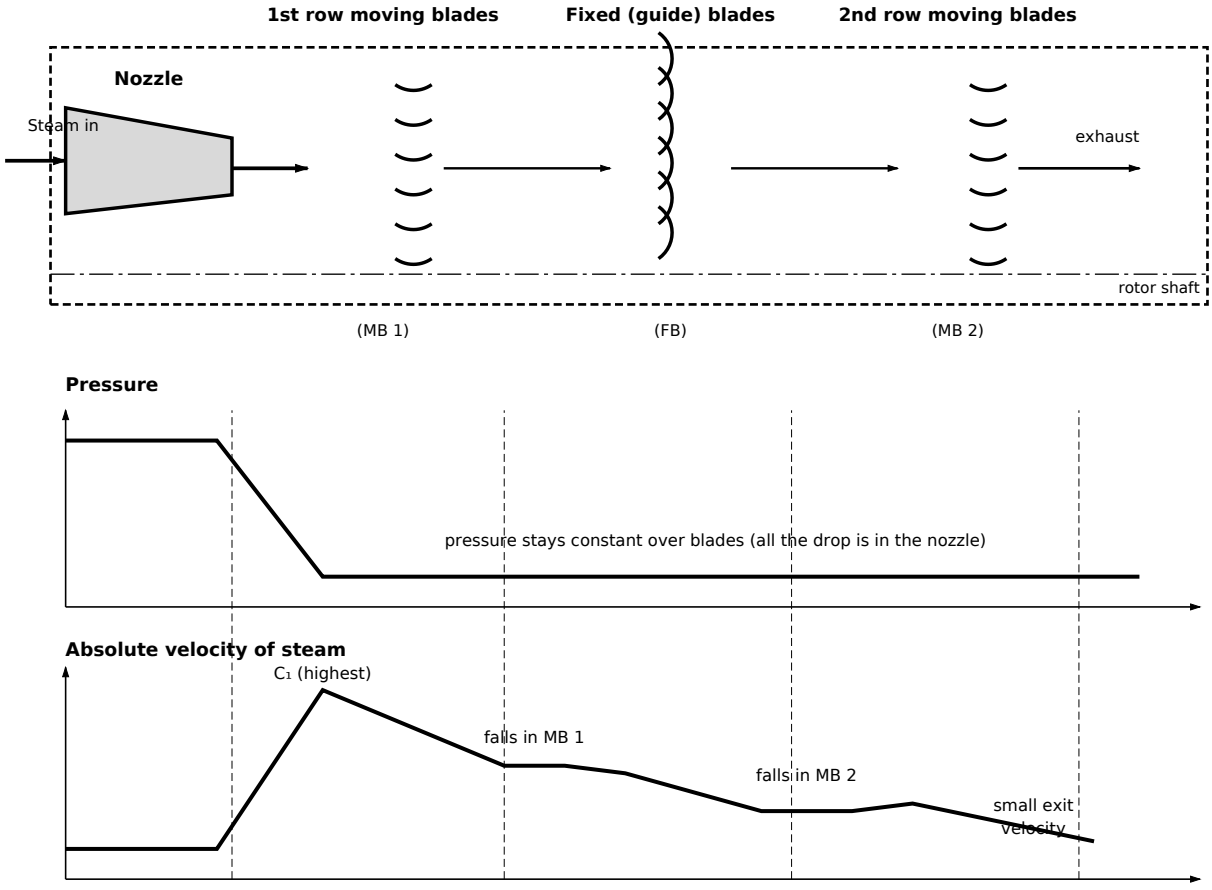
Component	Function
Coal circuit	Brings coal from the store through crusher and conveyor to the furnace; ash is removed to the ash pit.
Boiler	Burns coal and generates steam at high pressure (with superheater to give superheated steam).
Turbine	Converts heat energy of steam into mechanical work on the shaft.
Generator	Coupled to the turbine; converts mechanical energy into electrical energy.
Condenser	Condenses exhaust steam and creates vacuum at the turbine outlet.
Cooling tower	Cools the warm cooling water coming from the condenser so that it can be reused.
Feed pump	Pumps condensate (feed water) back to the boiler against boiler pressure.
Chimney	Discharges flue gases to the atmosphere at a safe height.

Marks split	Marks
Layout with correct flow arrows	2
Labelling of the eight components	1
Functions	1
Total	**4**

Q.3 (c) [4 Marks]

Explain the method of velocity compounding of an impulse turbine for achieving rotor speed reduction.

Ans.



Velocity compounding (Curtis) – two rows of moving blades

Need

In a simple impulse turbine the whole pressure drop takes place in one nozzle, so the steam leaves with a very high velocity and the rotor must run at a very high speed (the optimum blade speed is $u = C_1 \cos \alpha_1 / 2$). Such high speed is not practical for bearings and reduction gearing.

Method

1. The entire pressure drop of steam occurs in a set of nozzles, so steam leaves with a high velocity C_1.
2. This high-velocity steam passes through the **first row of moving blades**, which absorbs only part of the kinetic energy; the steam leaves with a reduced velocity.
3. The steam then passes through a row of **fixed (guide) blades** fixed to the casing, which only change its direction (no pressure drop) and guide it into the second row.
4. The **second row of moving blades** (on the same rotor) absorbs more of the remaining kinetic energy. More rows can be used.
5. The velocity is thus reduced in steps, while pressure remains constant over the blades.

Optimum blade speed $u = C_1 \cos \alpha_1 / (2\,z)$, where z = number of moving-blade rows

So with two rows the rotor speed is only half that of a single-stage turbine for the same nozzle velocity – this achieves the **speed reduction**.

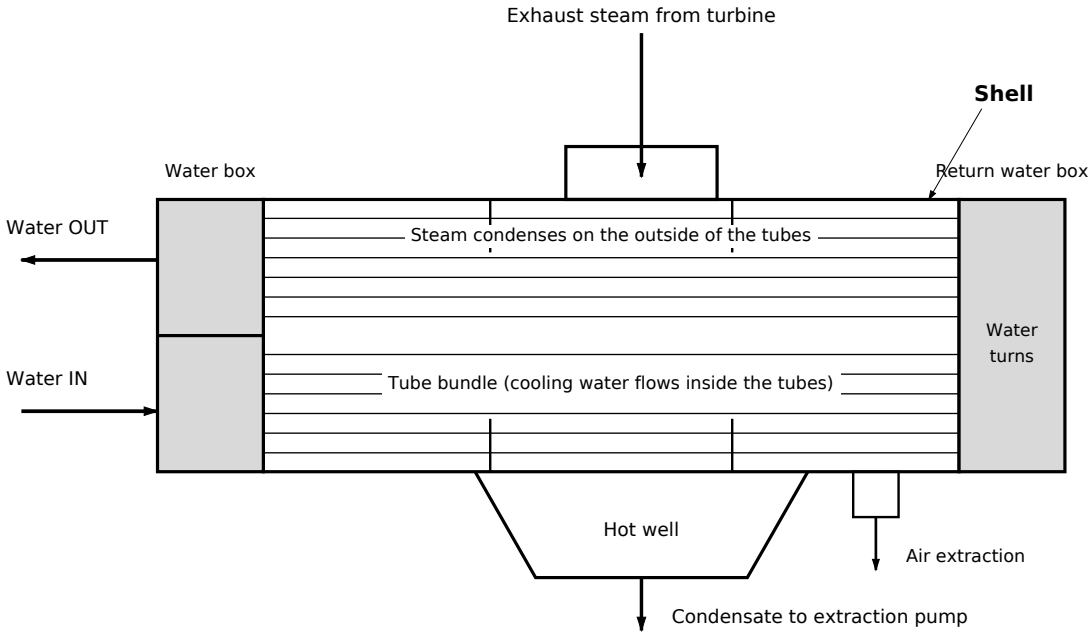
Marks split	Marks
Sketch with pressure and velocity variation	2
Explanation of method	2
Total	**4**

Q.3 (d) [4 Marks]

Suggest the non mixing type and high capacity condenser for power plant. Draw its labeled sketch and explain its working in brief.

Ans.

The non-mixing (surface) type condenser is suitable: the steam and cooling water do not mix, so the pure condensate is reused as boiler feed water. A shell-and-tube type with large tube area gives high capacity.



Surface condenser (shell and tube, two pass)

Working

1. Exhaust steam from the turbine enters the shell at the top and flows over the outside of the tubes.
2. Cooling water, pumped through the tubes in two passes, absorbs the latent heat of the steam and condenses it.
3. The condensate collects in the hot well and is pumped to the boiler as feed water.
4. An air extraction pump removes air and non-condensable gases to maintain a high vacuum.

Marks split	Marks
Suggestion – surface condenser with reason	1
Labelled sketch	2
Working	1
Total	**4**

Q.4 Attempt any THREE of the following : — 12 Marks (each sub-question = 4 marks)

Q.4 (a) [4 Marks]

Explain the Fourier's law and Newton's law of cooling related to heat transfer.

Ans.

- **Fourier's law (conduction):** the rate of heat flow by conduction through a material is directly proportional to the area A normal to the direction of flow and to the temperature gradient dT/dx.
- **Newton's law of cooling (convection):** the rate of heat transfer by convection from a surface to a fluid is directly proportional to the surface area A and to the temperature difference between the surface and the fluid.

Fourier: $Q = -\,k\,A\,(dT/dx)$ k = thermal conductivity, W/m K; the minus sign shows heat flows from higher to lower temperature

Newton: $Q = h\,A\,(T_s - T_f)$ h = convection heat transfer coefficient, W/m² K

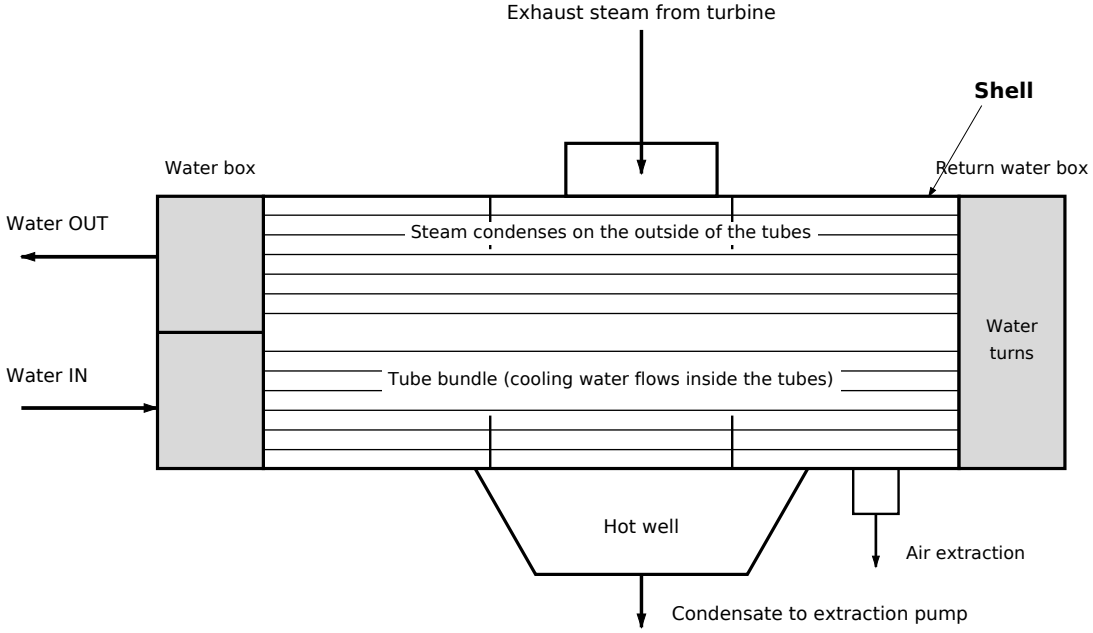
Marks split	Marks
Fourier’s law – statement and equation	2
Newton’s law of cooling – statement and equation	2
Total	**4**

Q.4 (b) [4 Marks]

Suggest the type of heat exchanger required for steam power plant. Explain its working with neat label sketch.

Ans.

A **shell-and-tube heat exchanger** is used in a steam power plant, for example as the **surface condenser** (also as closed feed water heater). It is a non-mixing (recuperative) type of heat exchanger.



Shell-and-tube heat exchanger used as a surface condenser

Working

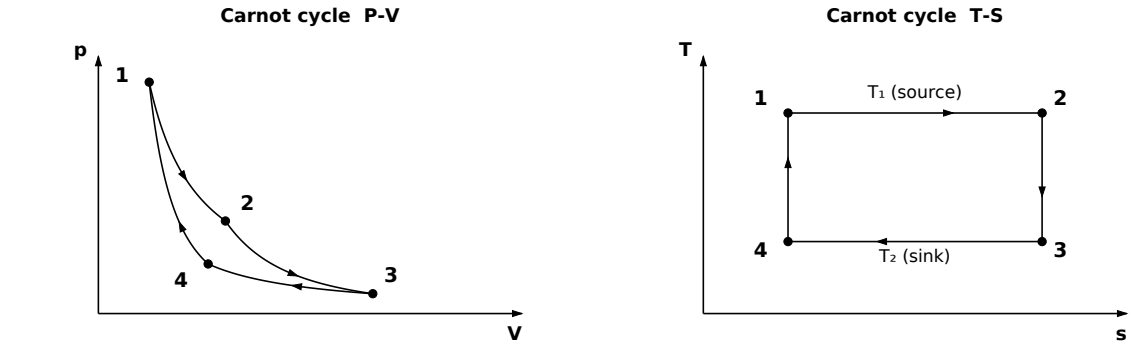
1. Exhaust steam (hot fluid) enters the shell and flows outside the tubes; cooling water (cold fluid) flows inside the tubes.
2. Heat passes from the steam through the tube wall to the cooling water; the steam condenses and the water gets warmer.
3. Condensate is collected in the hot well; the warm water goes to the cooling tower; the baffles increase the heat transfer.

Marks split	Marks
Type of heat exchanger	1
Sketch with labels	2
Working	1
Total	**4**

Q.4 (c) [4 Marks]

Represent the Carnot cycle on P-V and T-S diagram and label it. State its limitations.

Ans.



1-2 Isothermal expansion (heat added)

2-3 Isentropic expansion

3-4 Isothermal compression (heat rejected)

4-1 Isentropic compression

Carnot cycle (schematic, not to scale)

$$\eta_{Carnot} = 1 - T_2 / T_1$$

Limitations

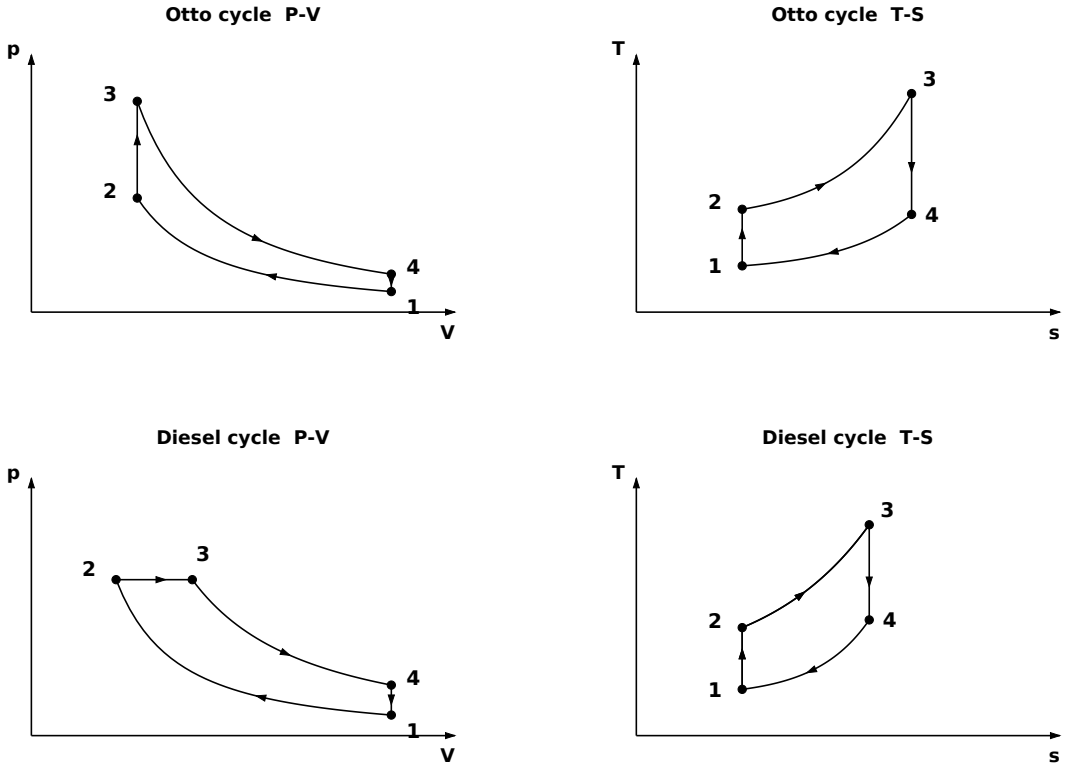
- Isothermal heat transfer needs an infinitely slow process, and adiabatic reversible expansion/compression needs frictionless working – both are impossible in practice.
- The two isothermal processes need a very large temperature difference to be kept constant; the cycle cannot be built with a gas as a practical engine.
- The work ratio and mean effective pressure are very low, so a very large engine is needed for small output.
- Efficiency depends only on source and sink temperatures; the maximum temperature is limited by materials.

Marks split	Marks
P-V diagram with labels	1
T-S diagram with labels	1
Limitations (any two)	2
Total	**4**

Q.4 (d) [4 Marks]

Compare the Otto cycle with Diesel cycle on following basis -
i) Representation on P-V and T-S Chart.
ii) Compression ratio.
iii) Heat addition
iv) Application

Ans.



Otto cycle (top) and Diesel cycle (bottom)

Basis	Otto cycle	Diesel cycle
Representation	Two isentropics and two constant-volume lines on P-V (see top diagrams)	Two isentropics, one constant-pressure and one constant-volume line (see bottom diagrams)
Compression ratio	Low: about 6 to 10	High: about 14 to 22
Heat addition	At constant volume (2–3)	At constant pressure (2–3)
Application	Petrol (S.I.) engines – cars, scooters, small generators	Diesel (C.I.) engines – trucks, buses, tractors, locomotives, ships

Marks split	Marks
Representation on P-V and T-S	1
Compression ratio	1
Heat addition	1
Application	1
Total	**4**

Q.4 (e) [4 Marks]

The initial volume of 0.17 kg of a certain gas was 0.14 m3 at a temperature of 15°C and a pressure of 1 bar. After adiabatic compression to 0.057m3, the pressure was found to be 4 bar.
Find -
i) Gas constant
ii) If ratio of specific heat is 1.407, find the value of CP and CV.
iii) Change in internal energy.

Ans.

Data note: The data are slightly inconsistent: with $\gamma = 1.407$ an adiabatic change from 0.14 m³ to 0.057 m³ would not give a final pressure of 4 bar (the end states imply $\gamma = 1.54$).
Assumption used: The end-state data are used as printed ($T_2 = p_2V_2/mR$); this is the usual method. The alternative T_2 from the adiabatic law would give a different ΔU.

Numerical: Adiabatic compression of 0.17 kg of gas

Given data	Value	Unit	Remark
Mass m	0.17	kg	
V_1	0.14	m³	
t_1	15	°C	$T_1 = 15 + 273 = 288$ K
p_1	1	bar	$= 10^5$ N/m²
V_2	0.057	m³	
p_2	4	bar	$= 4 \times 10^5$ N/m²
γ	1.407		

Formula and calculation:

1. (i) Gas constant
$R = p_1 V_1 / (m T_1) = (100000 \times 0.14) / (0.17 \times 288) =$ **285.95 J/kg K**

2. (ii) Specific heat at constant volume
$C_v = R / (\gamma - 1) = 285.95 / (1.407 - 1) =$ **0.7026 kJ/kg K**

3. Specific heat at constant pressure
$C_p = \gamma C_v = 1.407 \times 0.7026 =$ **0.9885 kJ/kg K**

4. Final temperature from end state
$T_2 = p_2 V_2 / (m R) = (400000 \times 0.057) / (0.17 \times 285.95) =$ **469.03 K**

5. (iii) Change in internal energy
$\Delta U = m C_v (T_2 - T_1) = 0.17 \times 0.7026 \times (469.03 - 288) =$ **21.62 kJ**

Final answer
Gas constant R = **285.95 J/kg K**
C_v = **0.7026 kJ/kg K**
C_p = **0.9885 kJ/kg K**
Change in internal energy ΔU (increase) = **21.62 kJ**

Checked in code by a second method: ✓ Adiabatic: $\Delta U = -W$, with $W = (p_1V_1 - p_2V_2)/(\gamma - 1)$: 21.6216 ≈ 21.6216; ✓ $C_p - C_v = R$: 0.2859 ≈ 0.2859

Note: Data check: the end states give γ(implied) = $\ln(p_2/p_1)/\ln(V_1/V_2)$ = 1.543, not 1.407. Using T_2 from the end state (as above) is the usual method.

Note: T_2 by the adiabatic law $T_1(V_1/V_2)^{(\gamma-1)}$ would be 415.2 K – it does not agree with 469.0 K, so the data are slightly inconsistent.

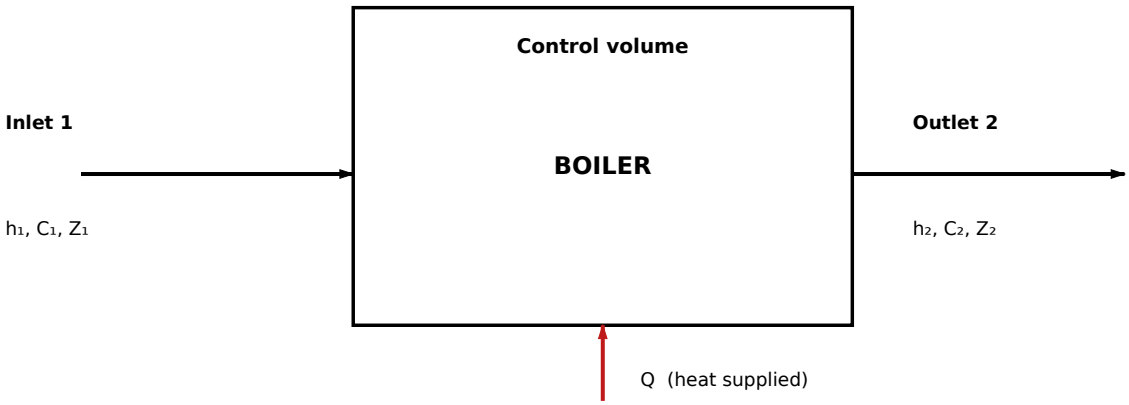
Marks split	Marks
Gas constant	1
C_p and C_v	1
Final temperature	1
Change in internal energy	1
Total	**4**

Q.5 Attempt any TWO of the following : — 12 Marks (each sub-question = 6 marks)

Q.5 (a) [6 Marks]

Apply the steady flow energy equation to boiler. Show that the heat supplied to boiler increases the enthalpy of system.

Ans.



Boiler as a steady-flow control volume

Steady flow energy equation (per kg)

$$h_1 + C_1^2/2 + g\,Z_1 + Q = h_2 + C_2^2/2 + g\,Z_2 + W$$

Application to a boiler

1. Feed water enters at state 1 and leaves as steam at state 2; the flow is steady (mass flow in = mass flow out).
2. A boiler has no moving parts delivering work: **W = 0**.
3. Inlet and outlet are at nearly the same level and the velocities are small: **ΔPE = 0** and **ΔKE ≈ 0**.
4. Substituting in S.F.E.E.: $h_1 + Q = h_2$

$$\mathbf{Q = h_2 - h_1}$$

So the heat supplied to the boiler per kg is exactly equal to the increase in enthalpy of the working fluid. Hence heat supplied increases the enthalpy of the system. Hence proved.

Marks split	Marks
S.F.E.E. with terms	2
Assumptions for boiler	2
Derivation and conclusion	2
Total	**6**

Q.5 (b) [6 Marks]

A closed vessel contains 0.4 m3 of steam at 10 bar and 300°C. Using following steam table -
i) Identify the quality of steam.
ii) Calculate the total enthalpy of steam
iii) Find specific volume of steam.
Take specific heat of sueperheated steam as 2.1KJ/kg K.

Absolute Pressure in bar: 10; Temp. in °C: 179.9; Sp. Volume (m3/kg): vf = 1.127, vg = 0.1943; Sp. Enthalpy (kJ/kg): hf = 763.2, hg = 2778; Sp. Entropy (kJ/kg K): sf = 2.140, sg = 6.586

Ans.

Data note: The table prints $v_f = 1.127$. The correct value is 0.001127 m³/kg (1.127 is in litres/kg, or the decimal is missing). "$s_f = 2.140$" is also not needed.
Assumption used: v_f, s_f and s_g are not needed for this question, so the answer is not affected. Only t_s, v_g and h_g are used.

Numerical: Superheated steam in a closed vessel (10 bar, 300 °C)

Given data	Value	Unit	Remark
Volume of vessel V	0.4	m³	
Pressure p	10	bar	

Given data	Value	Unit	Remark
Temperature t	300	°C	
From the table: t_s	179.9	°C	
v_g	0.1943	m³/kg	
h_g	2778	kJ/kg	
C_p of superheated steam	2.1	kJ/kg K	

Formula and calculation:

1. (i) Quality of steam

$t > t_s$? 300 °C > 179.9 °C

Steam temperature is higher than saturation temperature, so the steam is SUPERHEATED.

2. Degree of superheat

$t - t_s = 300 - 179.9 =$ **120.1 K**

3. (iii) Specific volume (Charles' law, p constant)

$v = v_g \times T / T_s = 0.1943 \times 573 / 452.9 =$ **0.2458 m³/kg**

4. Specific enthalpy of superheated steam

$h = h_g + C_p (t - t_s) = 2778 + 2.1 \times (300 - 179.9) =$ **3030.21 kJ/kg**

5. Mass of steam in vessel

$m = V / v = 0.4 / 0.2458 =$ **1.6272 kg**

6. (ii) Total enthalpy

$H = m\ h = 1.6272 \times 3030.21 =$ **4930.69 kJ**

Final answer
Quality of steam = **Superheated**
Total enthalpy H = **4930.69 kJ**
Specific volume v = **0.2458 m³/kg**

Checked in code by a second method: ✓ m × v = V: 0.4000 ≈ 0.4000; ✓ Charles-law v vs IAPWS-IF97 table value at 10 bar, 300 °C (approximate method, 6 %): 0.2458 ≈ 0.2580

Note: The table in the question gives vf = 1.127; the correct value is 0.001127 m³/kg (not needed here).

Marks split	Marks
Quality of steam	1
Specific volume	2
Specific enthalpy and mass	1
Total enthalpy	2
Total	**6**

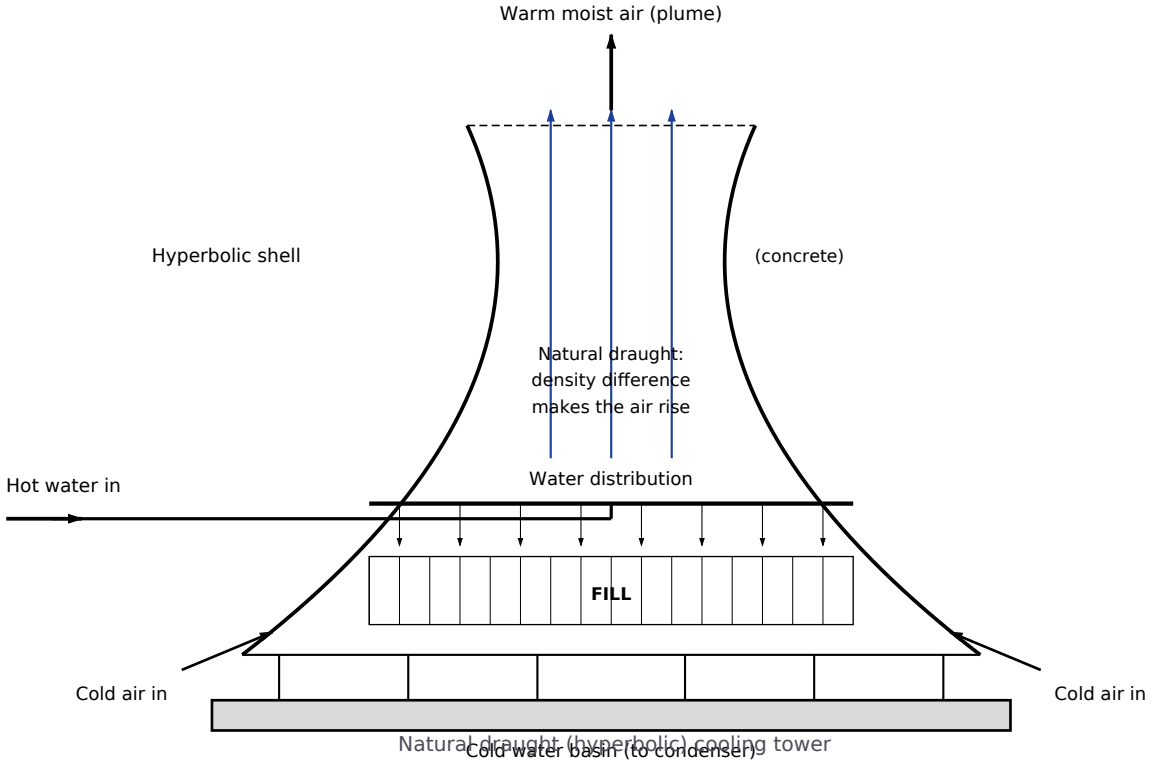
Q.5 (c) [6 Marks]

A steam power plant having power generation capacity 660 MW is to be installed where availability of water is limited. Suggest the type of cooling tower required for this plant. Draw the neat labeled sketch and explain its working principle in brief.

Ans.

Type suggested: natural draught (hyperbolic) cooling tower

- A 660 MW plant rejects a very large quantity of heat, so a large-capacity tower is needed. A natural draught tower handles very large water flows with no fan power.
- It is a **recirculating (closed-loop) tower**: the cooled water is used again and again. Only the evaporation, drift and blow-down loss (about 1–2 % of the flow) has to be made up – this suits a site with limited water, compared with once-through cooling.
- It has low running cost (no fan, no motor) and high reliability.



Natural draught (hyperbolic) cooling tower

Working principle

1. Hot water from the condenser is carried to the distribution system near the base of the tower and sprayed over the fill.
2. Atmospheric air enters through the open space at the base. Air in the tower is warm and humid, so it is lighter than the outside air; the density difference causes a **natural draught** (chimney effect) in the tall hyperbolic shell.
3. As air rises through the falling water, a small part of the water evaporates, taking latent heat from the rest and cooling it.
4. The cooled water falls into the basin and is pumped back to the condenser; the warm moist air leaves from the top.

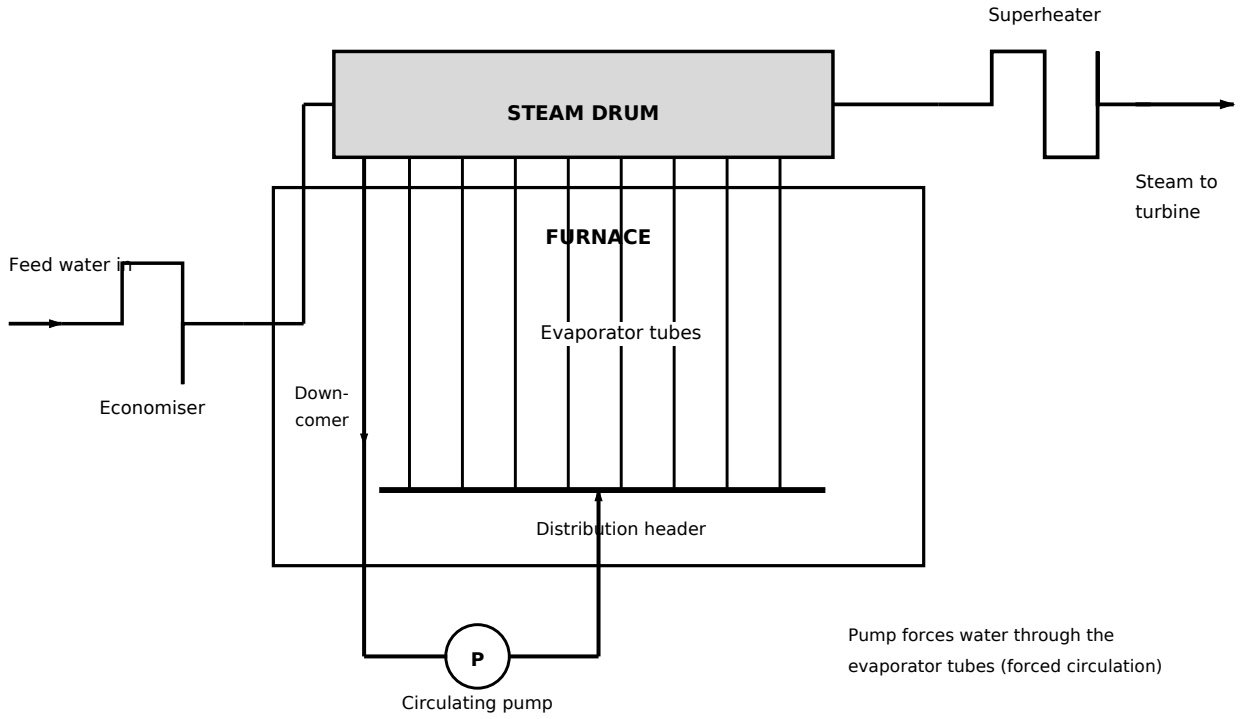
Marks split	Marks
Selection of the type with reasons	2
Neat labelled sketch	2
Working principle	2
Total	**6**

Q.6 Attempt any TWO of the following : — 12 Marks (each sub-question = 6 marks)

Q.6 (a) [6 Marks]

Explain with neat label sketch the construction and working of Lamont boiler.

Ans.



La-Mont boiler (forced circulation, water tube)

Construction

- The La-Mont boiler is a **high-pressure, water-tube, forced-circulation** boiler (up to about 170 bar).
- Main parts: a steam drum, a **circulating pump** that takes water from the drum, a distribution header, a bank of evaporator tubes in the furnace, an economiser, a superheater, and a feed pump.

Working

1. Feed water passes through the economiser and enters the steam drum.
2. The circulating pump takes water from the drum (through the downcomer) and forces it through the distribution header into the evaporator tubes.
3. Heat from the furnace converts part of the water into steam; the mixture of steam and water returns to the drum, where steam is separated.
4. The dry steam goes to the superheater and then to the turbine; the remaining water is circulated again.

Advantages

- Rapid steam generation and quick starting; tubes can be arranged freely.
- Forced circulation prevents overheating of tubes and scale deposit.

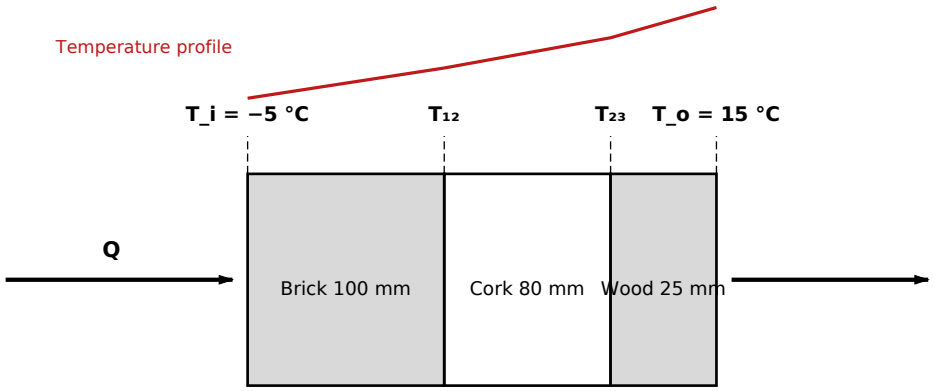
Marks split	Marks
Neat sketch with labels	2
Construction	2
Working	2
Total	**6**

Q.6 (b) [6 Marks]

A 100 mm thick brick wall measuring 5m × 3 m is insulated externally by cork 80 mm thick which is further protected by wood 25 mm thick. If the interior wall temperature is - 5°C and outer wall temperature is 15°C. Determine the heat transfer through the wall per hour. Take thermal conductivities of brick, cork and wood as 0.7, 0.05, 0.20 KW/m°C respectively. Also find temperature at inner surfaces.

Ans.

Data note: The thermal conductivities are printed as "0.7, 0.05, 0.20 KW/m°C". Brick, cork and wood have conductivities of 0.7, 0.05 and 0.20 **W**/m°C (a kilowatt value would be 1000 times too large). Also, the interior wall is colder (−5 °C) than the outer wall (15 °C), so heat flows **into** the room (cold store).
Assumption used: k is taken as 0.7, 0.05 and 0.20 W/m°C. "Temperature at inner surfaces" is taken as the temperatures of the brick–cork and cork–wood interfaces.



Brick, cork and wood wall

Numerical: Heat transfer through brick, cork and wood wall

Given data	Value	Unit	Remark
Wall area A = 5 × 3	15	m²	
Brick: L_1, k_1	0.1 m, 0.7	W/m °C	k printed as kW/m°C – taken as W/m°C
Cork: L_2, k_2	0.08 m, 0.05	W/m °C	
Wood: L_3, k_3	0.025 m, 0.2	W/m °C	
Inner wall temperature T_i	-5	°C	
Outer wall temperature T_o	15	°C	

Formula and calculation:

1. Resistance of layer 1 per m²
R1 = L1 / k1 = 0.1 / 0.7 = **0.14286 m² °C/W**

2. Resistance of layer 2 per m²
R2 = L2 / k2 = 0.08 / 0.05 = **1.60000 m² °C/W**

3. Resistance of layer 3 per m²
R3 = L3 / k3 = 0.025 / 0.2 = **0.12500 m² °C/W**

4. Total resistance per m²
$R = R_1 + R_2 + R_3 = 0.14286 + 1.60000 + 0.12500 =$ **1.86786 m² °C/W**

5. Heat flux
$q = (T_o - T_i) / R = (15 - (-5)) / 1.86786 =$ **10.7075 W/m²**

6. Heat flow through the wall
Q = q × A = 10.7075 × 15 = **160.612 W**

7. Heat transfer per hour
Q × 3600 = 160.612 × 3600 = **578.2 kJ/h**

8. Temperature at brick–cork surface
$T_{12} = T_i + q R_1 = -5 + 10.7075 \times 0.14286 =$ **-3.47 °C**

9. Temperature at cork–wood surface
$T_{23} = T_{12} + q R_2 = -3.47 + 10.7075 \times 1.60000 =$ **13.66 °C**

Final answer
Heat transfer per hour = **578.2 kJ/h**
Heat flow rate = **160.61 W**
Temperature at brick–cork surface = **-3.47 °C**
Temperature at cork–wood surface = **13.66 °C**

Checked in code by a second method: ✓ Outer wall: $T_{23} + q R_3 = T_o$: 15.0000 ≈ 15.0000

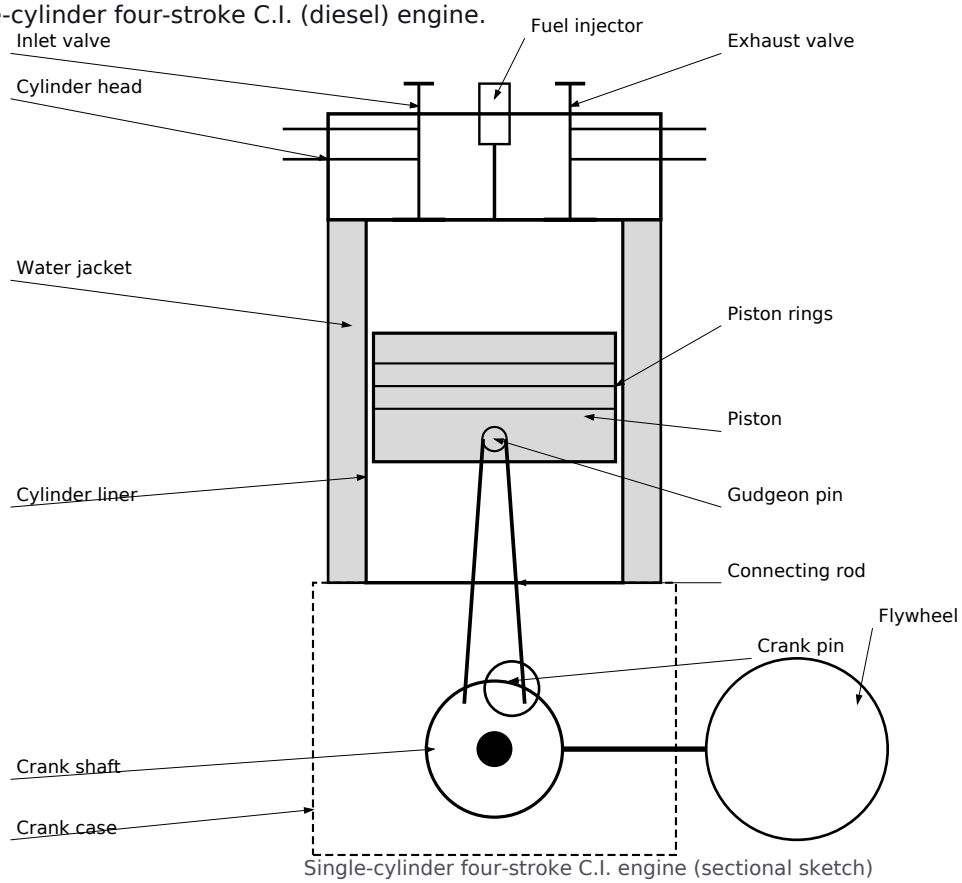
Marks split	Marks
Given data / sketch	1
Resistances and total resistance	2
Heat transfer per hour	2
Temperatures at the two inner surfaces	1
Total	**6**

Q.6 (c) [6 Marks]

Explain with neat sketch the construction and working of single cylinder four stroke C.I. engine available in your institute laboratory.

Ans.

Note: the engine in your laboratory may differ in details; the following is the general answer for a single-cylinder four-stroke C.I. (diesel) engine.



Single-cylinder four-stroke C.I. engine (sectional sketch)

Construction (main parts)

- **Cylinder** with liner and water jacket; **cylinder head** carrying inlet valve, exhaust valve and the **fuel injector** (no spark plug).
- **Piston** with piston rings, **gudgeon pin**, **connecting rod**, **crankshaft** with crank pin and **flywheel**; crankcase.
- **Valve gear:** camshaft driven at half the crankshaft speed by timing gears operates the valves. Fuel pump and injector supply fuel at about 200 bar.

Working – four strokes (two revolutions of the crankshaft)

Stroke	Piston	What happens
1. Suction	T.D.C. → B.D.C.	Inlet valve open. Only **air** is drawn into the cylinder.
2. Compression	B.D.C. → T.D.C.	Both valves closed. Air is compressed (ratio 14:1 to 22:1); its temperature rises above the self-ignition temperature of fuel (about 500 °C or more).
3. Power (expansion)	T.D.C. → B.D.C.	Near T.D.C. fuel is injected, ignites by itself and burns; hot gases expand and push the piston – work is produced.
4. Exhaust	B.D.C. → T.D.C.	Exhaust valve opens; the burnt gases are pushed out through the exhaust port.

Marks split	Marks
Neat sketch with labels	2
Construction	1
Working of four strokes	3
Total	**6**

Papers not received: Winter 2025 – not included

Not received, so not included: Winter 2025.

Questions and solutions for this paper will be added when the paper is supplied. Nothing has been guessed.

Chapter-wise marks table

Marks offered = marks of every sub-question printed in the paper, including the choice ('attempt any …'), so a column total can be more than 70. Chapter names are my own classification of each question (the syllabus was not supplied).

Basic concepts and laws of thermodynamics	**16**	**20**	**36**	**17.6 %**
Properties of steam	6	12	**18**	8.8 %
Perfect gases and gas processes	22	12	**34**	16.7 %
Steam power plant, condensers and cooling towers	10	18	**28**	13.7 %
Heat transfer and heat exchangers	16	16	**32**	15.7 %
Air standard cycles and I.C. engines	16	12	**28**	13.7 %
Steam boilers	10	8	**18**	8.8 %
Steam turbines	6	4	**10**	4.9 %
Total (marks offered)	**102**	**102**	**204**	100 %

Papers counted: Winter 2024, Summer 2025.

Top 10 repeated questions

Ranking rule: topics are ranked by the number of supplied sessions in which they appear (2 sessions supplied), then by total marks. 17 topic(s) appeared in more than one session. Topics that appeared only once are shown after them and are marked 'not repeated'.

#	Question / topic	Where asked	Marks	Sessions
1	Numerical on perfect gas (pV = mRT, work, ΔU)	W24: Q.2(d), Q.3(a); S25: Q.4(e)	12	2 (repeated)
2	Steady flow energy equation and applications	W24: Q.5(a); S25: Q.5(a)	12	2 (repeated)
3	Benson / La-Mont boiler	W24: Q.5(c); S25: Q.6(a)	12	2 (repeated)
4	Cooling tower – construction and working	W24: Q.3(c); S25: Q.5(c)	10	2 (repeated)
5	Numerical on composite wall (conduction)	W24: Q.4(b); S25: Q.6(b)	10	2 (repeated)
6	Two/four stroke engines, working of C.I. engine	W24: Q.4(c); S25: Q.6(c)	10	2 (repeated)
7	Numerical on wet / superheated steam	W24: Q.4(e); S25: Q.5(b)	10	2 (repeated)
8	Impulse vs reaction turbine / velocity compounding	W24: Q.6(a); S25: Q.3(c)	10	2 (repeated)
9	Heat exchangers – types and working	W24: Q.6(b); S25: Q.4(b)	10	2 (repeated)
10	Heat engine / heat pump / refrigerator, COP relation	W24: Q.2(a); S25: Q.2(a)	8	2 (repeated)

Last-minute revision sheet

Laws of thermodynamics

- Zeroth: A~C and B~C ⇒ A~B (basis of thermometer).
- Kelvin–Planck: no engine can convert all heat to work ($\eta < 100$ %).
- Clausius: heat cannot flow cold → hot without work.
- $\eta_{engine} = W/Q_1 = 1 - Q_2/Q_1$; $COP_R = Q_2/W$; $COP_{HP} = Q_1/W$; **$COP_{HP} = COP_R + 1$**.
- SFEE: $h_1 + C_1^2/2 + gZ_1 + Q = h_2 + C_2^2/2 + gZ_2 + W$. Turbine $W = h_1 - h_2$; boiler $Q = h_2 - h_1$.
- Flow work = pV; h = u + pv.
- Carnot $\eta = 1 - T_2/T_1$.

Perfect gas

- $pV = mRT$; $R = C_p - C_v$; $\gamma = C_p/C_v$; $C_v = R/(\gamma-1)$.
- $R_u = M\,R = 8.314$ kJ/kmol K.
- Dalton: $p = p_1 + p_2 + \ldots$ (partial pressures).
- Isobaric: $W = p(V_2-V_1)$, $Q = mC_p\Delta T$. Isochoric: $W = 0$, $Q = mC_v\Delta T$.
- Isothermal: $pV = C$, $W = p_1V_1 \ln(V_2/V_1)$, $\Delta U = 0$.
- Adiabatic: $pV^\gamma = C$, $W = (p_1V_1 - p_2V_2)/(\gamma-1)$, $Q = 0$.
- Always use T in kelvin (t + 273) and p in N/m² or kN/m² consistently.

Steam

- $x = m_g/(m_g+m_f)$; $h = h_f + x\,h_{fg}$; $v = x\,v_g$ (approx.).
- Superheated: $h = h_g + C_{ps}(t_{sup} - t_s)$; $v = v_g\,T_{sup}/T_s$.
- If $v > v_g$ at that pressure ⇒ superheated; if $v < v_g$ ⇒ wet.
- Steam generation: sensible heat h_f, latent heat h_{fg}, superheat.

Boilers

- Fire tube (hot gas in tubes: Cochran, Lancashire) vs water tube (water in tubes: Babcock & Wilcox, La-Mont, Benson).
- La-Mont: forced circulation, pump, up to 170 bar. Benson: once-through, no drum, ~225 bar (supercritical).
- Mountings: safety valve, pressure gauge, water level indicator, stop valve, fusible plug, blow-off cock.

Turbines, condensers, cooling towers

- Impulse: pressure drop only in nozzle, symmetrical blades. Reaction: drop in fixed + moving blades.
- Velocity compounding (Curtis): optimum $u = C_1\cos\alpha_1/(2z)$.
- Surface condenser: steam outside tubes, water inside, hot well, air pump (non-mixing). Jet condenser: mixing.
- Cooling tower: natural draught (hyperbolic, large plants), forced / induced draught (fan); make-up water replaces losses.
- Rankine: 1-2 pump, 2-3 boiler, 3-4 turbine, 4-1 condenser; $\eta = [(h_3-h_4)-(h_2-h_1)]/(h_3-h_2)$.

I.C. engines

- Otto: heat added at constant V, r = 6–10. Diesel: heat added at constant p, r = 14–22.
- Diesel $\eta = 1 - (1/r^{\gamma-1})[(\rho^\gamma-1)/(\gamma(\rho-1))]$.
- $r = (V_s+V_c)/V_c$; $V_s = (\pi/4)D^2L$; cut-off ratio $\rho = V_3/V_2$; m.e.p. $= W_{net}/V_s$.
- Four stroke: suction, compression, power, exhaust (2 revolutions). Two stroke: one power stroke per revolution, ports not valves.

Heat transfer

- Conduction (Fourier): $Q = -kA\,dT/dx$. Convection (Newton): $Q = hA(T_s-T_f)$. Radiation: $Q = \sigma AT^4$.
- Composite wall: $Q = \Delta T / \Sigma(L/kA)$. Interface: $T = T_1 - Q\,R_1$.
- k unit W/m K; h unit W/m² K.
- Heat exchangers: parallel, counter, cross flow; shell & tube, plate, double pipe; uses: condenser, radiator, economiser.

Production Drawing (313311)

Session	Status
Winter 2024	Included
Summer 2025	Included
Winter 2025	**Not received – not included**

Marks shown against sub-questions are the paper's group marks divided by the number of sub-questions to attempt (MSBTE equal split); the papers print only the group total.

Winter 2024

Subject	Production Drawing (313311)	Time / Marks	4 Hours / 70 Marks
Session	Winter 2024	Paper ID printed	12425
Scheme	K-scheme	Source file	Production_Drawing_Winter_2024_Question_Paper_First_K_Scheme_Paper.pdf

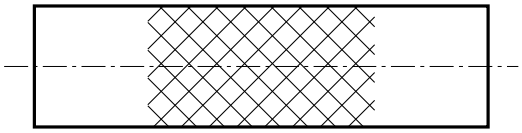
Q.1 Attempt any FIVE of the following : — 10 Marks (each sub-question = 2 marks)

Q.1 (a) [2 Marks]

Draw the conventional representation of the following :
(i) Diamond Knurling
(ii) Splined Shaft

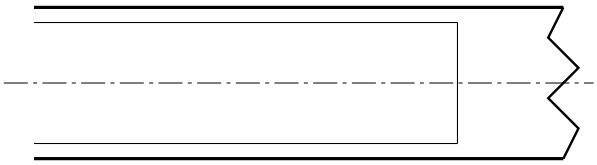
Ans.

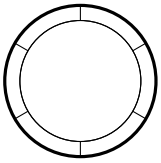
Diamond knurling (conventional)



Two sets of thin lines at 45° crossing each other, drawn over the knurled part

Splined shaft – front view





End view

- **Diamond knurling:** two sets of thin continuous lines at 45° to the axis, crossing each other to form diamonds, drawn over the knurled portion (it may be shown only over a small part of the length).
- **Splined shaft:** in the front view the major diameter is drawn with thick lines and the minor diameter with thin lines up to the end of the spline, the runout is shown by a thin sloping line; in the end view the shaft outline with teeth is drawn.

Marks split	Marks
Diamond knurling	1
Splined shaft	1
Total	**2**

Q.1 (b) [2 Marks]

Draw a sketch to represent taper 1 : 10 on a shaft of diameter 40 mm and length 110 mm.

Ans.

Numerical: Taper 1:10 on a shaft Ø40 × 110

Given data	Value	Unit	Remark
Large diameter D	40	mm	
Length L	110	mm	
Taper	1 : 10		taper = (D − d)/L

Formula and calculation:

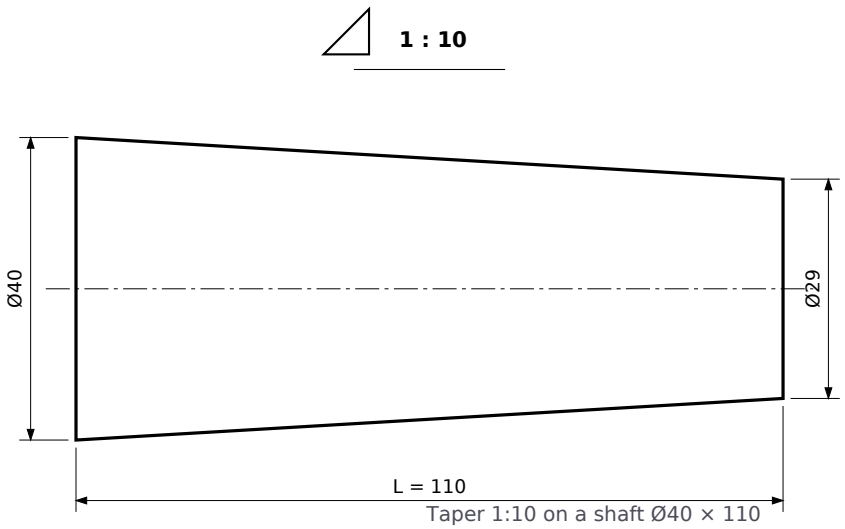
1. Small diameter

d = D − L / 10 = 40 − 110 / 10 = **29.00 mm**

Final answer
Small end diameter d = **29.00 mm**
Taper per unit length (D − d)/L = **1 : 10**

Checked in code by a second method: ✓ (D − d)/L = 1/k: 0.1000 ≈ 0.1000

Note: The taper is assumed to run over the whole length of 110 mm (it is not said otherwise).



Taper 1:10 on a shaft Ø40 × 110

Taper = (D − d)/L = 1/10

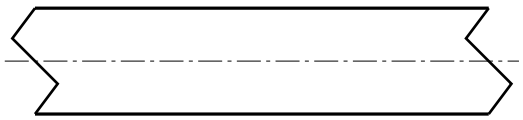
d = D − L/10 = 40 − 110/10 = 29 mm

Marks split	Marks
Calculation of small end diameter	1
Sketch with taper symbol and dimensions	1
Total	**2**

Q.1 (c) [2 Marks]

Draw the conventional representation of the following :
(i) Short break in pipe
(ii) Gate valve

Ans.

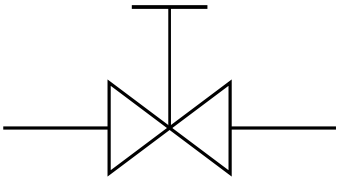


Short break in pipe

Pipe shown broken at both ends with S-shaped lines; long break uses thin zig-zag lines (free-hand).



Long break in pipe



Gate valve

Short break in pipe: the pipe is cut by two S-shaped (curved) thin lines so that the middle part need not be drawn. **Gate valve:** two triangles meeting at their apexes (the gate) between the pipe line, with a stem and handle on top.

Marks split	Marks
Short break in pipe	1
Gate valve	1
Total	**2**

Q.1 (d) **[2 Marks]**

Draw the symbol for representing the geometrical tolerance in case of the following :
(i) Circularity
(ii) Symmetry

Ans.


Circularity


Symmetry

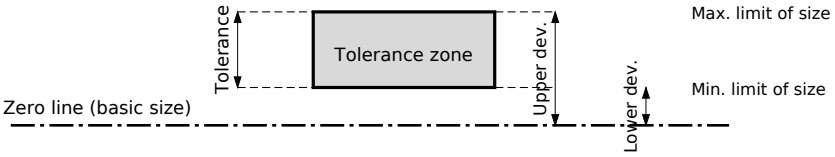
- **Circularity (roundness):** a circle ○.
- **Symmetry:** three horizontal parallel lines, the middle one longer ⌷.

Marks split	Marks
Circularity	1
Symmetry	1
Total	**2**

Q.1 (e) [2 Marks]

Draw a sketch showing basic size, lower deviation, upper deviation and tolerance.

Ans.



Basic size = nominal size, measured up to the zero line.

Upper deviation = max. limit − basic size; Lower deviation = min. limit − basic size.

Tolerance = max. limit − min. limit = upper deviation − lower deviation.

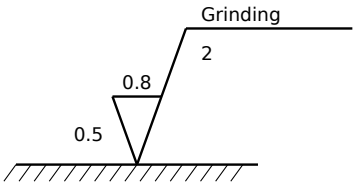
(Here both deviations are drawn above the zero line as an example.)

Marks split	Marks
Sketch showing zero line, basic size, upper and lower deviation, tolerance zone	2
Total	**2**

Q.1 (f) [2 Marks]

State the meaning of symbols shown in the Fig. No. 1.
[Fig. No. 1: machining symbol with 0.8 above the bar, 0.5 at left, “Grinding” above the extension line and 2 below it]

Ans.



0.8 = upper limit (maximum) of surface roughness Ra, in µm [on the bar]

0.5 = lower limit (minimum) of surface roughness Ra, in µm [left of the symbol]

Grinding = production (machining) method [above the extension line]

2 = sampling length / cut-off in mm [below the extension line]

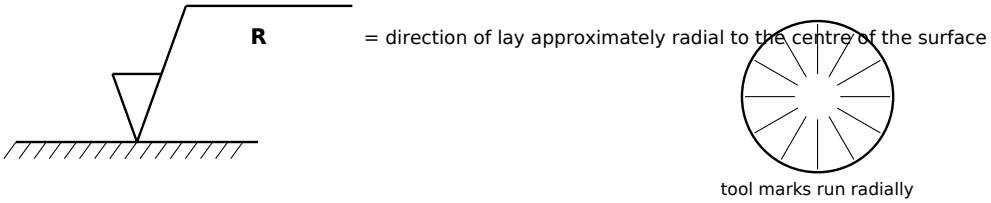
The symbol is the machining (surface texture) symbol: the surface is to be finished by **grinding**, the roughness value Ra must lie between **0.5 µm and 0.8 µm**, and the sampling length is **2 mm**.

Marks split	Marks
Roughness limits 0.8 and 0.5	1
Method “Grinding” and sampling length 2	1
Total	**2**

Q.1 (g) [2 Marks]

Show machining symbols to represent direction of lay for approximately radial relative to the center of the surface.

Ans.



The direction of lay is shown by the letter **R** placed at the right of the machining symbol: R = lay approximately radial relative to the centre of the surface.

Marks split	Marks
Machining symbol	1
Letter R for radial lay	1
Total	**2**

Q.2 Attempt any ONE of the following : — 8 Marks

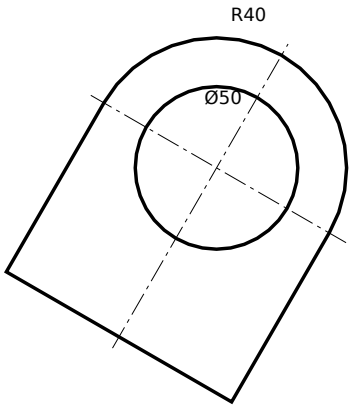
Q.2 (a) [8 Marks]

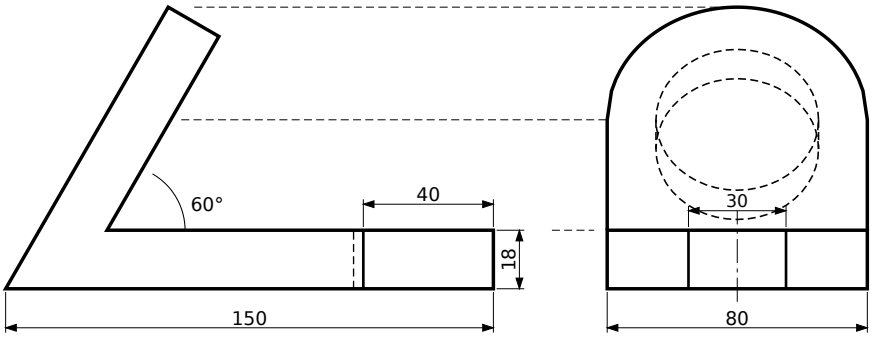
Fig. No. 2 shows front view, auxiliary view and incomplete side view of an object. Redraw the given views and complete the side view.
[Fig. No. 2: bent plate – base 150 × 18, arm 100 long at 60°, R40 end with Ø50 hole, plate width 80; incomplete S.V. 80 wide]

Ans.

Data note: The printed figure is small; a few details (the exact vertical lines in the base of the F.V., and the lines at 30 mm in the S.V.) are hard to read.
Assumption used: The drawing reproduces them as printed (a line at 40 mm from the right end in the F.V.; lines at ± 15 mm in the S.V.). Check against your paper.

Aux. view





F.V. **S.V. (completed)**

Front view, auxiliary view and completed side view (first angle)

Procedure

1. Draw the X–Y reference line. Draw the front view: base 150 × 18; arm of length 100 and thickness 18 inclined at 60° to the base at the left end; end of the arm cut square.
2. Draw the auxiliary view as given, projecting at right angles to the arm: plate width 80 (2 × R40), length 100 with R40 round end, Ø50 hole at 40 mm from the end (centre at 60 mm from the bend).
3. Draw the side view to the right of the front view. Base: rectangle 80 × 18 with the vertical lines at 30 mm apart as given.
4. Project the height of the arm from the F.V.: the arm rises to 100 sin 60° = 86.6 mm, the straight sides stop at 60 sin 60° = 52 mm, where the half-ellipse for the round end begins (semi-axes 40 and 34.6 mm).
5. Project the Ø50 hole: it appears as two dashed ellipses (semi-axes 25 and 25 sin 60° = 21.7 mm) joined by dashed lines at ± 25 mm.
6. Draw centre lines, show hidden lines dashed, and give the main dimensions.

Marks split	Marks
Redrawing front and auxiliary views	3
Projection of the arm in side view	3
Hole and hidden lines	1
Dimensions	1
Total	**8**

Q.2 (b) [8 Marks]

Fig. No. 3 shows incomplete front view, top view and auxiliary view of an object. Draw the given views and complete Front view.
[Fig. No. 3: base plate 15 thick with slot R12 and chamfered corners, upright block 50 high, auxiliary view with Ø40 hole]

Ans.

Data note: The figure is small and the angle of the inclined face, and the end radius of the slot are not clear: R12 does not match the slot width of 20 mm (15 + 20 + 15 = 50).
Assumption used: The inclined face is taken as 30° to the XY line and the slot width as 20 mm (end radius 10 mm). No finished view is drawn here because the shape cannot be established with certainty; the method and the calculated sizes are given.

Numerical: Completing the front view: size of the Ø40 hole in the inclined block

Given data	Value	Unit	Remark
Hole diameter	40	mm	
Angle of the inclined face with the XY line in plan	30	°	read from the auxiliary view of the figure

Formula and calculation:

1. Width of the hole's ellipse in the front view
$w = Ø \times \cos\theta = 40 \times \cos 30° =$ **34.64 mm**

2. Height of the ellipse in the front view
$h = Ø$ (true size, the vertical axis is parallel to the V.P.) $= 40 =$ **40.00 mm**

Final answer
Ellipse in front view: width × height = **34.64 × 40 mm**

Checked in code by a second method: ✓ $w^2 + (Ø \sin\theta)^2 = Ø^2$: 1600.0000 ≈ 1600.0000

Note: Slot check: the dimensions 15 + 20 + 15 = 50 mm give a slot width of 20 mm, so the end radius should be 10 mm, but the figure says R12.

Procedure (first angle method)

1. Draw the XY line. Draw the top view first, exactly as given: plate outline with the two 15 × 15 chamfers, slot with its end radii, and the inclined block at its angle.
2. Draw the given front view below it: base plate 15 high, block 50 high; project vertical lines from every corner of the top view down to the front view.
3. Draw the auxiliary view on the inclined reference line (parallel to the inclined face): the Ø40 hole appears as a true circle there.
4. Project the hole to the front view: its vertical axis stays 40 mm high; its horizontal width is 40 cos θ (see calculation). Draw the ellipse by the four-centre or trammel method, with the dash-dot axes.

5. Complete the front view: add the chamfer edges, the slot (hidden dashed lines below the plate top) and the vertical edges of the block from the projectors.
6. Show hidden lines by dashed lines, thick outlines, and dimension the main sizes.

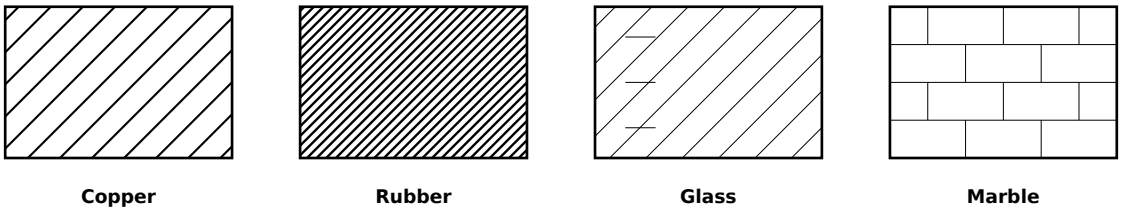
Marks split	Marks
Redraw given views	3
Projection to complete F.V.	3
Ellipse of the hole / hidden lines	1
Dimensions	1
Total	**8**

Q.3 Attempt any FIVE of the following : — 20 Marks (each sub-question = 4 marks)

Q.3 (a) [4 Marks]

Draw the material convention of the following :
(i) Copper
(ii) Rubber
(iii) Glass
(iv) Marble

Ans.



Material conventions (sectional view hatching)

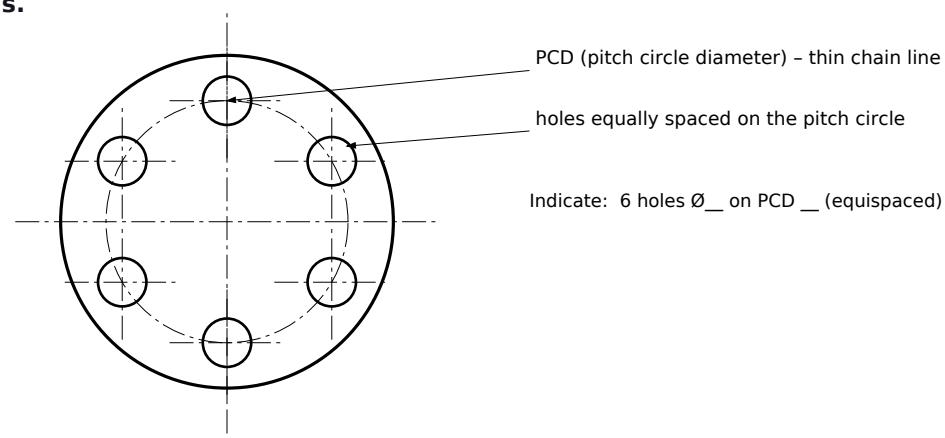
Copper (brass, bronze): inclined section lines at 45°. Rubber: close, heavy (almost solid) lines. Glass: thin lines at 45°, widely spaced with short thin dashes. Marble: thin horizontal and vertical lines forming a brick-like pattern.

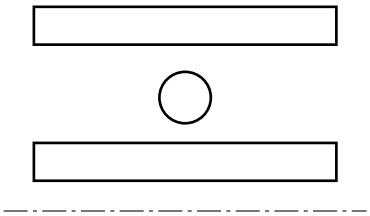
Marks split	Marks
Each convention, 1 mark	4
Total	**4**

Q.3 (b) [4 Marks]

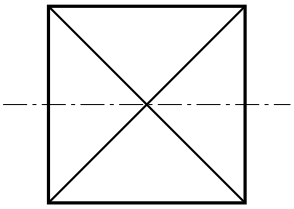
Draw the conventional representation of the following :
(i) Holes on circular pitch
(ii) Bearings

Ans.

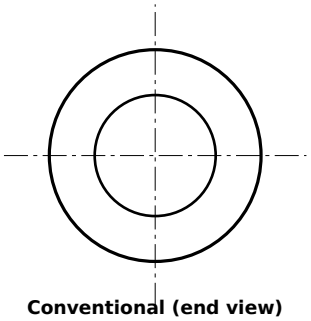




Ball bearing (detail)



Conventional (sectional view)



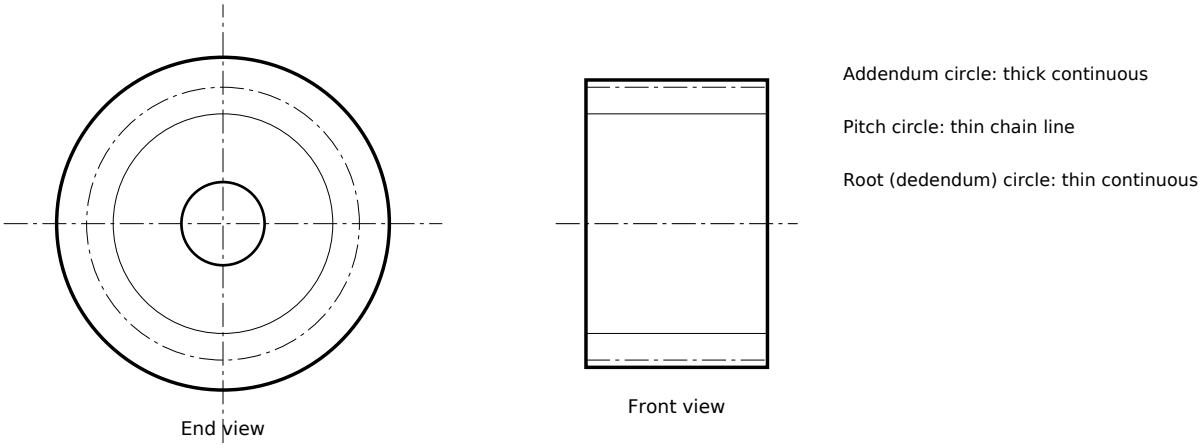
Conventional (end view)

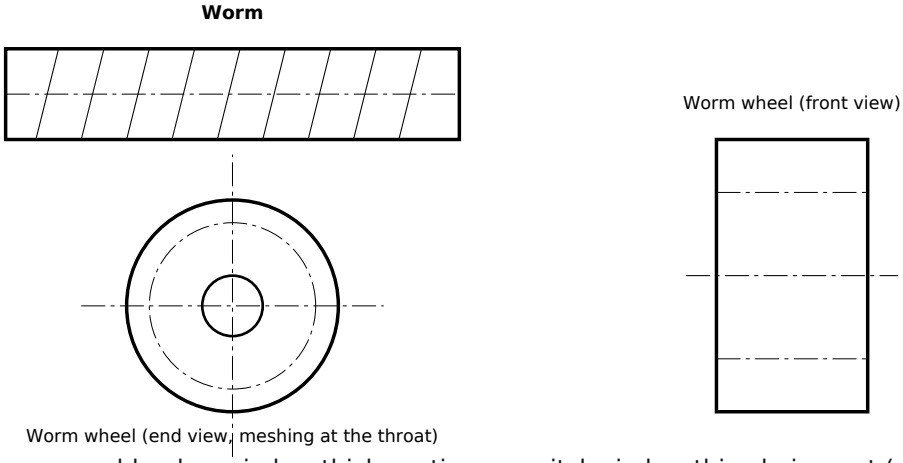
Marks split	Marks
Holes on circular pitch	2
Bearings	2
Total	**4**

Q.3 (c) [4 Marks]

Draw the conventional representation of the following :
(i) Spur gear
(ii) Worm and worm wheel

Ans.





- **Spur gear:** addendum circle – thick continuous; pitch circle – thin chain; root (dedendum) circle – thin continuous (not drawn in a sectional view). Teeth are not drawn.
- **Worm and worm wheel:** the worm is drawn as a shaft with pitch line (thin chain) and tooth lines; the wheel is drawn like a spur gear with addendum, pitch and root circles and a throat circle in the end view.

Marks split	Marks
Spur gear	2
Worm and worm wheel	2
Total	**4**

Q.3 (d) [4 Marks]

The shaft size is given as 40(−0.02/−0.04) and the hole size is 40(+0.02/−0.04). Determine the type of fit between them.

Ans.

Data note: The hole size is printed with a lower deviation of −0.04 (below the basic size). For a hole this is unusual (holes normally have a lower deviation of 0 or more).
Assumption used: The sizes are used exactly as printed, giving a transition fit.

Numerical: Fit between shaft 40(−0.02/−0.04) and hole 40(+0.02/−0.04)

Given data	Value	Unit	Remark
Basic size	40	mm	
Shaft deviations (es, ei)	−0.02, −0.04	mm	

Given data	Value	Unit	Remark
Hole deviations (ES, EI)	+0.02, −0.04	mm	as printed

Formula and calculation:

1. Fit: limits of hole

H_{max} = B + ES, H_{min} = B + EI = 40 + (0.02), 40 + (-0.04)

Hole: maximum = 40.020 mm, minimum = 39.960 mm

Shaft: maximum = 39.980 mm, minimum = 39.960 mm

2. Fit: maximum clearance

= $H_{max} - S_{min}$ = 40.020 − 39.960 = **0.060 mm**

3. Fit: minimum clearance

= $H_{min} - S_{max}$ = 39.960 − 39.980 = **-0.020 mm**

Fit: minimum clearance < 0 and maximum clearance > 0, so it is a TRANSITION fit.

Final answer
Maximum clearance = **0.060 mm**
Minimum clearance (negative = interference) = **-0.020 mm**
Type of fit = **Transition fit**

Checked in code by a second method: ✓ Fit: max clearance = ES − ei: 0.0600 ≈ 0.0600; ✓ Fit: min clearance = EI − es: -0.0200 ≈ -0.0200; ✓ Type is transition (clearance range straddles zero): 1.0000 ≈ 1.0000

Marks split	Marks
Limits of shaft and hole	1
Maximum and minimum clearance	2
Type of fit	1
Total	**4**

Q.3 (e) [4 Marks]

Refer Fig. No. 4 and state the meaning of symbol at X & Y.
[Fig. No. 4: X = ∥ 0.02 A, Y = ⊥ 0.03 A, datum A]

Ans.

 Parallelism

 Perpendicularity

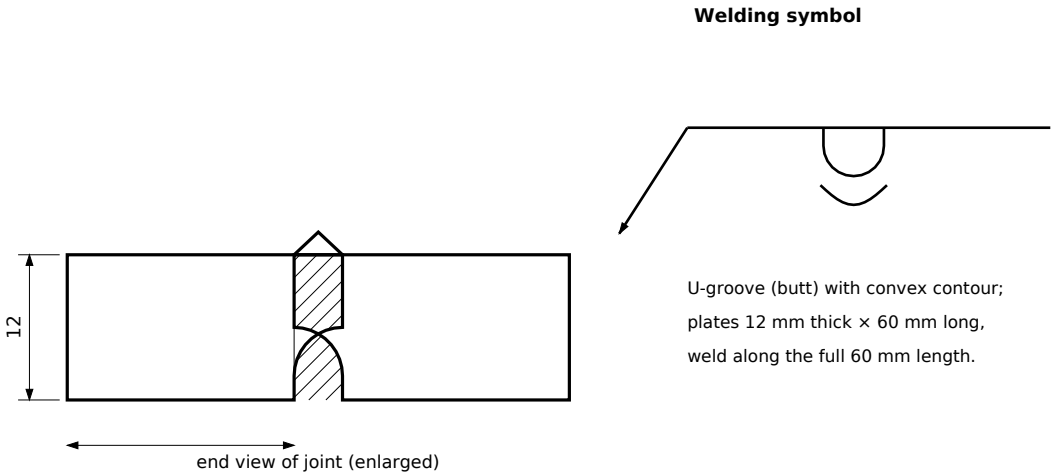
- **X (∥ 0.02 A):** parallelism – the surface must lie between two parallel planes 0.02 mm apart which are parallel to datum A.
- **Y (⊥ 0.03 A):** perpendicularity (squareness) – the surface must lie between two parallel planes 0.03 mm apart which are perpendicular to datum A.

Marks split	Marks
Meaning of X	2
Meaning of Y	2
Total	**4**

Q.3 (f) [4 Marks]

Two rectangular plates are to be welded with each other along the length. The thickness and length of both plates 12 mm and 60 mm respectively. The plates are to be U Butt welded with convex counter. Prepare welding drawing.

Ans.



U butt weld with convex contour

1. Draw the two plates, 12 mm thick and 60 mm long, with their edges prepared as a U groove (root face about 2 mm, root radius about 4 mm).
2. Show the weld metal (hatched) with a slightly convex reinforcement on the top.
3. Draw the arrow line touching the joint, the reference line, and the U-butt weld symbol below the line (arrow side).
4. Add the convex contour symbol (an arc) over the weld symbol, and the length 60 mm.

Marks split	Marks
Joint drawing with U preparation	2
Welding symbol with contour symbol	2
Total	**4**

Q.4 Fig. No. 5 shown the assembly of universal coupling. Attempt any TWO of the following : — 16 Marks (each sub-question = 8 marks)

Q.4 (a) [8 Marks]

Draw the sectional front view and top view of fork.

Ans.

Data note: Fig. No. 5 is a small scan; some sizes (the exact outline of the fork ears and the length of the collar) are hard to read.
Assumption used: The drawings use the sizes legible in the figure (Ø88, 86, 44, 28, Ø30, 106, Ø50, Ø6, 30) and are approximate in proportion. Take exact sizes from your paper's Fig. 5 in the exam.

Note: Fig. No. 5 (assembly of universal coupling). Part list printed in the figure and the proportions read from it are given once here; they apply to (a), (b) and (c).

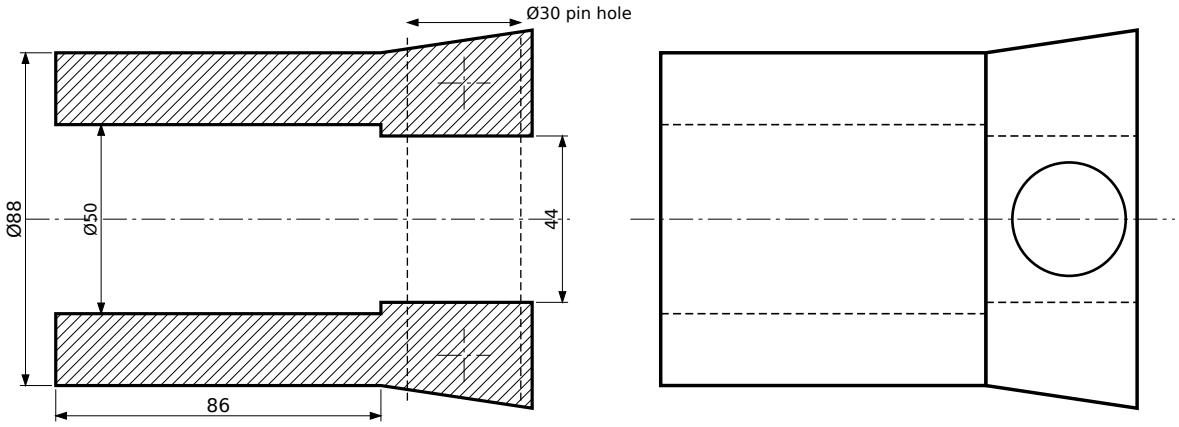
Part No.	Part name	Material	Qty.
1	Fork	C.I.	2
2	Centre block	C.I.	1
3	Pin	M.S.	2
4	Taper pin Ø6	M.S.	2
5	Collar	M.S.	2
6	Key	M.S.	2

Part No.	Part name	Material	Qty.
7	Shaft	M.S.	2

Feature (from Fig. 5)	Size	Ratio to shaft d = 50
Boss (hub) diameter of fork	Ø88	1.75 d
Length of boss	86	1.7 d
Gap between fork ears (centre block thickness)	44	0.9 d
Thickness of each ear	28	0.55 d
Pin diameter	Ø30	0.6 d
Pin length	106	2.1 d
Collar length	30	0.6 d
Taper pin	Ø6	0.12 d

Proportions are read from the figure and given as a ratio to the shaft diameter so that they can be checked quickly.

FORK – sectional front view **TOP VIEW**



Fork (C.I.): sectional front view and top view (approximate proportions)

Procedure

1. Draw the centre line. Draw the boss: rectangle 86 long × Ø88; the bore Ø50 through it.
2. At the right end draw the two ears: thickness 28 each with a gap of 44 between them; the ears taper to the end; hole Ø30 for the pin at the centre of the ears.
3. Hatch the cut surface at 45° (C.I.), leaving the bore and the gap clean.
4. Draw the top view by projection: rectangle 86 × 88 for the boss, the fork ends and the Ø30 hole as a circle; hidden bore and gap lines dashed.
5. Show the Ø6 taper-pin hole in the boss if required, centre lines and dimensions.

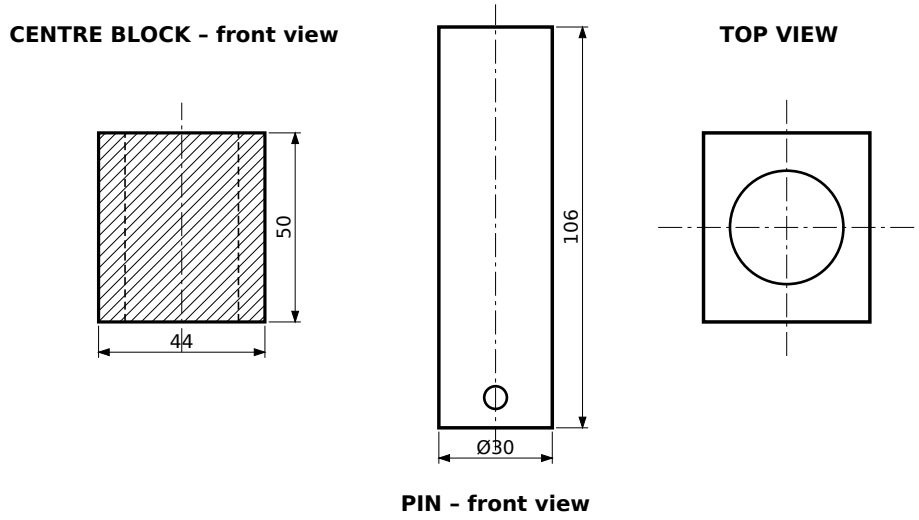
Marks split	Marks
Sectional front view	4
Top view	3
Dimensions, hatching, centre lines	1
Total	**8**

Q.4 (b) [8 Marks]

Draw the front view and top view of center block and pin.

Ans.

Data note: Fig. No. 5 is a small scan; some sizes (the exact outline of the fork ears and the length of the collar) are hard to read.
Assumption used: The drawings use the sizes legible in the figure (Ø88, 86, 44, 28, Ø30, 106, Ø50, Ø6, 30) and are approximate in proportion. Take exact sizes from your paper's Fig. 5 in the exam.



Centre block (C.I.) and pin (M.S.)

Procedure

1. Centre block: draw a block 44 thick and 50 × 50 (the width equals the pin length region); two holes Ø30 at right angles to each other at the centre.
2. Draw the front view in section (hatched) with the Ø30 hole as dashed lines, and the top view with the hole as a circle.
3. Pin: draw a cylinder Ø30 × 106; near one end a Ø6 hole for the taper pin; chamfer the ends 2 × 45°.
4. Draw the top view of the pin (a circle Ø30 with centre lines) and give dimensions.

Marks split	Marks
Centre block – both views	4
Pin – both views	3
Dimensions	1
Total	**8**

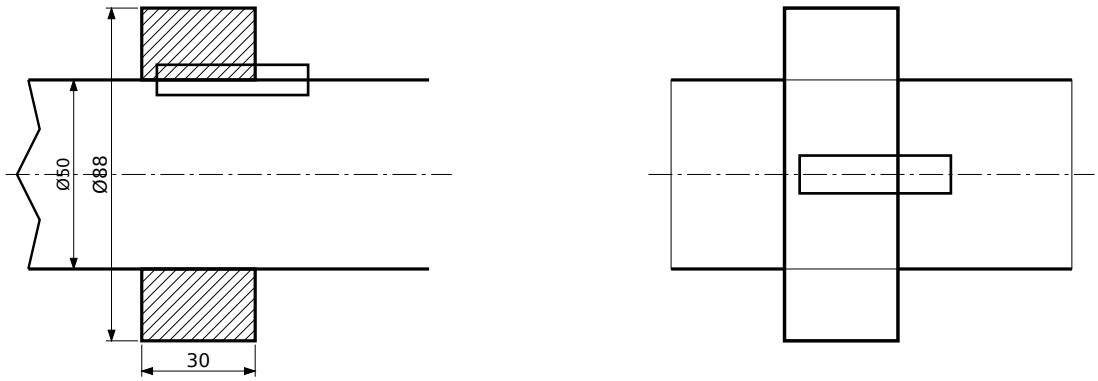
Q.4 (c) [8 Marks]

Draw the front view and top view of collar, shaft and key.

Ans.

Data note: Fig. No. 5 is a small scan; some sizes (the exact outline of the fork ears and the length of the collar) are hard to read.
Assumption used: The drawings use the sizes legible in the figure (Ø88, 86, 44, 28, Ø30, 106, Ø50, Ø6, 30) and are approximate in proportion. Take exact sizes from your paper's Fig. 5 in the exam.

COLLAR, SHAFT and KEY – front view (collar sectioned) **TOP VIEW**



Collar, shaft and key

Procedure

1. Shaft: a cylinder Ø50 with a broken end (free-hand break line). Draw the keyway for the key on the top of the shaft.
2. Collar: Ø88 × 30 with a Ø50 bore and a keyway; draw it in section (hatched at 45°).
3. Key: a rectangular key as in the part list (M.S.) placed in the keyway; draw it unsectioned in the front view when the cutting plane passes along its length.
4. Draw the top view by projection: rectangle for the collar, shaft lines and the key outline; show hidden bore with dashed lines.

Marks split	Marks
Collar and shaft – front view	3
Key	2
Top view	2
Dimensions	1
Total	**8**

Q.5 Attempt any ONE of the following : — 16 Marks

Q.5 (a) [16 Marks]

Fig. No. 6 shows the details of Oldham’s coupling. Draw the following views of the assembly :
(i) Sectional Front View
(ii) Left hand Side View
[Parts: 1 Flange C.I. 2 off; 2 Central disc C.I. 1 off; 3 Shaft M.S. 2 off; 4 Taper key M.S. 2 off]

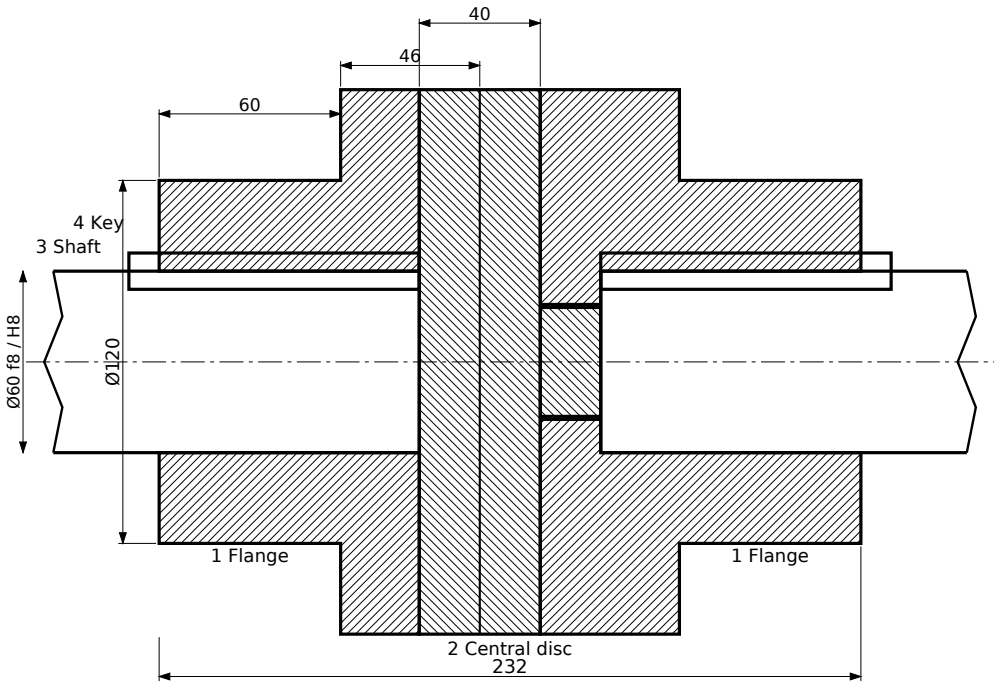
Ans.

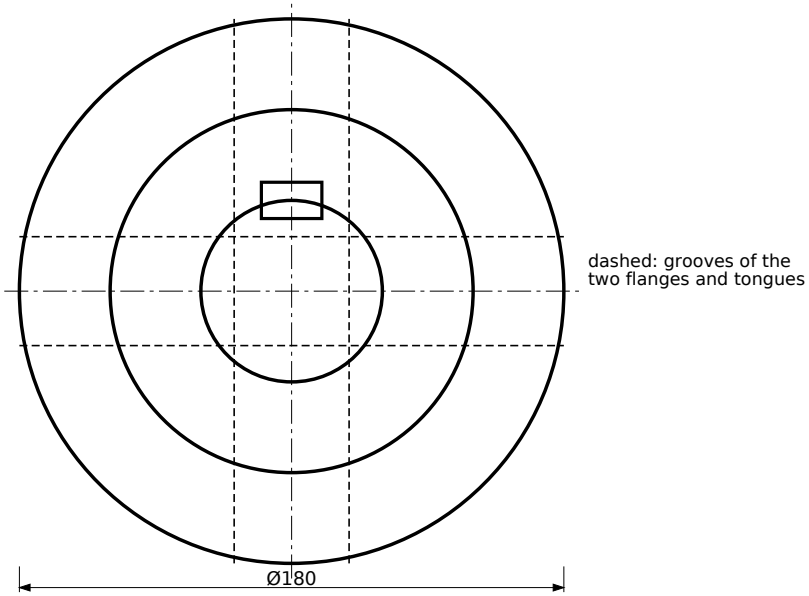
Data note: In Fig. 6, the hub diameter is printed as “φ12D” and the flange keyway width as “28H7”. The key is 20 mm wide (20n6) and the shaft keyway is 20H7.
Assumption used: “φ12D” is read as Ø120 (2 × shaft diameter), and the flange keyway width is taken as 20 mm to suit the 20 mm key. The key length 96 is more than the 86 mm bore, so the key is shown projecting 10 mm beyond the hub.

Part No.	Part name	Material	Qty.
1	Flange	C.I.	2
2	Central disc	C.I.	1
3	Shaft	M.S.	2
4	Taper key (20n6 × 12 × 96, taper 1 in 100)	M.S.	2

Feature (Fig. 6)	Size	Ratio to shaft d = 60
Flange head diameter	Ø180	3 d
Hub diameter	Ø120	2 d
Bore	Ø60 H8	d
Hub length	60	d
Head thickness (incl. 20 deep groove)	46	0.77 d
Groove width	38 H7	0.63 d
Disc diameter	Ø180	3 d
Disc plate thickness	20	d/3
Tongue width × height	36 g6 × 20	0.6 d × d/3
Key width	20	d/3
Keyway length in shaft	108	1.8 d

SECTIONAL FRONT VIEW (assembly)





LEFT HAND SIDE VIEW

Oldham's coupling – sectional front view and left hand side view

Procedure

1. Draw the centre line of the shafts. Draw the left flange: hub Ø120 × 60, head Ø180 × 46, bore Ø60; the 38 mm groove 20 deep on the head's face (vertical).
2. Draw the central disc Ø180 × 20 in the middle; its left tongue (36 × 20) enters the vertical groove of the left flange, its right tongue (turned through 90°) enters the horizontal groove of the right flange.
3. Draw the right flange as a mirror image, with its groove horizontal (so in section it is cut along its length and only 38 mm of material is removed at the centre).
4. Insert the shafts Ø60 up to the groove floors and the two taper keys 20 × 12 in the keyways (keys are shown unsectioned).
5. Hatch the flanges at 45° and the disc at 45° in the opposite direction; show break lines on the shafts.
6. Draw the left hand side view: circles Ø180, Ø120, Ø60, key outline, with dashed lines for the grooves. Add the overall dimensions and fits.

Marks split	Marks
Sectional front view	8
Left hand side view	5
Hatching and dimensions	2
Part names / fits	1
Total	**16**

Q.5 (b) [16 Marks]

Fig. No. 7 shows details of lathe tool post. Draw the following views of assembly :
(i) Sectional FV
(ii) Top view
(iii) Prepare bill of material and indicate type of fit.
[Parts: 1 Post C.I.; 2 Block C.I.; 3 Ring C.I.; 4 Wedge M.S.; 5 Screw C.I.; each 1 off]

Ans.

Data note: In the tolerance chart the entry for 24H7 is printed as “+0.021 / +40.000” and the wedge size “24n8” in the chart but “24n6” on the drawing. The post carries “φ60H7” and the block hole “φ60e8”, which is the reverse of normal practice (capital letters are for holes).
Assumption used: 24H7 is taken as +0.021 / 0.000 and 24n6 (+0.036 / +0.015). The Ø60 hole (block) is taken as H7 and the Ø60 post as e8, so that the fit is a clearance fit; if taken literally as printed the hole would be smaller than the shaft.

Part No.	Part name	Material	Qty.	Remark
1	Post	C.I.	1	Ø72 × 12 flange, Ø60 body, 144 high
2	Block	C.I.	1	90 square × 18, Ø74 recess, Ø60 hole
3	Ring	C.I.	1	Ø120 × 18, Ø60 bore, R144 seat
4	Wedge	M.S.	1	24 wide, 144 long, R144
5	Screw	C.I.	1	M18 × 72, square head 18, collar Ø36

Bill of material

Numerical: Lathe tool post geometry (R = 144)

Given data	Value	Unit	Remark
Wedge / ring seat radius R	144	mm	
Ring: outer radius, bore radius, thickness at outer edge	60, 30, 18	mm	

Formula and calculation:

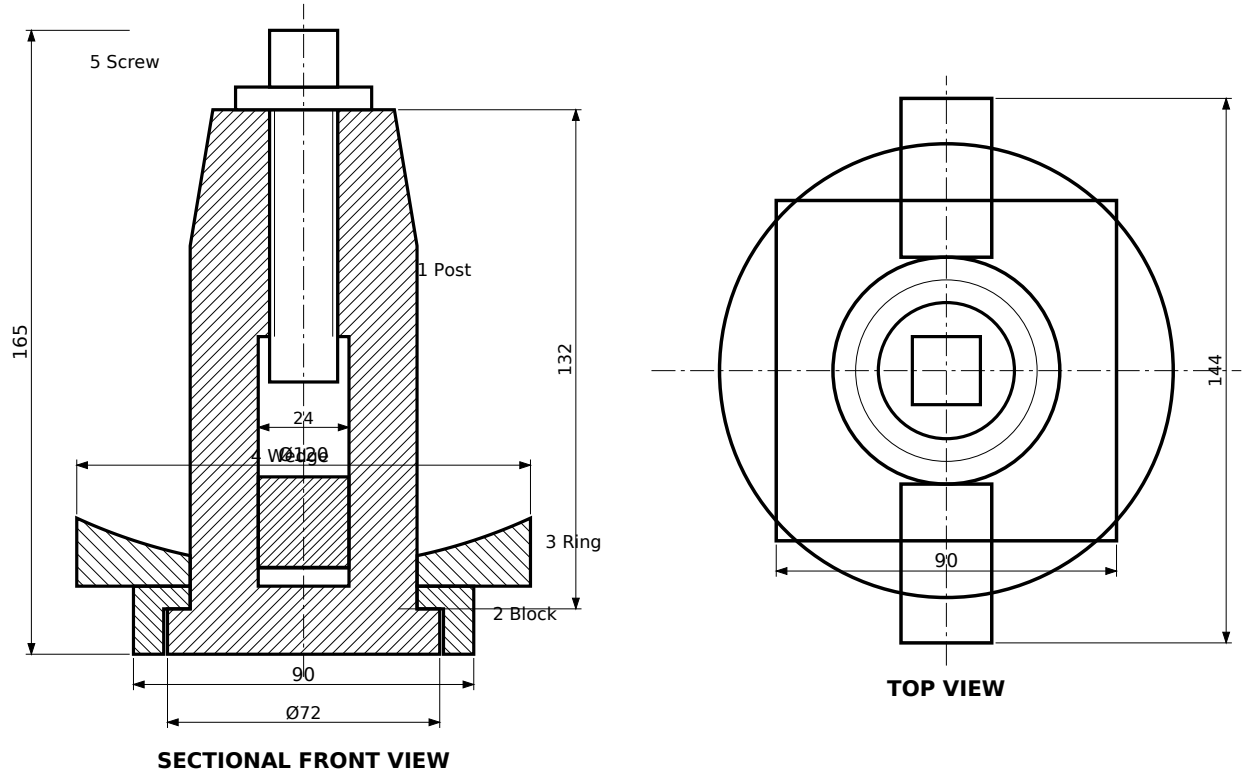
1. Fall of the concave seat from the ring edge to its centre
$s = R - \sqrt{(R^2 - 60^2)} = 144 - \sqrt{(144^2 - 60^2)} =$ **13.10 mm**

2. Seat height at the axis
$z_0 = 36 - s = 36 - 13.10 =$ **22.90 mm above the bottom of the block**

3. Wedge: depth of the curve at its ends (chord = R, as the wedge length equals R)
$s_w = R - \sqrt{(R^2 - (R/2)^2)} = 144 - \sqrt{(144^2 - 72^2)} =$ **19.29 mm**

Final answer
Seat height at the axis z_0 = **22.90 mm**
Depth of the wedge curve over its length = **19.29 mm**

Checked in code by a second method: ✓ Seat height + sag = ring edge height (36 mm): 36.0000 ≈ 36.0000



Lathe tool post – sectional front view and top view (assembly)

Procedure

1. Draw the centre line. Draw the block: 90 square × 18, with the Ø74 recess 12 deep from below and the Ø60 hole in the 6 mm top plate.
2. Draw the post: flange Ø72 × 12 in the recess, body Ø60 rising to 144, tapered to Ø48 over the top 36 mm; the 24-wide slot 66 long starting 60 below the top; the M18 hole at the top.
3. Place the ring Ø120 × 18 round the post on the block; its top is the concave R144 seat (calculated above).
4. Place the wedge (24 × 24 section) in the slot of the post resting on the ring seat.
5. Insert the screw M18 from the top: square head 18, collar Ø36 × 6, shank 72 long; show screw unsectioned.
6. Hatch the cut parts at 45° in different directions for adjacent parts; draw the top view by projection (wedge 24 × 144 across the post, ring Ø120, block 90 square, post Ø60, screw head).
7. Give the overall dimensions, part numbers and the bill of material.

Types of fit

Numerical: Fits in the lathe tool post (W24 tolerance chart)

Given data	Value	Unit	Remark
Tolerance chart (printed)	Ø60H7: +0.030/0; Ø60e8: −0.060/−0.106; 24H7: +0.021/0; 24n6: +0.036/+0.015	mm	

Formula and calculation:

1. Ø60 H7/e8 (block hole – post): limits of hole
H_{max} = B + ES, H_{min} = B + EI = 60 + (0.03), 60 + (0)
Hole: maximum = 60.030 mm, minimum = 60.000 mm

Shaft: maximum = 59.940 mm, minimum = 59.894 mm

2. Ø60 H7/e8 (block hole – post): maximum clearance
$= H_{max} - S_{min} = 60.030 - 59.894 =$ **0.136 mm**

3. Ø60 H7/e8 (block hole – post): minimum clearance
$= H_{min} - S_{max} = 60.000 - 59.940 =$ **0.060 mm**
Ø60 H7/e8 (block hole – post): minimum clearance ≥ 0 and maximum clearance > 0, so it is a CLEARANCE fit.

4. 24 H7/n6 (post slot – wedge): limits of hole
$H_{max} = B + ES$, $H_{min} = B + EI = 24 + (0.021), 24 + (0)$
Hole: maximum = 24.021 mm, minimum = 24.000 mm
Shaft: maximum = 24.036 mm, minimum = 24.015 mm

5. 24 H7/n6 (post slot – wedge): maximum clearance
$= H_{max} - S_{min} = 24.021 - 24.015 =$ **0.006 mm**

6. 24 H7/n6 (post slot – wedge): minimum clearance
$= H_{min} - S_{max} = 24.000 - 24.036 =$ **-0.036 mm**
24 H7/n6 (post slot – wedge): minimum clearance < 0 and maximum clearance > 0, so it is a TRANSITION fit.

Final answer
Ø60 H7/e8 – min / max clearance = **0.060 / 0.136 mm**
Ø60 H7/e8 = **CLEARANCE fit**
24 H7/n6 – max clearance / max interference = **0.006 / 0.036 mm**
24 H7/n6 = **TRANSITION fit**

Checked in code by a second method: ✓ Ø60 H7/e8 (block hole – post): max clearance = ES − ei: 0.1360 ≈ 0.1360; ✓ Ø60 H7/e8 (block hole – post): min clearance = EI − es: 0.0600 ≈ 0.0600; ✓ 24 H7/n6 (post slot – wedge): max clearance = ES − ei: 0.0060 ≈ 0.0060; ✓ 24 H7/n6 (post slot – wedge): min clearance = EI − es: -0.0360 ≈ -0.0360

Marks split	Marks
Sectional front view	6
Top view	4
Bill of material	2
Types of fit	2
Dimensions, hatching, finish	2
Total	**16**

Summer 2025

Subject	Production Drawing (313311)	Time / Marks	4 Hours / 70 Marks
Session	Summer 2025	Paper ID printed	24225
Scheme	K-scheme	Source file	Production_Drawing_Summer_2025_Question_Paper_Second_K_Scheme_Paper.pdf

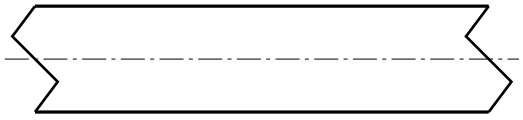
Q.1 Attempt any FIVE of the following: — 10 Marks (each sub-question = 2 marks)

Q.1 (a) [2 Marks]

Draw the conventional representation of the following:
i) Long Break in pipe
ii) I-section pipe.

Ans.

Data note: "I-section pipe" is not a standard item; an I-section is a rolled steel section, not a pipe.
Assumption used: It is answered as the conventional representation of a break in an I-section (the usual BIS SP-46 item "break in rolled sections").

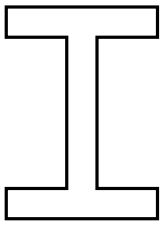


Short break in pipe

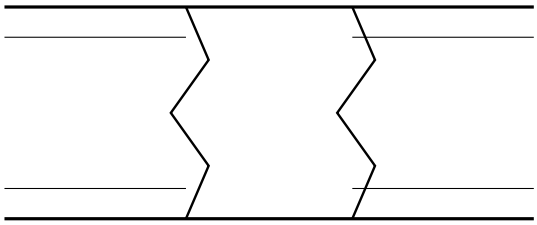
Pipe shown broken at both ends with S-shaped lines; long break uses thin zig-zag lines (free-hand).



Long break in pipe



I-section (end view)



Break shown with wavy free-hand lines

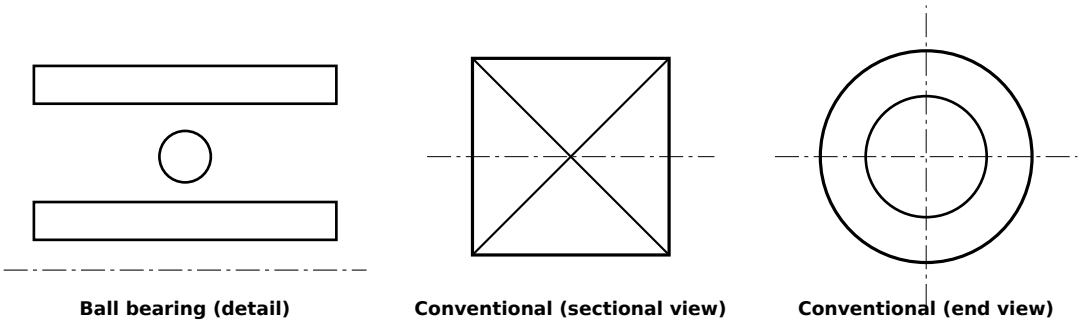
- **Long break in pipe:** the middle portion is omitted and the cut ends are shown by thin zig-zag lines (free-hand), with the centre line continuing across.
- **I-section:** a rolled I-section is shown with its outline in the end view; when a long part is broken, wavy free-hand break lines are used across the section.

Marks split	Marks
Long break in pipe	1
I-section	1
Total	**2**

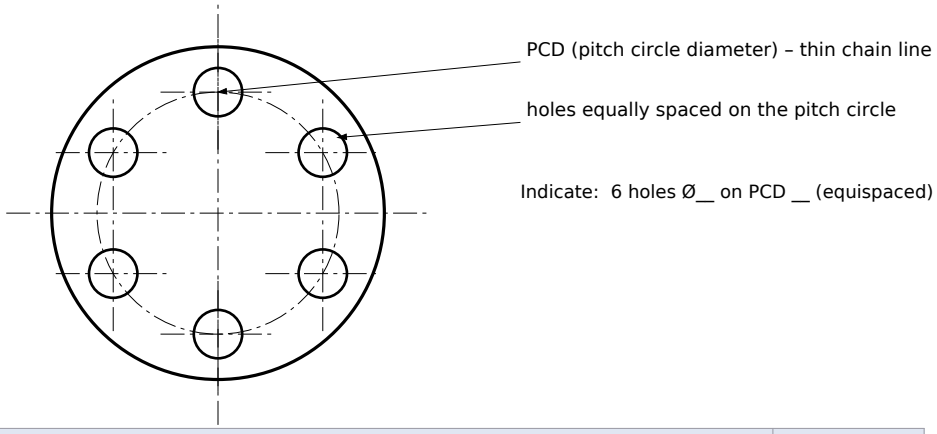
Q.1 (b) [2 Marks]

Draw the conventional representations of following:
i) Bearings
ii) Holes on circular pitch.

Ans.



Ball bearing (detail) **Conventional (sectional view)** **Conventional (end view)**

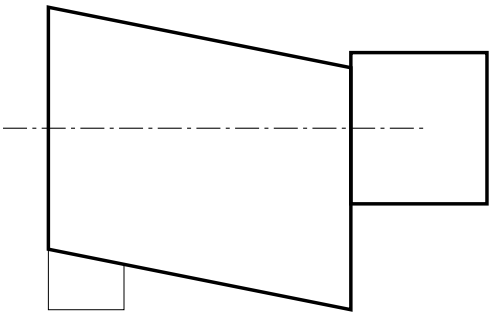


Marks split	Marks
Bearings	1
Holes on circular pitch	1
Total	**2**

Q.1 (c) [2 Marks]

Draw the conventional representations of following machine components:
i) Bevel Gear
ii) Spur Gear

Ans.

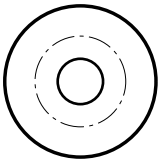


Sectional front view

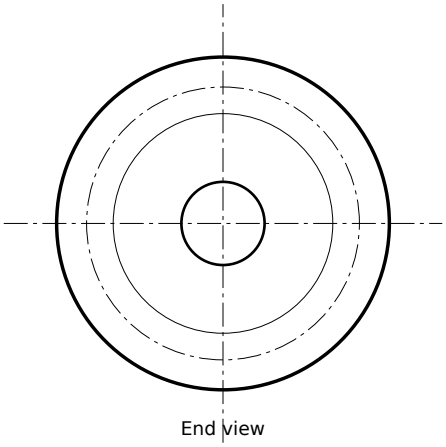
Outer (back) cone and face lines thick;

pitch cone line thin chain;

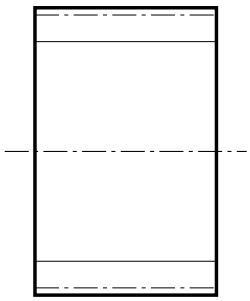
teeth not drawn in section.



End view



End view



Front view

Addendum circle: thick continuous

Pitch circle: thin chain line

Root (dedendum) circle: thin continuous

In both gears the teeth are not drawn: addendum (outer) circle/lines thick, pitch circle/lines thin chain, root circle thin continuous (omitted in sections).

Marks split	Marks
Bevel gear	1
Spur gear	1
Total	**2**

Q.1 (d) [2 Marks]

Define “Taper”. Draw its standard convention.

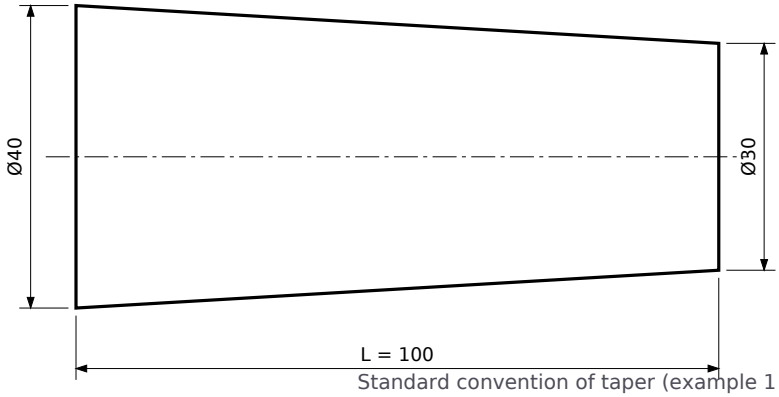
Ans.

Taper is the ratio of the difference between the diameters of the two ends of a cone (frustum) to its length:

$$\text{Taper} = (D - d) / L = 1 : k$$

It is shown by the taper symbol (a right-angled triangle whose sloping side points the same way as the taper) followed by the ratio, drawn on a leader or on a reference line parallel to the axis, as in the figure for a 1 : 10 taper (example).





Standard convention of taper (example 1 : 10)

Taper = (D − d)/L = 1/10

d = D − L/10 = 40 − 100/10 = 30 mm

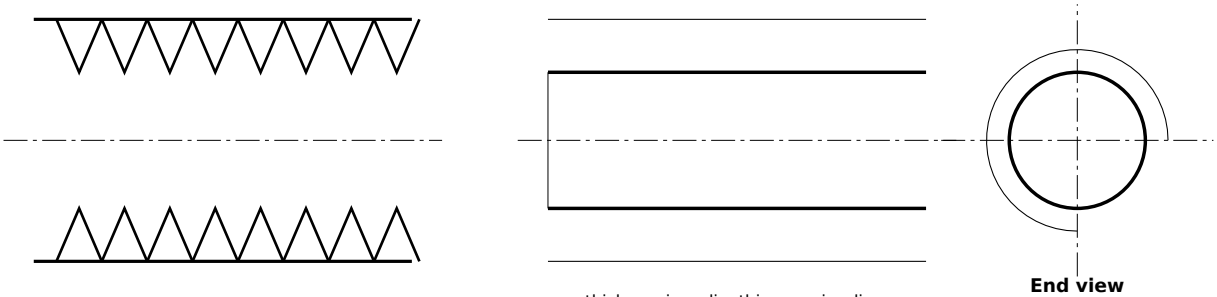
Marks split	Marks
Definition with formula	1
Convention (symbol and ratio)	1
Total	**2**

Q.1 (e) [2 Marks]

Draw the actual view and conventional representation of:
i) Internal Screw thread
ii) Semi-elliptic leaf spring.

Ans.

Actual view **Conventional (sectional)**



thick = minor dia, thin = major dia

End view



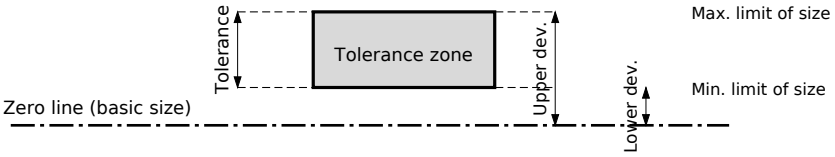
Actual view (leaves + clip) **Conventional (single bow line + cross)**

Marks split	Marks
Internal screw thread – actual and conventional	1
Leaf spring – actual and conventional	1
Total	**2**

Q.1 (f) [2 Marks]

Draw a sketch showing basic size, lower deviation, upper deviation and tolerance.

Ans.



Basic size = nominal size, measured up to the zero line.

Upper deviation = max. limit − basic size; Lower deviation = min. limit − basic size.

Tolerance = max. limit − min. limit = upper deviation − lower deviation.

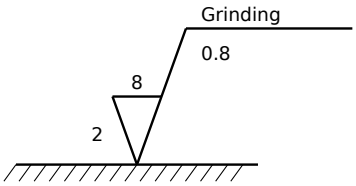
(Here both deviations are drawn above the zero line as an example.)

Marks split	Marks
Sketch with zero line, basic size and zone	2
Total	**2**

Q.1 (g) [2 Marks]

State the meaning of the symbol shown in Figure No. 1.
[Fig. No. 1: machining symbol with 8 above the bar, 2 at left, “Grinding” above the extension line and 0.8 below it]

Ans.



8 = upper limit (maximum) of surface roughness Ra, in µm [on the bar]

2 = lower limit (minimum) of surface roughness Ra, in µm [left of the symbol]

Grinding = production (machining) method [above the extension line]

0.8 = sampling length / cut-off in mm [below the extension line]

The surface is to be finished by **grinding**; the roughness Ra must lie between **2 µm and 8 µm**; the sampling length is **0.8 mm**.

Marks split	Marks
Roughness limits 8 and 2	1
Grinding and sampling length 0.8	1
Total	**2**

Q.2 Attempt any ONE of the following: — 8 Marks

Q.2 (a) [8 Marks]

Figure No. 2 shows front view, incomplete top view and partial auxiliary view of an object. Redraw the given views and complete the top view (use first angle method of projection.)
[Fig. No. 2: bent plate – base 75 × 6, arm 60 long at 120° with the base, square hole 20 in the rounded end of the arm, top view with rounded end R10 and slot 20 wide]

Ans.

Data note: The printed figure is small. The exact shape of the arm end (two 10 mm layers and the square hole turned 45°) cannot be read with certainty.
Assumption used: Only the lengths that can be calculated are given (above); no finished view is drawn here, because it would need guessing the shape of the object. Check them against your figure.

Numerical: Completing the top view of the bent plate (inclined arm at 60°)

Given data	Value	Unit	Remark
Length of inclined arm (true length, seen in the F.V.)	60	mm	
Inclination of the arm with the horizontal	60	°	bend angle 120° between the base and the arm
Plate thickness	6	mm	

Formula and calculation:

1. Length of the arm in the top view
$l = L \cos\theta = 60 \times \cos 60° =$ **30.00 mm**

2. Height of the arm end in the front view
$h = L \sin\theta = 60 \times \sin 60° =$ **51.96 mm**

3. A 20 mm square hole on the arm: length in the top view
$a' = 20 \cos\theta = 20 \times \cos 60° =$ **10.00 mm**

Final answer
Foreshortened length of arm in top view = **30.00 mm**
Height of arm end in F.V. = **51.96 mm**
20 mm square hole appears as 20 × a' = **10.00 mm**

Checked in code by a second method: ✓ $l^2 + h^2 = L^2$: 3600.0000 ≈ 3600.0000

Procedure (first angle method)

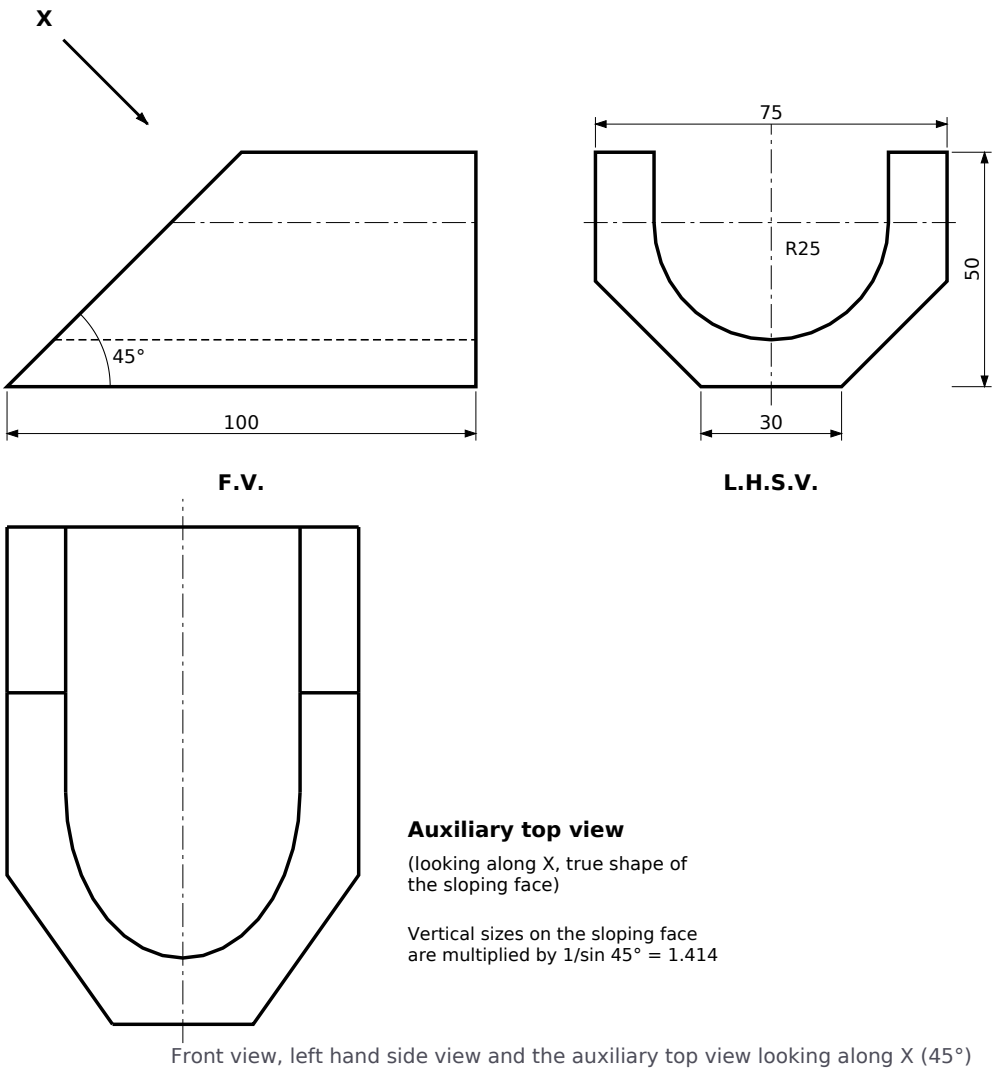
1. Draw the XY line. Draw the front view as given: base 75 long × 6 thick, with the arm 60 long rising at 120° to the base (60° to the horizontal), two 10 mm layers at the arm end as shown.
2. Draw the top view below the front view: project all vertical lines from the base. The base shows its outline with the rounded end and the 20 mm slot (R10 ends) as given.
3. The inclined arm is foreshortened in the top view: its true length 60 becomes 60 cos 60° = 30 mm. Project the arm ends from the F.V. to get its length and keep its width 40 as in the auxiliary view.
4. Project the square hole of the aux. view: its side across the arm stays 20 mm, the side along the arm becomes 20 cos 60° = 10 mm in the top view.
5. Complete the top view: draw hidden edges (dashed), centre lines, and the thick outline; give the main dimensions.

Marks split	Marks
Redraw front and auxiliary views	3
Projection of base and arm in top view	3
Hole, hidden lines and dimensions	2
Total	**8**

Q.2 (b) [8 Marks]

Figure No. 3 shows front view and left hand side view of an object. Draw the given views and project an auxiliary top view looking in the direction of X.

Ans.



Front view, left hand side view and the auxiliary top view looking along X (45°)

Procedure

1. Draw the front view: base 100, left face sloping at 45°, height 50. Draw the left hand side view: channel 75 wide and 50 high, bottom 30 wide with 45° chamfers, groove R25 with centre 15 below the top.
2. Draw the auxiliary plane (reference line) at 90° to the direction of viewing X, that is parallel to the sloping face of the front view.
3. Project every point of the sloping face and of the top face from the F.V. parallel to the direction X (at 45° to the base) up to the new reference line.
4. Measure the distances from the L.H.S.V. (the widths: ± 37.5, ± 25, ± 15) and mark them on the projectors, using the centre line as datum.
5. The groove R25 appears on the sloping face as an ellipse (semi-axes 25 and 25/sin 45° = 35.4 mm). Join the points with a smooth curve.
6. Draw the top face as two strips and the groove edges (± 25), complete the outline, add centre lines and dimensions.

Marks split	Marks
Redraw the given views	3
Auxiliary reference line and projectors	2
Elliptical groove and outline in auxiliary view	2
Dimensions / line work	1
Total	**8**

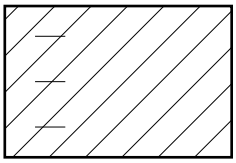
Q.3 Attempt any FIVE of the following: — 20 Marks (each sub-question = 4 marks)

Q.3 (a) [4 Marks]

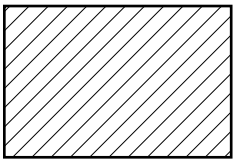
Draw conventional representation of any four of the following:
i) Glass
ii) Pipe or tubing
iii) Cast iron
iv) Splined shaft
v) Globe valve
vi) Counter sunk

Ans.

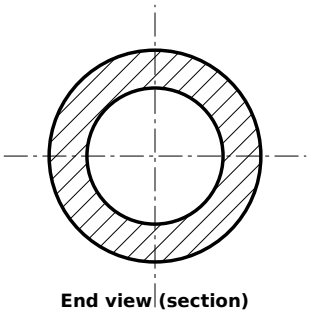
All six are shown; any four are to be attempted (1 mark each).



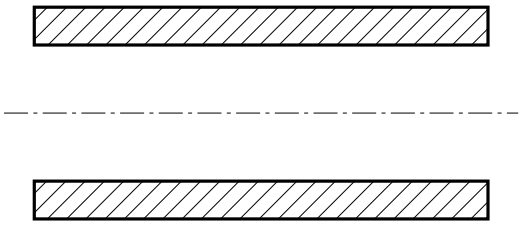
Glass



Cast iron

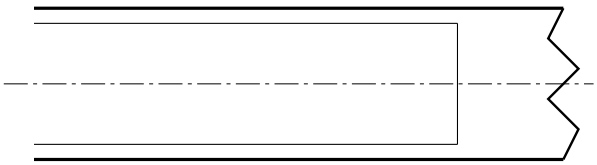


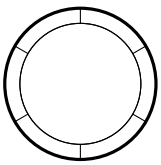
End view (section)



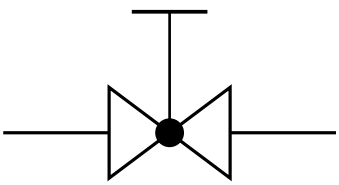
Sectional front view (pipe or tubing)

Splined shaft – front view

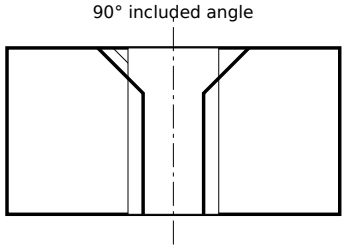




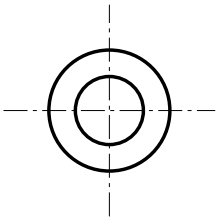
End view



Globe valve



Countersunk hole – section



Plan (two concentric circles)

Marks split	Marks
Any four items, 1 mark each	4
Total	**4**

Q.3 (b) [4 Marks]

Draw a sketch to represent taper 1:20 on a shaft of φ40mm and length 120mm.

Ans.

Numerical: Taper 1:20 on a shaft Ø40 × 120

Given data	Value	Unit	Remark
Large diameter D	40	mm	
Length L	120	mm	
Taper	1 : 20		taper = (D − d)/L

Formula and calculation:

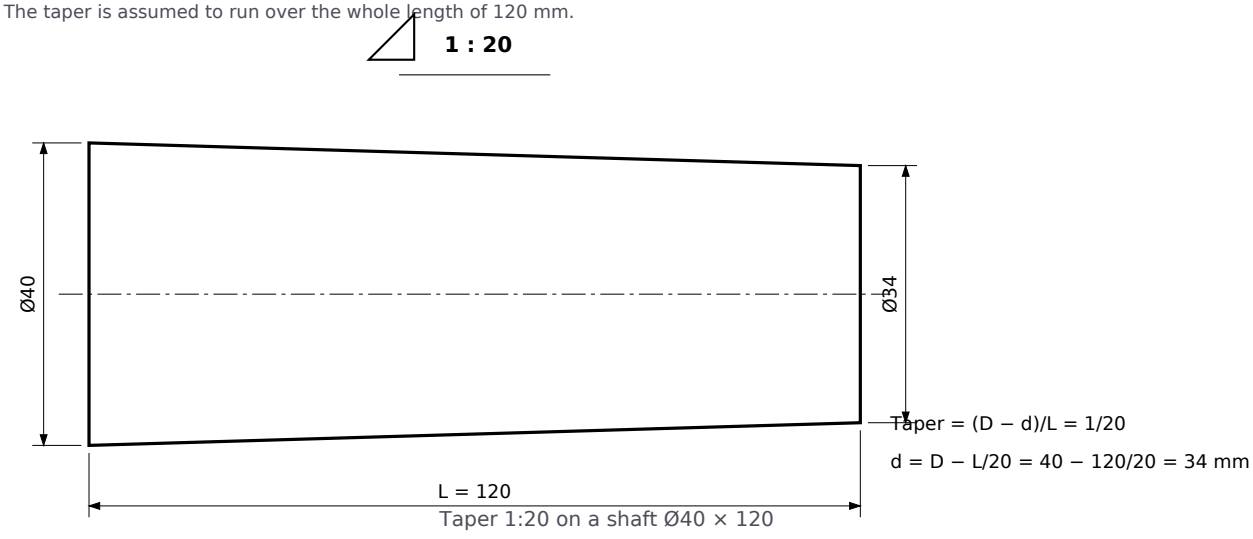
1. Small diameter

d = D − L / 20 = 40 − 120 / 20 = **34.00 mm**

Final answer
Small end diameter d = **34.00 mm**
Taper per unit length (D − d)/L = **1 : 20**

Checked in code by a second method: ✓ (D − d)/L = 1/k: 0.0500 ≈ 0.0500

Note: The taper is assumed to run over the whole length of 120 mm.



Marks split	Marks
Calculation of small end diameter	1
Sketch with taper symbol and dimensions	3
Total	**4**

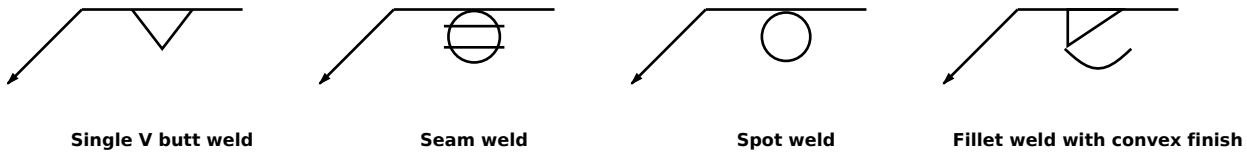
Q.3 (c) [4 Marks]

Draw the symbols for the any two of following:
i) Single V butt weld
ii) Seam weld
iii) Spot weld
iv) Fillet weld with convex finish.

Ans.

All four are shown; any two are to be attempted (2 marks each).

Arrow points to the joint; symbol below the reference line = weld on the arrow side.



Single V butt weld **Seam weld** **Spot weld** **Fillet weld with convex finish**

Marks split	Marks
Any two symbols, 2 marks each	4
Total	**4**

Q.3 (d) [4 Marks]

Draw the symbols of following features which are controlled in geometrical tolerance.
i) Symmetry
ii) Angularity
iii) Concentricity
iv) Flatness.

Ans.



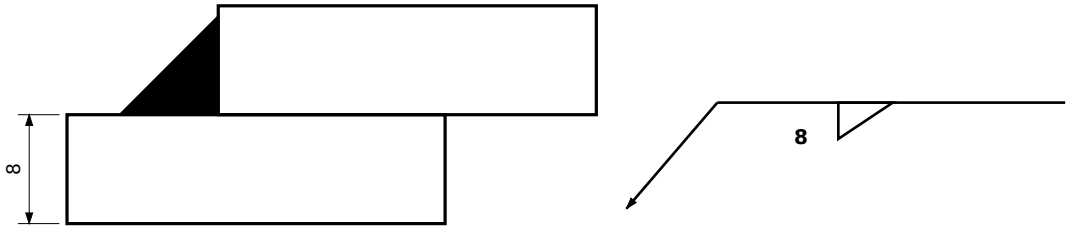
Marks split	Marks
Each symbol, 1 mark	4
Total	**4**

Q.3 (e) [4 Marks]

Two mild steel plants of 8 mm thickness are to be welded to have a lap joint by a fillet weld of leg length 8 mm. Represent the weld on drawing with proper symbol.

Ans.

Data note: The question says “plants”; plates are meant.
Assumption used: Read as “plates”.



Two M.S. plates, 8 mm thick, lap joint

Fillet weld, leg length 8 mm,
on the arrow side (symbol below line).

Lap joint with fillet weld, leg length 8 mm

1. Draw the two plates of 8 mm thickness overlapping (lap joint).
2. Draw the fillet weld (triangle with 8 mm legs) at the edge of the upper plate.
3. Draw the arrow line to the joint, the reference line and the fillet weld symbol (triangle) below the line, with the leg length 8 to its left.

Marks split	Marks
Lap joint drawing with weld	2
Weld symbol with 8 mm leg	2
Total	**4**

Q.3 (f) [4 Marks]

What is the meaning of symbol at x and y as shown in Figure No. 4.
[Fig. No. 4: X = ∥ 0.03 A, Y = ⊥ 0.03 A, datum A]

Ans.


Parallelism


Perpendicularity

- **X (∥ 0.03 A):** parallelism – the toleranced surface must lie between two parallel planes 0.03 mm apart that are parallel to datum A.
- **Y (⊥ 0.03 A):** perpendicularity – the toleranced surface must lie between two parallel planes 0.03 mm apart that are perpendicular to datum A.

Marks split	Marks
Meaning of X	2
Meaning of Y	2
Total	**4**

Q.3 (g) [4 Marks]

The shaft is given as φ20(+0.023/+0.012) and hole size is φ20(+0.018/+0.000). Determine the types of fit between them.

Ans.

Numerical: Fit between shaft Ø20(+0.023/+0.012) and hole Ø20(+0.018/0)

Given data	Value	Unit	Remark
Basic size	20	mm	
Shaft deviations (es, ei)	+0.023, +0.012	mm	
Hole deviations (ES, EI)	+0.018, +0.000	mm	

Formula and calculation:

1. Fit: limits of hole
H_{max} = B + ES, H_{min} = B + EI = 20 + (0.018), 20 + (0)
Hole: maximum = 20.018 mm, minimum = 20.000 mm
Shaft: maximum = 20.023 mm, minimum = 20.012 mm

2. Fit: maximum clearance
= $H_{max} - S_{min}$ = 20.018 − 20.012 = **0.006 mm**

3. Fit: minimum clearance
= $H_{min} - S_{max}$ = 20.000 − 20.023 = **-0.023 mm**
Fit: minimum clearance < 0 and maximum clearance > 0, so it is a TRANSITION fit.

Final answer
Maximum clearance = **0.006 mm**
Minimum clearance (negative = interference) = **-0.023 mm**
Type of fit = **Transition fit**

Checked in code by a second method: ✓ Fit: max clearance = ES − ei: 0.0060 ≈ 0.0060; ✓ Fit: min clearance = EI − es: -0.0230 ≈ -0.0230; ✓ Type is transition: 1.0000 ≈ 1.0000

Marks split	Marks
Limits of shaft and hole	1
Maximum and minimum clearance	2
Type of fit	1
Total	**4**

Q.4 Attempt the following: — 16 Marks

Q.4 [16 Marks]

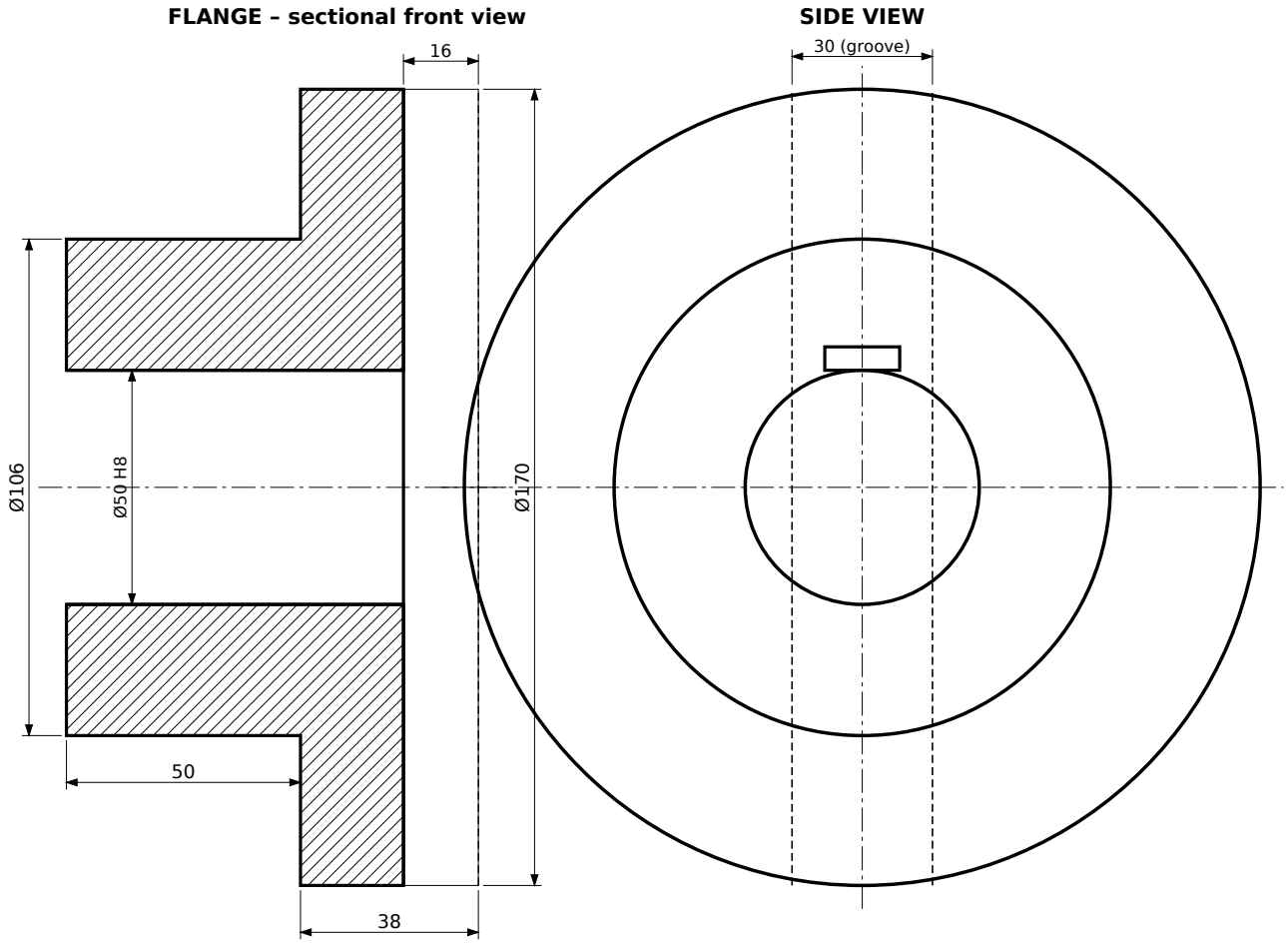
Figure No. 5 shows sectional F.V. and S.V. of the assembly of 'OLDHAM'S COUPLING'. Draw the following details by using first angle method:
i) Flange: Front view and S.V.
ii) Central disc: Front view and S.V.
iii) Shaft: Front view and S.V.
iv) Key: F.V., T.V and S.V.
[Part list: 1 Flange C.I. ×2; 2 Centre block C.I. ×1; 3 Keys M.S. ×2; 4 Shaft M.S. ×2. Fit chart: 16H7/n6 transition, 30H7/g6 clearance, 50H8/g8 clearance]

Ans.

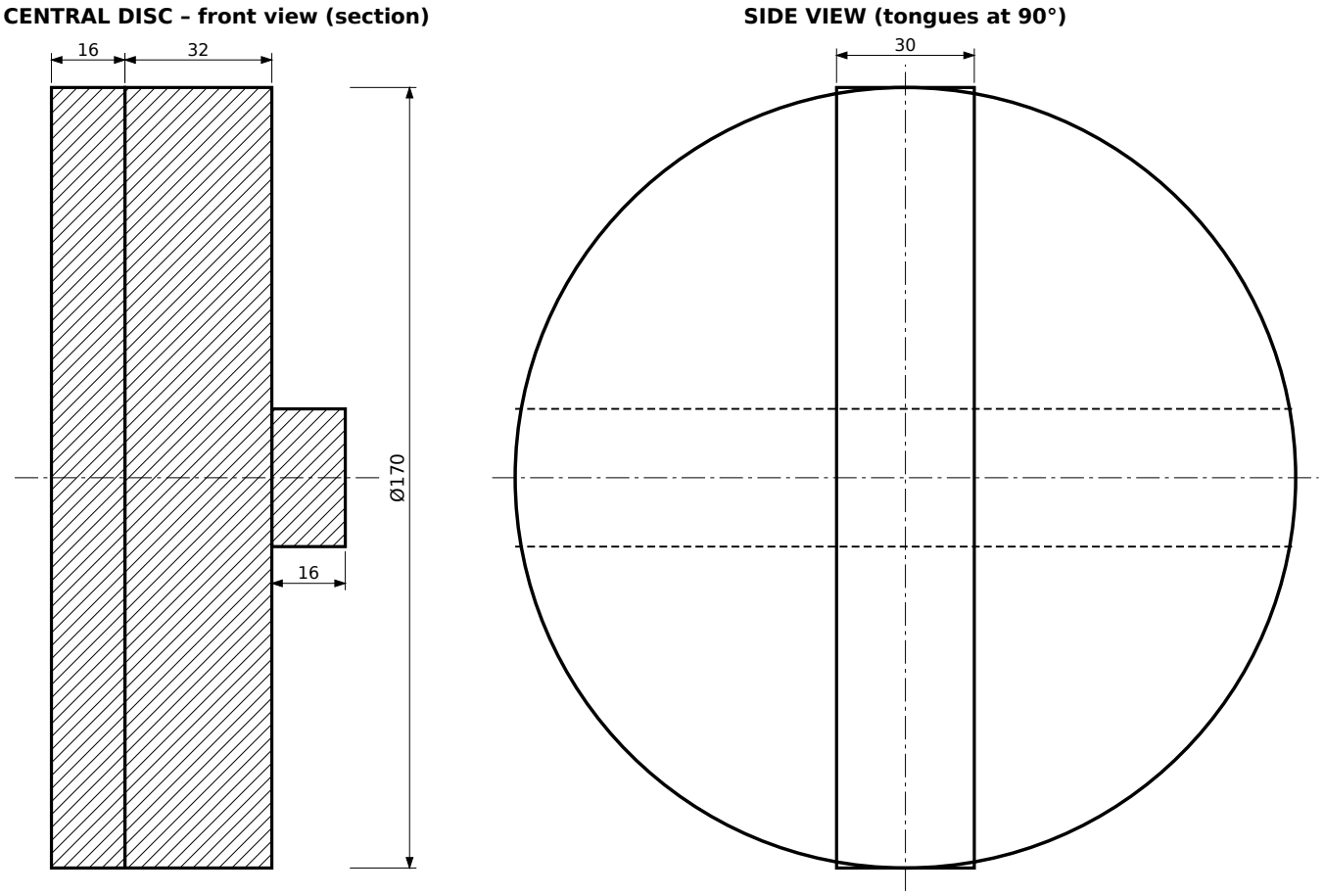
Data note: Fig. No. 5 is a rotated, low-resolution scan. The label "20" beside the groove, the exact length of the shaft and key, and the fit label printed on the drawing ("50H6/f6") differ from the fit chart ("50H8/g8").
Assumption used: Sizes in the table are those legible in the figure; the fit chart is used for the fits. The shaft and key length are drawn to suit the hub (not given).

Part No.	Part name	Material	Qty.
1	Flange	C.I.	2
2	Centre block (disc)	C.I.	1
3	Keys (taper 16 × 10)	M.S.	2
4	Shaft Ø50	M.S.	2

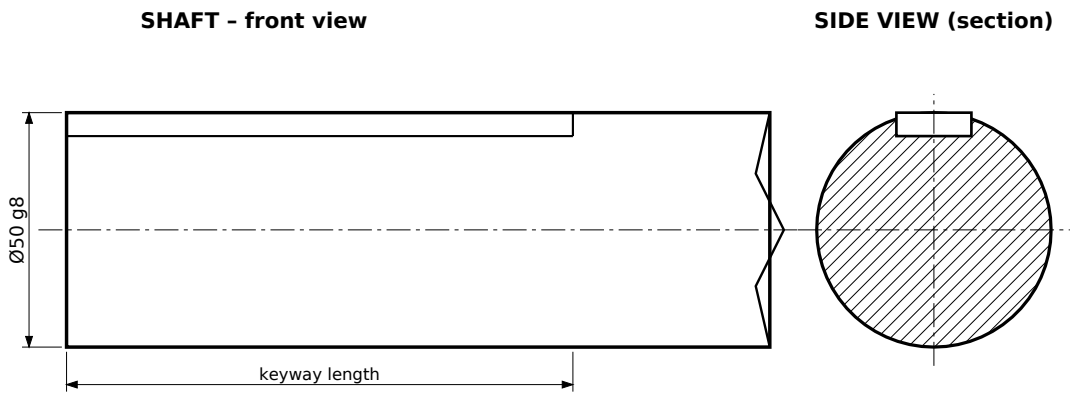
Feature (Fig. 5)	Size	Ratio to shaft d = 50
Flange head diameter	Ø170	3.4 d
Hub diameter	Ø106	2.1 d
Hub length	50	d
Head thickness (with 16 deep groove)	38	0.76 d
Groove (tongue) width	30	0.6 d
Groove depth = tongue height	16	0.32 d
Disc thickness at centre	16 + 16 = 32	0.64 d
Key	16 × 10, taper	0.32 d × 0.2 d
Overall length of assembly	50 + 38 + 16 + 16 + 38 + 50 = 208	–



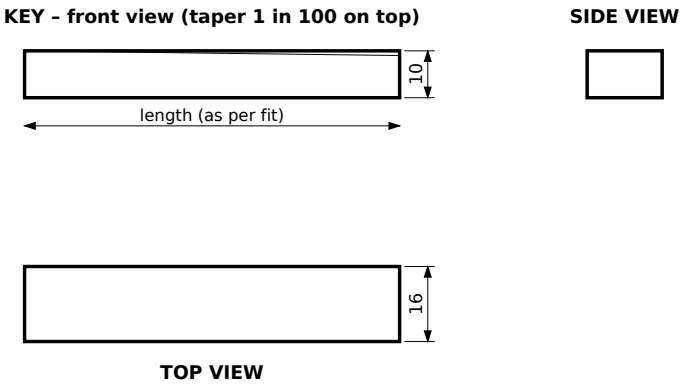
Flange – sectional front view and side view



Central disc – front view and side view



Shaft – front view and side view



Key – front view, top view and side view

Fits

Numerical: Fits in the Oldham's coupling (ISO 286 limits)

Given data	Value	Unit	Remark
Fit chart (printed)	16H7/n6 transition; 30H7/g6 clearance; 50H8/g8 clearance		
Limits used (standard ISO 286 values – not printed in the paper)	16H7 +0.018/0, n6 +0.023/+0.012; 30H7 +0.021/0, g6 −0.007/−0.020; 50H8 +0.039/0, g8 −0.009/−0.048	mm	

Formula and calculation:

1. 16 H7/n6 (key – keyway): limits of hole
H_{max} = B + ES, H_{min} = B + EI = 16 + (0.018), 16 + (0)
Hole: maximum = 16.018 mm, minimum = 16.000 mm
Shaft: maximum = 16.023 mm, minimum = 16.012 mm

2. 16 H7/n6 (key – keyway): maximum clearance
= $H_{max} - S_{min}$ = 16.018 − 16.012 = **0.006 mm**

3. 16 H7/n6 (key – keyway): minimum clearance
= $H_{min} - S_{max}$ = 16.000 − 16.023 = **-0.023 mm**
16 H7/n6 (key – keyway): minimum clearance < 0 and maximum clearance > 0, so it is a TRANSITION fit.

4. 30 H7/g6 (groove – tongue): limits of hole
H_{max} = B + ES, H_{min} = B + EI = 30 + (0.021), 30 + (0)
Hole: maximum = 30.021 mm, minimum = 30.000 mm
Shaft: maximum = 29.993 mm, minimum = 29.980 mm

5. 30 H7/g6 (groove – tongue): maximum clearance
$= H_{max} - S_{min} = 30.021 - 29.980 =$ **0.041 mm**

6. 30 H7/g6 (groove – tongue): minimum clearance
$= H_{min} - S_{max} = 30.000 - 29.993 =$ **0.007 mm**
30 H7/g6 (groove – tongue): minimum clearance ≥ 0 and maximum clearance > 0, so it is a CLEARANCE fit.

7. 50 H8/g8 (flange bore – shaft): limits of hole
$H_{max} = B + ES$, $H_{min} = B + EI = 50 + (0.039), 50 + (0)$
Hole: maximum = 50.039 mm, minimum = 50.000 mm
Shaft: maximum = 49.991 mm, minimum = 49.952 mm

8. 50 H8/g8 (flange bore – shaft): maximum clearance
$= H_{max} - S_{min} = 50.039 - 49.952 =$ **0.087 mm**

9. 50 H8/g8 (flange bore – shaft): minimum clearance
$= H_{min} - S_{max} = 50.000 - 49.991 =$ **0.009 mm**
50 H8/g8 (flange bore – shaft): minimum clearance ≥ 0 and maximum clearance > 0, so it is a CLEARANCE fit.

Final answer
16 H7/n6 = **TRANSITION fit**
30 H7/g6 = **CLEARANCE fit**
50 H8/g8 = **CLEARANCE fit**

Checked in code by a second method: ✓ 16 H7/n6 (key – keyway): max clearance = ES − ei: 0.0060 ≈ 0.0060; ✓ 16 H7/n6 (key – keyway): min clearance = EI − es: -0.0230 ≈ -0.0230; ✓ 30 H7/g6 (groove – tongue): max clearance = ES − ei: 0.0410 ≈ 0.0410; ✓ 30 H7/g6 (groove – tongue): min clearance = EI − es: 0.0070 ≈ 0.0070; ✓ 50 H8/g8 (flange bore – shaft): max clearance = ES − ei: 0.0870 ≈ 0.0870; ✓ 50 H8/g8 (flange bore – shaft): min clearance = EI − es: 0.0090 ≈ 0.0090; ✓ Types agree with the printed fit chart (transition, clearance, clearance): 1.0000 ≈ 1.0000

Procedure

1. Study the assembly: note the part sizes (the dimension string 50, 38, 16, 16, 38, 50) and the fits.
2. Flange: draw the centre line, hub Ø106 × 50, head Ø170 × 38 with a 30 wide × 16 deep groove across the face, bore Ø50 with the keyway for the 16 × 10 key; hatch at 45° in the front view; side view: circles Ø170, Ø106, Ø50 and the groove as dashed lines.
3. Central disc: Ø170 plate, 32 thick, with a 30 wide × 16 high tongue on each face, the two tongues at 90° to each other; side view shows the disc, one tongue as a rectangle and the other as dashed lines.
4. Shaft: cylinder Ø50 with a broken end, keyway 16 wide for the key; side view in section with the keyway.
5. Key: rectangular 16 × 10, length to suit (hub bore), with the 1 in 100 taper on the top face; draw the three views.
6. Give main dimensions, fits and tolerances shown in the fit chart.

Marks split	Marks
Flange – F.V. and S.V.	5
Central disc – F.V. and S.V.	4
Shaft – F.V. and S.V.	3
Key – F.V., T.V., S.V.	2
Dimensions and fits	2
Total	**16**

Q.5 Attempt any ONE of the following: — 16 Marks

Q.5 (a) [16 Marks]

Figure No. 6 shows the details of ‘FOOT STEP BEARING’. Draw the following views of the assembly:
i) Section Front view
ii) Top view
Use suitable scale
Give overall dimension. Show assembly fits.
[Parts: 1 Body C.I.; 2 Bush G.M.; 3 Shaft M.S.; 4 Disc G.M.; 5 Pin M.S.; each 1 off]

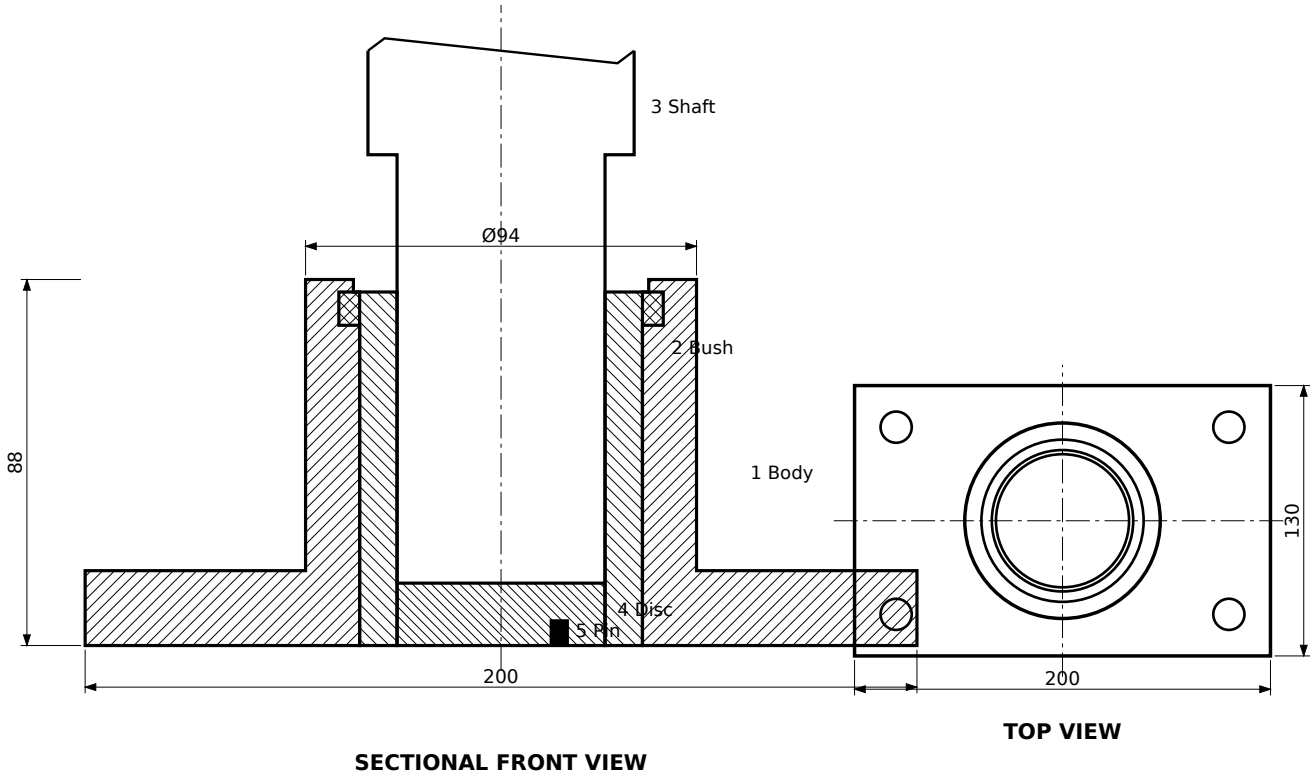
Ans.

Data note: Fig. 6 is small and several labels are doubtful: the body bore is printed “Ø68Hz” (read as H7); the bush is “68g6” on the part but the tolerance chart gives “68h6”; the bush bore “50H7” has the chart value +0.039 (that is H8); signs of e8 and g6 are lost in the scan.
Assumption used: H7 is used for the body bore; the chart values (68H7/68h6, 50H7 +0.039/0 with 50e8 −0.050/−0.089, 8H7/8g6) are used for the fits. The assembly sketch shows the main arrangement with approximate proportions.

Part No.	Part name	Material	Qty.
1	Body	C.I.	1
2	Bush	G.M. (gun metal)	1
3	Shaft	M.S.	1
4	Disc	G.M. (gun metal)	1
5	Pin	M.S.	1

Bill of material

Part	Main sizes (read from Fig. 6)
Body	Base 200 × 130 (4 slotted holes 15 wide); boss Ø94; height 88; bore Ø68 H7; sunk recess Ø71, 3 deep; pin hole Ø4.6
Bush	Ø68 outside, bore Ø50 H7, collar Ø78, 8 thick; length about 70; 8 wide oil groove
Shaft	Ø64 upper part, Ø50 H8 lower part 72 long; end R60
Disc	Ø50 e8 × 15, dished R60, pin hole Ø4, 6 deep
Pin	Ø4 × 12



Foot step bearing – sectional front view and top view (approximate proportions)

Procedure

1. Choose a scale (1 : 1 or 1 : 2). Draw the centre line and the base line. Draw the body: base 200 × 18, boss Ø94 up to 88, bore Ø68 with the Ø71 sunk recess 3 deep.
2. Draw the bush Ø68 outside inside the bore, bore Ø50, with the collar Ø78 × 8 resting in the recess.
3. Draw the disc (Ø50 × 15) at the bottom of the bush bore with the pin (Ø4) locating it; then the shaft Ø50 resting on the dished disc and its Ø64 upper portion above the bush.
4. Hatch the cut parts at 45° in different directions for each part; the shaft, pin and disc are shown according to the sectioning rules (shaft and pin are not hatched).
5. Draw the top view: rectangle 200 × 130, four slotted holes, circles Ø94, Ø78, Ø68, Ø64; centre lines.
6. Give overall dimensions (200, 130, 88) and the fits (Ø68 H7/h6, Ø50 H7/e8, Ø8 H7/g6) with part numbers.

Fits

Numerical: Fits in the foot step bearing (tolerance chart)

Given data	Value	Unit	Remark
Tolerance chart (printed)	68H7: +0.030/0; 68h6: 0/−0.019; 50H7: +0.039/0; 50e8: −0.050/−0.089; 8H7: +0.015/0; 8g6: −0.005/−0.014 (signs read from the scan)	mm	

Formula and calculation:

1. Ø68 H7/h6 (body bore – bush): limits of hole

H_{max} = B + ES, H_{min} = B + EI = 68 + (0.03), 68 + (0)

Hole: maximum = 68.030 mm, minimum = 68.000 mm

Shaft: maximum = 68.000 mm, minimum = 67.981 mm

2. Ø68 H7/h6 (body bore – bush): maximum clearance
$= H_{max} - S_{min} = 68.030 - 67.981 =$ **0.049 mm**

3. Ø68 H7/h6 (body bore – bush): minimum clearance
$= H_{min} - S_{max} = 68.000 - 68.000 =$ **0.000 mm**
Ø68 H7/h6 (body bore – bush): minimum clearance ≥ 0 and maximum clearance > 0, so it is a CLEARANCE fit.

4. Ø50 H7/e8 (bush bore – shaft/disc): limits of hole
$H_{max} = B + ES$, $H_{min} = B + EI = 50 + (0.039), 50 + (0)$
Hole: maximum = 50.039 mm, minimum = 50.000 mm
Shaft: maximum = 49.950 mm, minimum = 49.911 mm

5. Ø50 H7/e8 (bush bore – shaft/disc): maximum clearance
$= H_{max} - S_{min} = 50.039 - 49.911 =$ **0.128 mm**

6. Ø50 H7/e8 (bush bore – shaft/disc): minimum clearance
$= H_{min} - S_{max} = 50.000 - 49.950 =$ **0.050 mm**
Ø50 H7/e8 (bush bore – shaft/disc): minimum clearance ≥ 0 and maximum clearance > 0, so it is a CLEARANCE fit.

7. Ø8 H7/g6 (pin): limits of hole
$H_{max} = B + ES$, $H_{min} = B + EI = 8 + (0.015), 8 + (0)$
Hole: maximum = 8.015 mm, minimum = 8.000 mm
Shaft: maximum = 7.995 mm, minimum = 7.986 mm

8. Ø8 H7/g6 (pin): maximum clearance
$= H_{max} - S_{min} = 8.015 - 7.986 =$ **0.029 mm**

9. Ø8 H7/g6 (pin): minimum clearance
$= H_{min} - S_{max} = 8.000 - 7.995 =$ **0.005 mm**
Ø8 H7/g6 (pin): minimum clearance ≥ 0 and maximum clearance > 0, so it is a CLEARANCE fit.

Final answer
Ø68 H7/h6 = **CLEARANCE fit (clearance 0.000 to 0.049 mm)**
Ø50 H7/e8 = **CLEARANCE fit (clearance 0.050 to 0.128 mm)**
Ø8 H7/g6 = **CLEARANCE fit (clearance 0.005 to 0.029 mm)**

Checked in code by a second method: ✓ Ø68 H7/h6 (body bore – bush): max clearance = ES − ei: 0.0490 ≈ 0.0490; ✓ Ø68 H7/h6 (body bore – bush): min clearance = EI − es: 0.0000 ≈ 0.0000; ✓ Ø50 H7/e8 (bush bore – shaft/disc): max clearance = ES − ei: 0.1280 ≈ 0.1280; ✓ Ø50 H7/e8 (bush bore – shaft/disc): min clearance = EI − es: 0.0500 ≈ 0.0500; ✓ Ø8 H7/g6 (pin): max clearance = ES − ei: 0.0290 ≈ 0.0290; ✓ Ø8 H7/g6 (pin): min clearance = EI − es: 0.0050 ≈ 0.0050

Marks split	Marks
Sectional front view	7
Top view	4
Overall dimensions	2
Assembly fits	2
Part numbers / hatching	1
Total	**16**

Q.5 (b) [16 Marks]

Figure No. 7 shows the details of ‘LATHE TOOL POST’. Draw the following views of the assembly:
i) Sectional front view
ii) Top view
iii) Prepare bill of material. Indicate types of fit.
[Parts: 1 Post C.I.; 2 Block C.I.; 3 Ring C.I.; 4 Wedge C.I.; 5 Screw C.I.; each 1 off]

Ans.

Data note: The post carries “φ60H7” and the block hole “φ60e8”, which is the reverse of normal practice (capital letters are for holes). The wedge shows “24 h6” in its view while the chart gives “24n6”.
Assumption used: The Ø60 hole (block) is taken as H7 and the Ø60 post as e8 (clearance fit) – if taken literally the hole would be smaller than the shaft. The wedge is taken as 24n6 as in the chart.

Part No.	Part name	Material	Qty.	Remark
1	Post	C.I.	1	Ø72 × 12 flange, Ø60 body, 144 high

Part No.	Part name	Material	Qty.	Remark
2	Block	C.I.	1	90 square × 18, Ø74 recess, Ø60 hole
3	Ring	C.I.	1	Ø120 × 18, Ø60 bore, R146 seat
4	Wedge	C.I.	1	24 wide, 146 long, R146
5	Screw	C.I.	1	M18 × 72, square head 18, collar Ø36

Bill of material

Numerical: Lathe tool post geometry (R = 146)

Given data	Value	Unit	Remark
Wedge / ring seat radius R	146	mm	
Ring: outer radius, bore radius, thickness at outer edge	60, 30, 18	mm	

Formula and calculation:

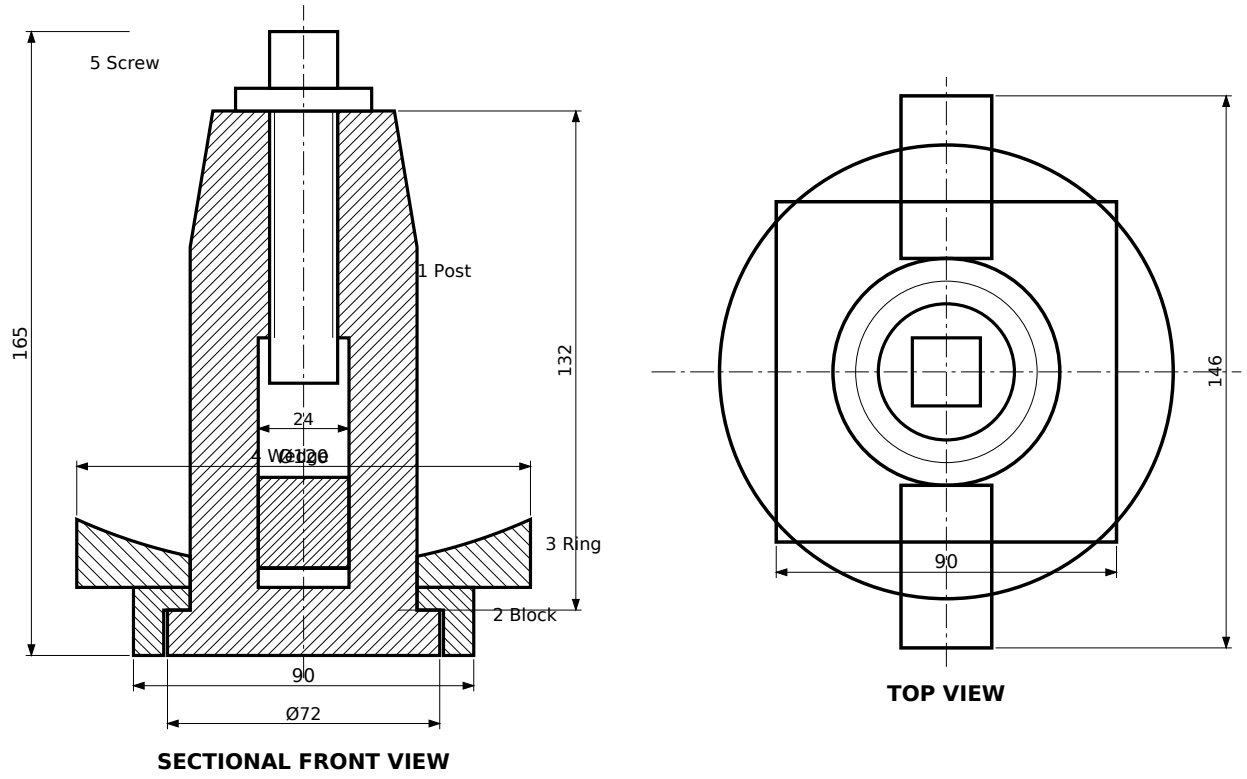
1. Fall of the concave seat from the ring edge to its centre
$s = R - \sqrt{(R^2 - 60^2)} = 146 - \sqrt{(146^2 - 60^2)} =$ **12.90 mm**

2. Seat height at the axis
$z_0 = 36 - s = 36 - 12.90 =$ **23.10 mm above the bottom of the block**

3. Wedge: depth of the curve at its ends (chord = R, as the wedge length equals R)
$s_w = R - \sqrt{(R^2 - (R/2)^2)} = 146 - \sqrt{(146^2 - 73^2)} =$ **19.56 mm**

Final answer
Seat height at the axis z_0 = **23.10 mm**
Depth of the wedge curve over its length = **19.56 mm**

Checked in code by a second method: ✓ Seat height + sag = ring edge height (36 mm): 36.0000 ≈ 36.0000



Lathe tool post – sectional front view and top view (assembly)

Procedure

1. Draw the centre line. Draw the block: 90 square × 18, with the Ø74 recess 12 deep from below and the Ø60 hole in the 6 mm top plate.
2. Draw the post: flange Ø72 × 12 in the recess, body Ø60 rising to 144 high, tapered to Ø48 over the top 36 mm; the slot 24 wide and 66 long, starting 60 below the top; the M18 hole at the top.

3. Place the ring Ø120 × 18 round the post, resting on the block; its top is the concave R146 seat (calculated above).
4. Place the wedge (24 × 24 section) in the slot of the post on the ring seat.
5. Insert the screw M18: square head 18, collar Ø36 × 6, shank 72 long; show the screw unsectioned.
6. Hatch the cut parts at 45° (adjacent parts in opposite directions). Draw the top view by projection.
7. Add overall dimensions, part numbers, bill of material and the types of fit.

Types of fit

Numerical: Fits in the lathe tool post (S25 tolerance chart)

Given data	Value	Unit	Remark
Tolerance chart (printed)	Ø60H7: +0.000/+0.030; Ø60e8: −0.060/−0.106; 24H7: +0.021/+0.000; 24n6: +0.036/+0.015	mm	

Formula and calculation:

1. Ø60 H7/e8 (block hole – post): limits of hole
H_{max} = B + ES, H_{min} = B + EI = 60 + (0.03), 60 + (0)
Hole: maximum = 60.030 mm, minimum = 60.000 mm
Shaft: maximum = 59.940 mm, minimum = 59.894 mm

2. Ø60 H7/e8 (block hole – post): maximum clearance
= $H_{max} - S_{min}$ = 60.030 − 59.894 = **0.136 mm**

3. Ø60 H7/e8 (block hole – post): minimum clearance
= $H_{min} - S_{max}$ = 60.000 − 59.940 = **0.060 mm**
Ø60 H7/e8 (block hole – post): minimum clearance ≥ 0 and maximum clearance > 0, so it is a CLEARANCE fit.

4. 24 H7/n6 (post slot – wedge): limits of hole
H_{max} = B + ES, H_{min} = B + EI = 24 + (0.021), 24 + (0)
Hole: maximum = 24.021 mm, minimum = 24.000 mm
Shaft: maximum = 24.036 mm, minimum = 24.015 mm

5. 24 H7/n6 (post slot – wedge): maximum clearance
= $H_{max} - S_{min}$ = 24.021 − 24.015 = **0.006 mm**

6. 24 H7/n6 (post slot – wedge): minimum clearance
= $H_{min} - S_{max}$ = 24.000 − 24.036 = **-0.036 mm**
24 H7/n6 (post slot – wedge): minimum clearance < 0 and maximum clearance > 0, so it is a TRANSITION fit.

Final answer
Ø60 H7/e8 – min / max clearance = **0.060 / 0.136 mm**
Ø60 H7/e8 = **CLEARANCE fit**
24 H7/n6 – max clearance / max interference = **0.006 / 0.036 mm**
24 H7/n6 = **TRANSITION fit**

Checked in code by a second method: ✓ Ø60 H7/e8 (block hole – post): max clearance = ES − ei: 0.1360 ≈ 0.1360; ✓ Ø60 H7/e8 (block hole – post): min clearance = EI − es: 0.0600 ≈ 0.0600; ✓ 24 H7/n6 (post slot – wedge): max clearance = ES − ei: 0.0060 ≈ 0.0060; ✓ 24 H7/n6 (post slot – wedge): min clearance = EI − es: -0.0360 ≈ -0.0360

Marks split	Marks
Sectional front view	6
Top view	4
Bill of material	2
Types of fit	2
Dimensions, hatching, finish	2
Total	**16**

Papers not received: Winter 2025 – not included

Not received, so not included: Winter 2025.

Questions and solutions for this paper will be added when the paper is supplied. Nothing has been guessed.

Chapter-wise marks table

Marks offered = marks of every sub-question printed in the paper, including the choice ('attempt any …'), so a column total can be more than 70. Chapter names are my own classification of each question (the syllabus was not supplied).

Conventions and standard representations (BIS SP-46)	**18**	**18**	**36**	**16.7 %**
Limits, fits, tolerances, surface finish and welding symbols	20	24	**44**	20.4 %
Orthographic and auxiliary views	16	16	**32**	14.8 %
Assembly and detail drawings	56	48	**104**	48.1 %
Total (marks offered)	**110**	**106**	**216**	100 %

Papers counted: Winter 2024, Summer 2025.

Top 10 repeated questions

Ranking rule: topics are ranked by the number of supplied sessions in which they appear (2 sessions supplied), then by total marks. 14 topic(s) appeared in more than one session. Topics that appeared only once are shown after them and are marked 'not repeated'.

#	Question / topic	Where asked	Marks	Sessions
1	Orthographic / auxiliary view completion	W24: Q.2(a), Q.2(b); S25: Q.2(a), Q.2(b)	32	2 (repeated)
2	Oldham's coupling – assembly / details	W24: Q.5(a); S25: Q.4	32	2 (repeated)
3	Lathe tool post – assembly, bill of material, fits	W24: Q.5(b); S25: Q.5(b)	32	2 (repeated)
4	Taper – calculation, sketch and convention	W24: Q.1(b); S25: Q.1(d), Q.3(b)	8	2 (repeated)
5	Material and component conventions	W24: Q.3(a); S25: Q.3(a)	8	2 (repeated)
6	Type of fit from limits (numerical)	W24: Q.3(d); S25: Q.3(g)	8	2 (repeated)
7	Meaning of geometrical tolerance frames	W24: Q.3(e); S25: Q.3(f)	8	2 (repeated)
8	Welding drawing and symbol for a joint	W24: Q.3(f); S25: Q.3(e)	8	2 (repeated)
9	Geometrical tolerance symbols	W24: Q.1(d); S25: Q.3(d)	6	2 (repeated)
10	Holes on circular pitch and bearings	W24: Q.3(b); S25: Q.1(b)	6	2 (repeated)

Last-minute revision sheet

Lines and conventions

- Thick continuous: outlines; thin continuous: dimensions, hatching; dashed: hidden edges; chain thin: centre/pitch lines; chain thick at ends: cutting plane.
- Section hatching 45° thin lines; adjacent parts opposite direction; C.I. close thin lines; copper/brass alternate thick-thin; rubber almost solid.
- Shafts, keys, bolts, pins, ribs are not sectioned when the cutting plane passes along their axis.
- Spur gear: addendum thick, pitch thin chain, root thin (not in section). Teeth not drawn.

Taper, thread, springs

- Taper = $(D - d)/L = 1 : k$; $d = D - L/k$. Taper symbol: triangle pointing towards small end, with ratio.
- External thread: major dia thick, minor dia thin; internal thread: minor dia thick, major dia thin; end view 3/4 thin circle.
- Short break: S-curves; long break: thin zig-zag.
- Semi-elliptic leaf spring conventional: single arc with a cross at the centre.

Limits, fits, tolerances

- Upper limit = basic + upper deviation; lower limit = basic + lower deviation. Tolerance = upper − lower limit.
- Max clearance = hole max − shaft min; Min clearance = hole min − shaft max.
- Min clearance ≥ 0: clearance fit; max clearance ≤ 0: interference fit; otherwise transition fit.
- Capital letters (H) for holes, small letters (e, g, h, n) for shafts. H7/e8 clearance; H7/n6 transition; H7/p6 interference.
- Hole-basis: H hole, EI = 0.

Geometrical tolerance symbols

- Form: straightness —, flatness ▱, circularity ○, cylindricity.
- Orientation: parallelism ∥, perpendicularity ⊥, angularity ∠.
- Location: position, concentricity ◎, symmetry ⌷.
- Frame: symbol | tolerance value | datum letter.

Surface finish and welding

- Machining symbol: Ra (µm) on top; lower limit left; method above extension line; sampling length below; lay letter R, C, M, X, = on the right.
- Weld symbols below the reference line = arrow side; above = other side.
- Single V butt ∨, fillet ◺ (leg length at left), spot ○, seam ○ with two lines, U butt ∪; convex contour = arc over the symbol.

Projection (first angle)

- FV above TV; LHSV to the right of FV.
- Auxiliary view: reference line ⌷ to direction of view; heights/widths measured from the reference line.
- True length of an inclined line in FV: L; its top view = $L \cos\theta$; height = $L \sin\theta$.
- A circle on an inclined plane appears as an ellipse: width = Ø $\cos\theta$.

Assembly drawings

- Steps: study details → list parts and materials → choose scale → main body first → inner parts → hatch → dimensions (overall) → bill of material.
- Oldham's coupling: 2 flanges + central disc with two tongues at 90°, taper keys 1 in 100.
- Lathe tool post: post + block + ring (concave seat) + wedge + screw.
- Universal coupling: 2 forks + centre block + 2 pins + taper pins + keys.
- Foot step bearing: body + bush + shaft + disc + pin.